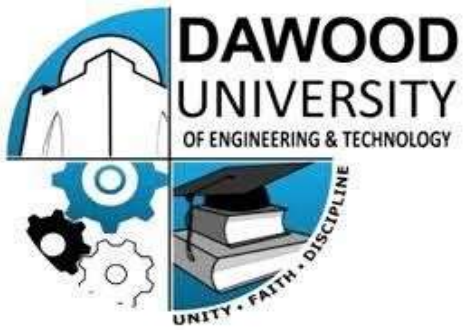
AI-3206 - Computer Vision

(Practical Lab Manual)



6th Semester, 3rd Year
BATCH -2023

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BS ARTIFICIAL INTELLIGENCE

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

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BS (ARTIFICIAL INTELLIGENCE)

CERTIFICATE

This is to certify that Ms./Mr **Mehak Faheem** Roll No **23-AI-47** has satisfactorily completed the Lab course of "**AI-3206 - Computer Vision"** approved by Lab Committee of the Department of **BS (ARTIFICIAL INTELLIGENCE)** for Batch 2023F in the academic year **2026.**

Date: **13th June 2026**

Signature,
Lab Instructor

LAB RULES AND OPERATING PROCEDURES

Before starting a lab, you must read the following instructions. Failure to conform to any of the below Instructions may result in not being allowed to participate in the laboratory experiment.

Respect the Lab Rules:
Respect the specific guidelines provided by the lab supervisor or instructor.

Quiet Environment:
Keep noise levels to a minimum to create a conducive learning environment for everyone.

No Food or Drinks:
Do not bring food or drinks into the computer lab to prevent spills and maintain cleanliness.

Log in with Your Own Credentials:
Use only your assigned login credentials, and do not attempt to access another student's account.

Log Out Properly:
Always log out of the computer and any applications or platforms you use before leaving the computer.

Respect Equipment:
Treat computer equipment and peripherals with care. Report any malfunctioning equipment to lab staff.

No Unauthorized Software Installation:
Do not install or attempt to install any software on the lab computers without permission from lab staff or instructors.

No Tampering with Hardware:
Do not tamper with computer hardware or cables. Report any issues to lab staff.

Internet Usage:
Use the internet for educational purposes only. Avoid accessing inappropriate or non-educational websites.

Be Mindful of Time:
Be aware of the lab's opening and closing hours. Finish your work and leave the lab on time.

Personal Belongings:
Keep personal belongings secure. Do not leave valuables unattended.

Emergency Procedures:
Familiarize yourself with emergency procedures, including the location of exits and emergency contacts.

I have read and understand these rules and procedures. I agree to abide by these rules and procedures at all times while using these facilities. I understand that failure to follow these rules and procedures will result in my immediate dismissal from the laboratory and additional disciplinary action may be taken.

_____
Student’s Signature

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LAB # 01

DATA PREPROCESSING

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Demonstrates clear understanding of how OpenCV represents images as NumPy arrays, BGR colour order, and coordinate system; strongly relates to lab objectives.	Basic understanding of image representation and OpenCV functions; partial connection to lab objectives.	Little or no understanding of how images are stored or processed; unclear relevance to lab topic.	
Code Implementation (6 marks)	All operations (read, resize, rotate, flip, grayscale, blur, save) are correctly implemented, error-free, and well-structured.	Most operations work with minor errors or missing steps; partial implementation.	Code is incorrect, incomplete, or does not perform the required image operations.	
Use of Programming Features/Tools (4 marks)	Efficiently uses cv2.imread, cv2.resize, cv2.rotate, cv2.flip, cv2.cvtColor, cv2.GaussianBlur, and cv2.imwrite with correct parameters throughout.	Uses most functions with limited parameter control or minor incorrect usage.	Minimal or incorrect use of OpenCV functions; relies on hard-coded or trivial approaches.	
Results and Report (6 marks)	All output images are correctly saved and displayed; clear side-by-side comparisons with brief written observations for each operation.	Most outputs are correct; report is adequate but lacks visual comparisons or written observations.	Outputs are missing, incorrect, or not saved; no meaningful report.	

LAB # 01

INTRODUCTION TO COMPUTER VISION WITH OPENCV

OBJECTIVE

To understand and implement Data preprocessing in preparing real-world datasets for machine learning.

LAB OUTCOMES

- Import and utilize OpenCV in Python
- Read and display images using OpenCV
- Perform basic image operations, such as resizing, rotating, and flipping
- Apply image filtering techniques, including blurring and grayscale conversion
- Draw on images to annotate and highlight features
- Save processed images to the local filesystem

THEORY

What is Computer Vision?

Computer Vision is a field of Artificial Intelligence that enables machines to interpret and understand visual information from images and videos. Applications include medical imaging, autonomous vehicles, face recognition, and industrial quality inspection.

What is OpenCV?

OpenCV (Open Source Computer Vision Library) is an open-source library originally developed by Intel, providing over 2,500 optimised algorithms for image and video processing. In Python it is imported as cv2. Every image loaded by OpenCV is stored internally as a **NumPy array**, meaning all standard NumPy operations can be applied directly to images.

How Images Are Represented

A digital image is a grid of pixel values:

- A **grayscale image** has shape (Height, Width) — each pixel holds a single intensity value from 0 (black) to 255 (white).
- A **colour image** has shape (Height, Width, 3) — each pixel holds three channel values.

OpenCV reads colour images in **BGR order** (Blue, Green, Red) rather than the more common RGB. This means images displayed directly via matplotlib without colour conversion will appear with red and blue channels swapped.

Basic Image Operations

Operation	Function	Key Note
Read image	cv2.imread()	Returns NumPy array
Display image	cv2.imshow()	Use with cv2.waitKey(0)
Resize	cv2.resize(img, (W, H))	Width before Height
Rotate	cv2.rotate()	Three fixed angles available
Flip	cv2.flip(img, flipCode)	1=horizontal, 0=vertical, -1=both
Convert colour	cv2.cvtColor()	e.g. BGR → Grayscale
Save image	cv2.imwrite()	Format set by file extension

Gaussian Blur

Blurring reduces high-frequency noise by replacing each pixel with a weighted average of its neighbourhood. Gaussian blur uses a kernel whose weights follow a bell-curve distribution — pixels closer to the centre contribute more. cv2.GaussianBlur(image, (ksize, ksize), 0) applies it; kernel size must be an odd integer (3, 5, 7...). Larger kernels produce stronger smoothing. This is used as a standard preprocessing step before edge detection and filtering in later labs.

SOLVED LAB ACTIVITES:

Activity 1: Import and Utilize OpenCV in Python

Solution:

1. **Open Python IDE:**
   a. Launch your Python IDE (e.g., IDLE or Jupyter Notebook).
2. **Import OpenCV:**
   a. Type **import cv2** and press Enter.
   b. Ensure there are no errors, indicating successful import.

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
```

Activity 2: Read and Display Images using OpenCV

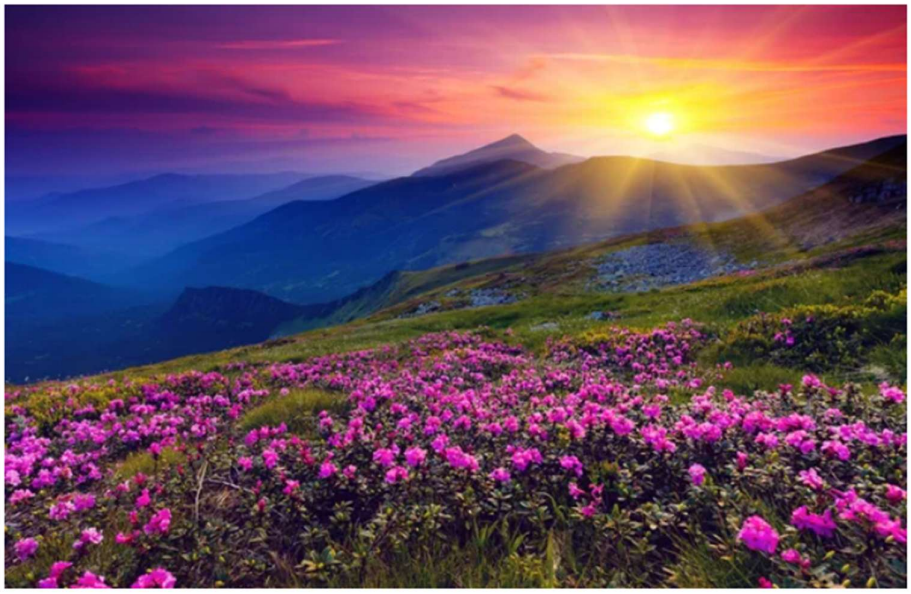
Solution:

1. **Read an Image:**
   a. Type image = cv2.imread('path_to_image.jpg') and press Enter.
   b. Replace 'path_to_image.jpg' with the actual path of an image file.

```
img = cv2.imread('image.jpg')
```

2. **Display the Image:**
   a. Type cv2.imshow('Image', image) and press Enter.
   b. Add cv2.waitKey(0) and cv2.destroyAllWindows() to keep the window open until a key is pressed.

```
cv2_imshow(img)
```

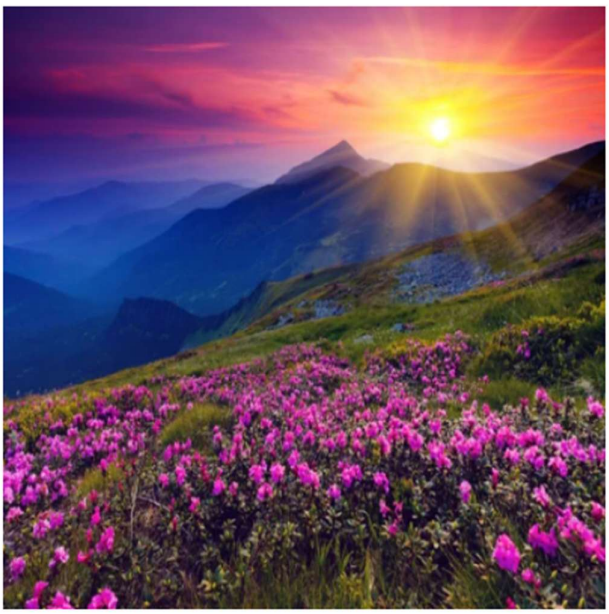


Activity 3: Perform Basic Image Operations

Solution:

1. **Resize the Image:**
    a. Type resized_image = cv2.resize(image, (new_width, new_height)) and press Enter.

```
from google.colab.patches import cv2_imshow
resized_image = cv2.resize(img, (400, 400))
cv2_imshow(resized_image)
```



2. Rotate the Image:

a. Type rotated_image = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE) and press Enter.

```
rotated_image = cv2.rotate(resized_image, cv2.ROTATE_90_CLOCKWISE)
cv2_imshow(rotated_image)
```

3. Flip the Image Horizontally:

a. Type flipped_image = cv2.flip(image, 1) and press Enter.

```
flipped_image = cv2.flip(rotated_image, 1)
cv2_imshow(flipped_image)
```

Activity 4: Apply Image Filtering Techniques

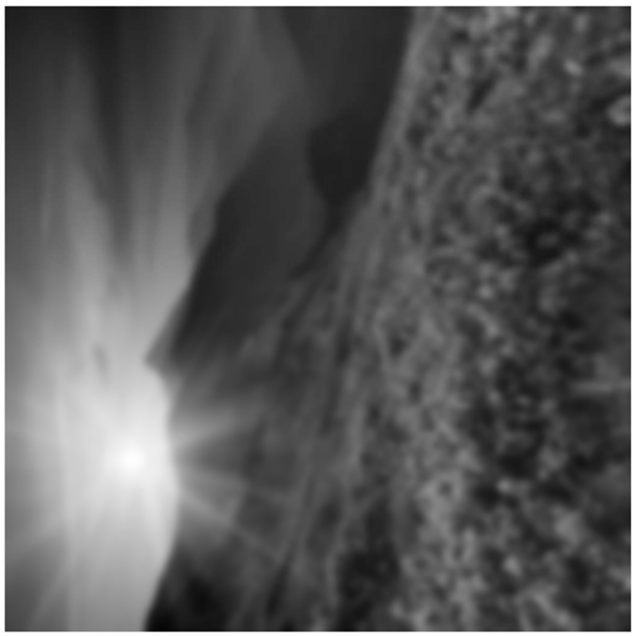
Solution:

1. **Apply Gaussian Blur:**
    a. Type blurred_image = cv2.GaussianBlur(image, (kernel_size, kernel_size), 0) and press Enter.

```
blurred_image = cv2.GaussianBlur(flipped_image, (15, 15), 0)
cv2_imshow(blurred_image)
```

2. **Convert Image to Grayscale:**
    a. Type gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY) and press Enter.

```
gray_image = cv2.cvtColor(blurred_image, cv2.COLOR_BGR2GRAY)
cv2_imshow(gray_image)
```



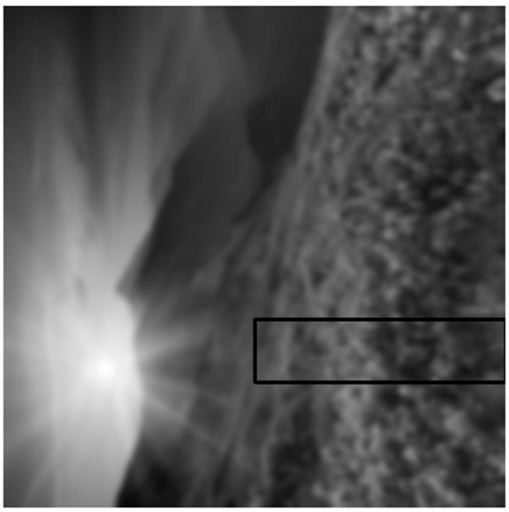
Activity 5: Draw on Images to Annotate and Highlight Features

Solution:

1. **Draw a Rectangle:**
   a. Type cv2.rectangle(image, (start_x, start_y), (end_x, end_y), (0, 255, 0), 2) and press Enter.

```
image_with_rectangle = gray_image.copy()
```

```
rectangle = cv2.rectangle(image_with_rectangle, (400, 250), (200, 300) , (0, 255, 0), 2)
cv2_imshow(rectangle)
```



2. **Add Text:**
   a. Type cv2.putText(image, 'Hello, OpenCV!', (start_x, start_y), cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 0, 0), 2) and press Enter.

```
text = cv2.putText(resized_image, 'Hello, OpenCV!', (100, 250), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
cv2_imshow(text)
```

Activity 6: Save Processed Images to the Local Filesystem

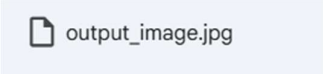
Solution:

1. **Save Image:**
   a. Type cv2.imwrite('output_image.jpg', image) and press Enter.

```
cv2.imwrite('output_image.jpg', img)
```

```
True
```

   b. Check your directory for the newly saved image.



LAB TASKS

Before performing, please read the material below.

https://medium.com/nerd-for-tech/using-opencv-with-python-to-perform-basic-operations-on-imageshttps://medium.com/nerd-for-tech/using-opencv-with-python-to-perform-basic-operations-on-images-bb396153ae2abb396153ae2a

Lab Task 1:

Basic Image Operations

Instructions:

- Read any open-source image.

```
img = cv2.imread('image.jpg')
```

- Resize the image to half its original dimensions.

```
img_resize = cv2.resize(img, (img.shape[1]//2, img.shape[0]//2))
cv2_imshow(img_resize)
```

- Rotate the resized image 180 degrees clockwise.

```
rotated_image = cv2.rotate(img_resize, cv2.ROTATE_180)
cv2_imshow(rotated_image)
```

- Flip the rotated image vertically.

```
flipped_image = cv2.flip(rotated_image, 0)
cv2_imshow(flipped_image)
```

Lab Task 2:

Image Filtering Techniques

Instructions:

- Take the image from Exercise 2.
- Blur the image using a Gaussian blur with a kernel size of 5x5.

```
blurred_image = cv2.GaussianBlur(flipped_image, (5, 5), 0)
cv2_imshow(blurred_image)
```

- Convert the blurred image to grayscale.

```
gray_image = cv2.cvtColor(blurred_image, cv2.COLOR_BGR2GRAY)
cv2_imshow(gray_image)
```

- Save the image

```
cv2.imwrite("final_output.jpg", gray_image)
```

```
True
```

LAB # 02

DRAWING FUNCTIONS IN OPENCV

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly understands OpenCV's coordinate system, BGR colour format, and the role of drawing functions in CV pipelines; strongly relates to lab objectives.	Basic understanding of drawing functions and colour format; minimal connection to lab objectives.	Little or no understanding of coordinate system or colour representation; unclear relevance.	
Code Implementation (6 marks)	All eight tasks correctly implemented — ellipses, polygon conversion, filled polygons, rectangles, lines, arrowed lines, polylines, and text — in a single well-structured script with comments.	Most tasks implemented with minor errors or missing tasks; partial completion of the script.	Code is incorrect, incomplete, or fewer than four tasks are implemented.	
Use of Programming Features/Tools (4 marks)	Correctly and efficiently uses cv2.ellipse, cv2.ellipse2Poly, cv2.fillConvexPoly, cv2.rectangle, cv2.line, cv2.arrowedLine, cv2.polylines, and cv2.putText with varied parameters.	Uses most functions with limited parameter variation or partial correct usage.	Minimal use of drawing functions; most tasks rely on a single repeated function or incorrect syntax.	
Results and Report (6 marks)	All eight task outputs clearly displayed with distinct parameters; combined script is well-commented and submitted with documentation explaining each task.	Most task outputs are correct; script is partially commented; documentation is basic.	Fewer than four task outputs shown; script lacks comments or documentation.	

LAB # 02

DRAWING FUNCTIONS IN OPENCV

OBJECTIVE

Drawing functions in OpenCV facilitate the manipulation of images and matrices, providing the ability to create and visualize various shapes, lines, and text. These functions are designed to work with matrices or images of arbitrary depth, allowing for a wide range of applications in computer vision and image processing..

LAB OUTCOMES

- Understand OpenCV's coordinate system and BGR colour representation used in all drawing functions
- Draw geometric shapes including circles, ellipses, rectangles, lines, arrowed lines, and polylines on images using OpenCV
- Convert ellipses to polygon point arrays using `cv2.ellipse2Poly` and render them as polylines
- Fill convex polygons with solid colours using `cv2.fillConvexPoly`
- Add text annotations on images with control over font, size, colour, and position using `cv2.putText`
- Combine multiple drawing functions into a single annotated image to simulate real-world CV output visualisation

THEORY

Why Drawing Functions Matter

In Computer Vision, drawing functions are not just for aesthetics — they are an essential tool for visualising results. When a model detects an object, we draw a bounding box around it. When a tracking algorithm follows a path, we draw that trail on screen. When we debug a pipeline, we annotate frames with labels and measurements. Every lab from Lab 8 onwards will use these functions extensively to display detection and tracking results.

Coordinate System

OpenCV uses a **pixel coordinate system** where the origin `(0, 0)` is at the **top-left corner** of the image. The x-axis increases to the right and the y-axis increases downward. All drawing functions accept coordinates as `(x, y)` tuples — remember this is the opposite of NumPy's `(row, column)` convention which corresponds to `(y, x)`.

Colour Representation

OpenCV represents colours in **BGR format** (Blue, Green, Red) as a tuple of three integers, each between 0 and 255. Common colours:

Colour	BGR Tuple
Red	(0, 0, 255)

Colour	BGR Tuple
Green	(0, 255, 0)
Blue	(255, 0, 0)
White	(255, 255, 255)
Black	(0, 0, 0)
Yellow	(0, 255, 255)

Common Drawing Functions

Shapes:

- `cv2.circle(img, center, radius, color, thickness)` — draws a circle. Setting `thickness=-1` fills the shape completely.
- `cv2.rectangle(img, pt1, pt2, color, thickness)` — draws a rectangle defined by its top-left and bottom-right corners.
- `cv2.ellipse(img, center, axes, angle, startAngle, endAngle, color, thickness)` — draws an ellipse. `axes` is a tuple `(majorAxis, minorAxis)` and `angle` rotates the entire ellipse.
- `cv2.line(img, pt1, pt2, color, thickness)` — draws a straight line between two points.
- `cv2.arrowedLine(img, pt1, pt2, color, thickness)` — same as line but adds an arrowhead at `pt2`.
- `cv2.polylines(img, [pts], isClosed, color, thickness)` — draws a series of connected line segments. Setting `isClosed=True` closes the shape by connecting the last point back to the first.
- `cv2.fillConvexPoly(img, pts, color)` — fills a convex polygon defined by an array of points with a solid colour.

Text:

- `cv2.putText(img, text, origin, fontFace, fontScale, color, thickness)` — renders text on the image at the specified origin point. `fontFace` selects the font style (e.g., `cv2.FONT_HERSHEY_SIMPLEX`). `fontScale` controls the size.

The `thickness` Parameter

All drawing functions share the `thickness` parameter which controls line width in pixels. A value of `1` draws a thin single-pixel line while larger values produce thicker strokes. Passing `-1` as thickness fills the shape with the specified colour instead of drawing only the outline.

SOLVED LAB ACTIVITES:

Activity 1: Drawing Circles

- **Load an Image:**
  - Choose an image (e.g., 'sample_image.jpg').
- **Read and Display the Image:**

```
import cv2

image = cv2.imread('sample_image.jpg')
cv2.imshow('Original Image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

- **Draw a Circle:**

```
import cv2

# Coordinates and parameters for the circle
center_coordinates = (150, 100)
radius = 20
color = (0, 255, 0)  # Green color in BGR
thickness = 2

# Draw the circle on the image
image_with_circle = cv2.circle(image, center_coordinates, radius, colo

# Display the image with the circle
cv2.imshow('Image with Circle', image_with_circle)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Activity 2: Drawing Lines

- **Load an Image:**
  - Choose an image (e.g., 'sample_image.jpg').
- **Read and Display the Image:**

```
import cv2

image = cv2.imread('sample_image.jpg')
cv2.imshow('Original Image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

- **Draw a Line:**

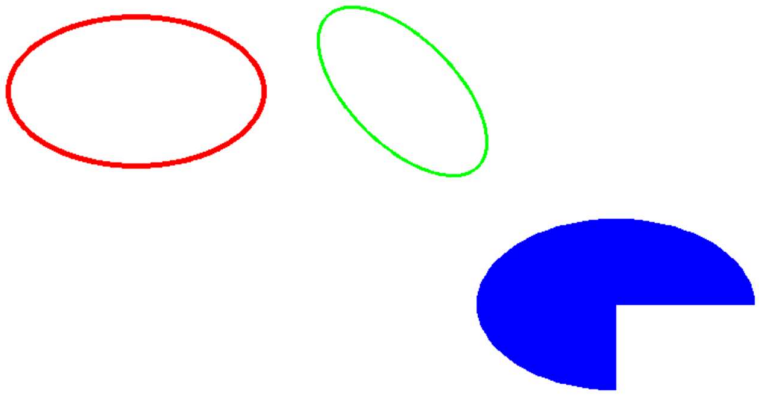
```
import cv2

# Coordinates and parameters for the line
start_point = (50, 50)
end_point = (200, 100)
color = (0, 255, 0)  # Green color in BGR
thickness = 2

# Draw the line on the image
image_with_line = cv2.line(image, start_point, end_point, color, thickn

# Display the image with the line
cv2.imshow('Image with Line', image_with_line)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

LAB TASKS

Task-1: Drawing Ellipses

- Load an image of your choice.
- Draw at least three ellipses with different parameters (varying center, axes lengths, angles, and colors). ○ Display the image with the drawn ellipses.

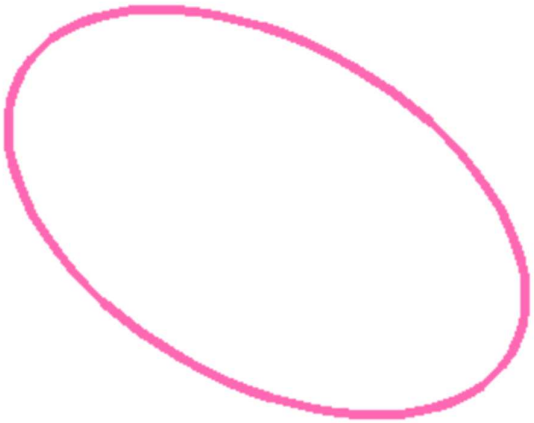
```
import cv2
import numpy as np

img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255) # white background

# Ellipse 1: Red, tilted 0 degrees, thick outline
cv2.ellipse(img, center=(200, 200), axes=(120, 70),
        angle=0, startAngle=0, endAngle=360,
        color=(0, 0, 255), thickness=3)

# Ellipse 2: Green, rotated 45 degrees
cv2.ellipse(img, center=(450, 200), axes=(100, 50),
        angle=45, startAngle=0, endAngle=360,
        color=(0, 255, 0), thickness=2)

# Ellipse 3: Blue, filled (-1 thickness), partial arc
cv2.ellipse(img, center=(650, 400), axes=(80, 130),
        angle=90, startAngle=0, endAngle=270,
        color=(255, 0, 0), thickness=-1)

cv2.imshow("Task 1 - Ellipses", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Task 2: Converting Ellipses to Polygons

- ○ Choose an ellipse from Task 1.
- ○ Use **ellipse2Poly** to convert the ellipse to a polygon. ○ Display the image with the original ellipse and the polygonized version.

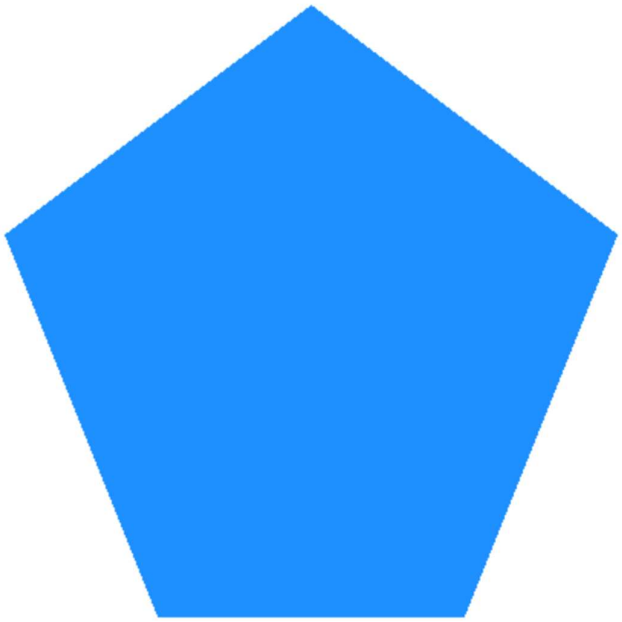
```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# Define ellipse parameters
center = (400, 300)
axes = (150, 90)
angle = 30
arc_start = 0
arc_end = 360
delta = 5 # Angle sibling spacing between sequential points (lower = smoother)

# Convert ellipse parameters into a sequence of points forming a polygon
points = cv2.ellipse2Poly(center, axes, angle, arc_start, arc_end, delta)

# Reshape the points array to the format required by cv2.polylines
points = points.reshape((-1, 1, 2))

# Draw the converted polygon onto the image (Color: Purple, Thickness: 3)
cv2.polylines(img, [points], isClosed=True, color=(180, 105, 255), thickness=3)

# Display the final output
cv2.imshow("Task 2 - Ellipse to Polygon", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Task 3: Filling Convex Polygons

- Select a convex polygon (you can use the polygonized ellipse from Task 2).
- Fill the convex polygon with a color of your choice. ○ Display the image with the filled convex polygon.

```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# Define the vertices of a convex polygon (an asymmetric diamond/shield shape)
# Coordinates must be in a specific order around the perimeter (clockwise or counter-clockwise)
points = np.array([
    [400, 100],  # Top vertex
    [600, 250],  # Right vertex
    [500, 500],  # Bottom-right vertex
    [300, 500],  # Bottom-left vertex
    [200, 250]   # Left vertex
], dtype=np.int32)

# Fill the convex polygon (Color: Cyan/Teal-ish Blue)
cv2.fillConvexPoly(img, points, color=(255, 144, 30))

# Display the final output
cv2.imshow("Task 3 - Fill Convex Polygon", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



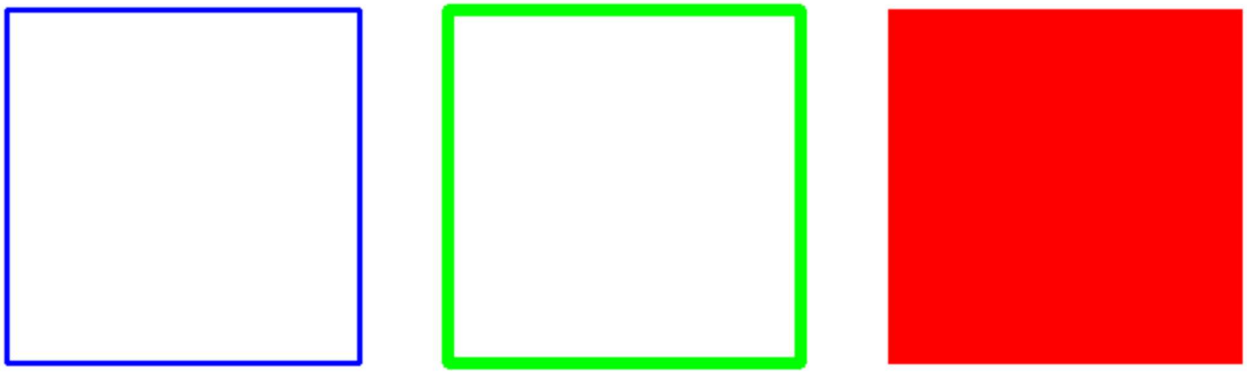
Task 4:

Drawing Rectangles

- Choose an image.
- Draw at least two rectangles with different parameters (varying start and end points, colors, and thickness).
- Display the image with the drawn rectangles.

```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# Rectangle 1: Thin blue outline
# pt1 = top-left corner, pt2 = bottom-right corner
cv2.rectangle(img, pt1=(50, 100), pt2=(250, 300),
        color=(255, 0, 0), thickness=2)

# Rectangle 2: Thick green outline
cv2.rectangle(img, pt1=(300, 100), pt2=(500, 300),
        color=(0, 255, 0), thickness=6)

# Rectangle 3: Completely filled red rectangle (thickness=-1 or cv2.FILLED)
cv2.rectangle(img, pt1=(550, 100), pt2=(750, 300),
        color=(0, 0, 255), thickness=-1)

# Display the final output
cv2.imshow("Task 4 - Rectangles", img)
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```



Task 5: Drawing Lines and Arrowed Lines

- Select two points and draw a line connecting them.
- Draw an arrowed line starting from one point and ending at another.
- Display the image with the drawn lines.

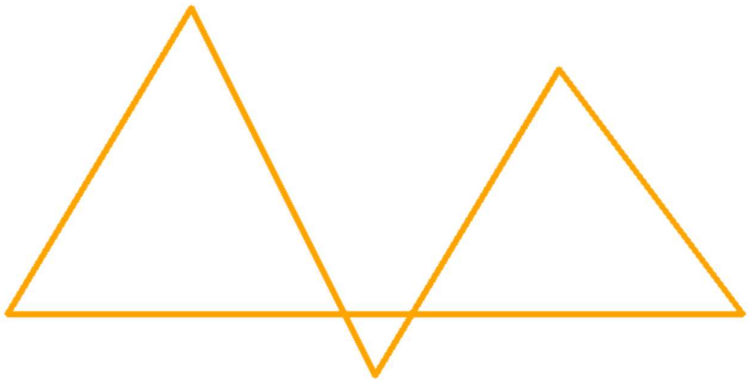
```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# 1. Draw a simple thin diagonal line (Color: Blue, Thickness: 2)
# pt1 = start point, pt2 = end point
cv2.line(img, pt1=(50, 100), pt2=(350, 100), color=(255, 0, 0), thickness=2)

# 2. Draw a thick segment with anti-aliasing (Color: Green, Thickness: 8)
cv2.line(img, pt1=(50, 250), pt2=(350, 250), color=(0, 255, 0), thickness=8,
lineType=cv2.LINE_AA)

# 3. Draw a default directional arrow (Color: Red, Thickness: 3)
cv2.arrowedLine(img, pt1=(450, 100), pt2=(750, 100), color=(0, 0, 255), thickness=3)

# 4. Draw a customized arrow with a larger tip (Color: Purple, Tip Size: 30% of length)
cv2.arrowedLine(img, pt1=(450, 250), pt2=(750, 250), color=(180, 105, 255), thickness=4,
tipLength=0.3)

# Display the final output
cv2.imshow("Task 5 - Lines and Arrows", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Task 6: Drawing Polylines

- Define a set of points to create a polyline (a shape with multiple connected line segments).
- Draw the polyline on the image.
- Display the image with the drawn polyline.

```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# Define a set of coordinate points for a zig-zag / wave shape
# Shape must be formatted as an int32 array of shape (Number of points, 1, 2)
points = np.array([
    [100, 400],
    [250, 150],
    [400, 450],
    [550, 200],
    [700, 400]
], dtype=np.int32)

# Reshape the array to match OpenCV's expected format for polylines
points = points.reshape((-1, 1, 2))

# 1. Draw an OPEN polyline (isClosed=False) -> Color: Blue, Thickness: 4
# The first point and last point will NOT connect.
img_open = img.copy()
cv2.polylines(img_open, [points], isClosed=False, color=(255, 0, 0), thickness=4)

# 2. Draw a CLOSED polyline (isClosed=True) -> Color: Red, Thickness: 4
# OpenCV automatically draws a line connecting the last point back to the first point.
img_closed = img.copy()
cv2.polylines(img_closed, [points], isClosed=True, color=(0, 0, 255), thickness=4)

# Combine both examples side-by-side or show the closed version
```

```
# For your lab manual, we will display the closed configuration as the final canvas layout
cv2.polylines(img, [points], isClosed=True, color=(0, 165, 255), thickness=4)

# Display the final output
cv2.imshow("Task 6 - Polylines", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Task 7: Putting Text on Images

- o Choose an image.
- o Add text with different font sizes, colors, and positions.
- o Display the image with the added text.

```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# 1. Standard text using FONT_HERSHEY_SIMPLEX (Color: Blue, Scale: 1.0)
cv2.putText(img, text="1. Simplex Font Example", org=(50, 100),
        fontFace=cv2.FONT_HERSHEY_SIMPLEX, fontScale=1.0,
        color=(255, 0, 0), thickness=2)

# 2. Elegant script style using FONT_HERSHEY_SCRIPT_SIMPLEX (Color: Green, Scale:
1.2)
cv2.putText(img, text="2. Script Elegance Style", org=(50, 220),
        fontFace=cv2.FONT_HERSHEY_SCRIPT_SIMPLEX, fontScale=1.2,
        color=(0, 200, 0), thickness=2)

# 3. Bold/Thick text using FONT_HERSHEY_COMPLEX (Color: Red, Scale: 1.0,
Thickness: 3)
cv2.putText(img, text="3. Complex Bold Styling", org=(50, 340),
        fontFace=cv2.FONT_HERSHEY_COMPLEX, fontScale=1.0,
        color=(0, 0, 255), thickness=3)
```

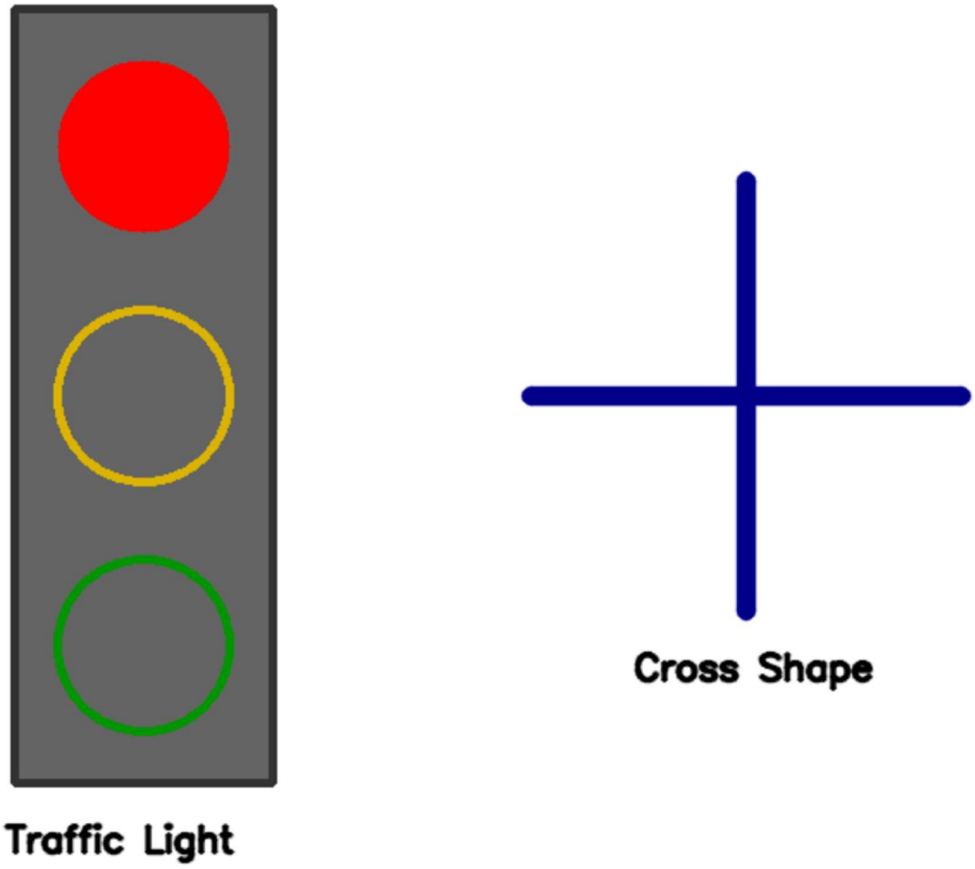
```
# 4. Clean anti-aliased look using LINE_AA flag (Color: Purple, Scale: 1.1)
cv2.putText(img, text="4. Clean Text with Anti-Aliasing", org=(50, 460),
        fontFace=cv2.FONT_HERSHEY_DUPLEX, fontScale=1.1,
        color=(180, 105, 255), thickness=2, lineType=cv2.LINE_AA)

# Display the final output
cv2.imshow("Task 7 - Text Fonts", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Task 8: Additional Challenges

- o Implement any two additional drawing functions.
- o Experiment with various parameters and observe the effects.
- o Document your findings and display the image with the applied functions.

```
import cv2
import numpy as np

# Create a clean white canvas
img = np.zeros((600, 800, 3), dtype=np.uint8)
img[:] = (255, 255, 255)

# --- PART 1: THE TRAFFIC LIGHT ---
# Draw the main backing frame (Grey Rectangle)
cv2.rectangle(img, pt1=(150, 50), pt2=(300, 500), color=(100, 100, 100), thickness=-1)
cv2.rectangle(img, pt1=(150, 50), pt2=(300, 500), color=(50, 50, 50), thickness=3)

# Red Light (Top) - Active/Filled
cv2.circle(img, center=(225, 130), radius=50, color=(0, 0, 255), thickness=-1)
# Yellow Light (Middle) - Dim/Outline
cv2.circle(img, center=(225, 275), radius=50, color=(0, 180, 220), thickness=4)
# Green Light (Bottom) - Dim/Outline
cv2.circle(img, center=(225, 420), radius=50, color=(0, 150, 0), thickness=4)
```

```
# --- PART 2: THE CROSS SHAPE ---
# Horizontal bar of the cross (Color: Dark Blue, Thickness: 10)
cv2.line(img, pt1=(450, 275), pt2=(700, 275), color=(139, 0, 0), thickness=10,
lineType=cv2.LINE_AA)
# Vertical bar of the cross (Color: Dark Blue, Thickness: 10)
cv2.line(img, pt1=(575, 150), pt2=(575, 400), color=(139, 0, 0), thickness=10,
lineType=cv2.LINE_AA)


# --- LABELS ---
cv2.putText(img, "Traffic Light", (145, 540), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0,
0), 2, cv2.LINE_AA)
cv2.putText(img, "Cross Shape", (510, 440), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0,
0), 2, cv2.LINE_AA)

# Display the final output
cv2.imshow("Task 8 - Additional Challenges", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



LAB # 03

VIDEO PROCESSING WITH OPENCV

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly understands the frame-by-frame nature of video, FPS, VideoCapture properties, and the role of VideoWriter; strongly relates to lab objectives.	Basic understanding of video as a frame sequence; partial connection to lab objectives.	Little or no understanding of how OpenCV reads or writes video; unclear relevance to lab topic.	
Code Implementation (6 marks)	Video is loaded, frames processed with at least three image processing techniques, processed video saved with correct codec/FPS/resolution, and dynamic text overlaid correctly.	Video loads and basic processing works; VideoWriter or text overlay partially implemented or missing.	Code fails to load video, process frames, or save output; major components missing.	
Use of Programming Features/Tools (4 marks)	Correctly uses cv2.VideoCapture, cap.get(), cap.read(), cap.release(), cv2.VideoWriter, and cv2.putText with appropriate parameters for FPS, codec, and resolution.	Most functions used correctly with minor parameter errors or incomplete usage.	Minimal or incorrect use of VideoCapture/VideoWriter; output video not produced or corrupted.	
Results and Report (6 marks)	Original and processed frames displayed side by side; output video saved and playable; frame number and processing status overlaid correctly; clear written observations.	Output video saved but lacks text overlay or side-by-side comparison; report is basic.	Output video missing or unplayable; no visual comparisons or written observations.	

LAB # 03

VIDEO PROCESSING WITH OPENCV

OBJECTIVE

- To understand how OpenCV handles video files and camera streams as sequences of individual frames
- To load, display, and process video files frame by frame using `cv2.VideoCapture`
- To apply image processing techniques learned in previous labs to each frame of a video
- To create a new video file from a sequence of processed frames using `cv2.VideoWriter`

LAB OUTCOMES

- Understand the frame-by-frame structure of video and how OpenCV reads it as a sequence of NumPy arrays
- Load and play video files using `cv2.VideoCapture` and control playback speed using `cv2.waitKey`
- Access video metadata such as FPS, frame dimensions, and total frame count using `cap.get()`
- Apply image processing operations from previous labs (resizing, blurring, colour conversion) to individual video frames
- Create a new video file from processed frames using `cv2.VideoWriter` with correct codec, FPS, and resolution settings
- Overlay dynamic text information (frame number, processing status) on video frames using `cv2.putText`

THEORY

How Video Works in OpenCV

A video is simply a sequence of images (frames) displayed at a fixed rate measured in **frames per second (FPS)**. OpenCV treats video as a stream — it reads one frame at a time, processes it, and moves to the next. This frame-by-frame approach is the foundation of every real-time CV pipeline, from object detection to motion analysis.

Reading Video — cv2.VideoCapture

`cv2.VideoCapture` is used to open both video files and live camera feeds:

```
cap = cv2.VideoCapture('video.mp4')   # from file
cap = cv2.VideoCapture(0)          # from webcam (index 0)
```

Once opened, useful properties can be retrieved using `cap.get()`:

Property	Code
Frame width	`cap.get(cv2.CAP_PROP_FRAME_WIDTH)`
Frame height	`cap.get(cv2.CAP_PROP_FRAME_HEIGHT)`
FPS	`cap.get(cv2.CAP_PROP_FPS)`

Property	Code
Total frames	`cap.get(cv2.CAP_PROP_FRAME_COUNT)`

Each call to `cap.read()` returns two values: a boolean `ret` indicating whether the frame was read successfully, and the `frame` itself as a NumPy array. When the video ends, `ret` becomes `False`. Always call `cap.release()` after processing to free the resource.

Writing Video — cv2.VideoWriter

`cv2.VideoWriter` saves a sequence of frames as a video file:

```
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out    = cv2.VideoWriter('output.mp4', fourcc, fps, (width, height))
out.write(frame)   # called once per frame
out.release()      # called after all frames are written
```

The `fourcc` (Four Character Code) specifies the video codec. Common options are `mp4v` for `.mp4` and `XVID` for `.avi`. The output resolution and FPS must be set at initialisation and kept consistent across all written frames.

Frame Delay and Real-Time Display

When displaying video with `cv2.imshow`, `cv2.waitKey(delay)` controls the pause between frames in milliseconds. Setting `delay=1` gives the fastest possible display. Setting `delay = int(1000/fps)` matches the original video playback speed. Pressing `q` while the window is focused can be used to exit the loop cleanly by checking `cv2.waitKey(1) & 0xFF == ord('q')`.

SOLVED LAB ACTIVITES:

Before performing the lab, please read the material below.

https://www.geeksforgeeks.org/python-create-video-using-multiple-images-usinghttps://www.geeksforgeeks.org/python-create-video-using-multiple-images-using-opencv/?ref=lbpopencv/?ref=lbp

Activity 1:

Play a Video using OpenCV

- **Load a Video File:**
  - Choose a video file (e.g., 'sample_video.mp4').
  - Load the video using OpenCV.

```python
import cv2

# Video file path
video_path = 'sample_video.mp4'

# Open the video file
cap = cv2.VideoCapture(video_path)
```

- **Play the Video:**
  - Display each frame of the video. ○ Add a delay between frames.

```python
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    # Display the frame
    cv2.imshow('Video Playback', frame)

    # Exit on 'q' key press
    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

# Release the video capture object
cap.release()
cv2.destroyAllWindows()
```

Activity 2:

Create Video using Multiple Images

- **Load Images:**
  - ○ Choose a set of images (e.g., 'image1.jpg', 'image2.jpg', ...).
  - Read each image using OpenCV.
- **Create a Video:**
  - Combine the images to create a video.
  - Set video parameters such as frame width, height, and frames per second (fps).

LAB TASKS

Please read the following material before starting the lab. https://www.geeksforgeeks.org/python-process-images-of-a-video-using-opencv/?ref=lbp https://www.geeksforgeeks.org/python-opencv-capture-video-from-camera/?ref=lbp

Helping materials: https://www.geeksforgeeks.org/python-process-images-of-a-video-using-opencv/?ref=lbp https://www.geeksforgeeks.org/python-opencv-capture-video-from-camera/?ref=lbp

Lab Task 1: Processing Images of a Video

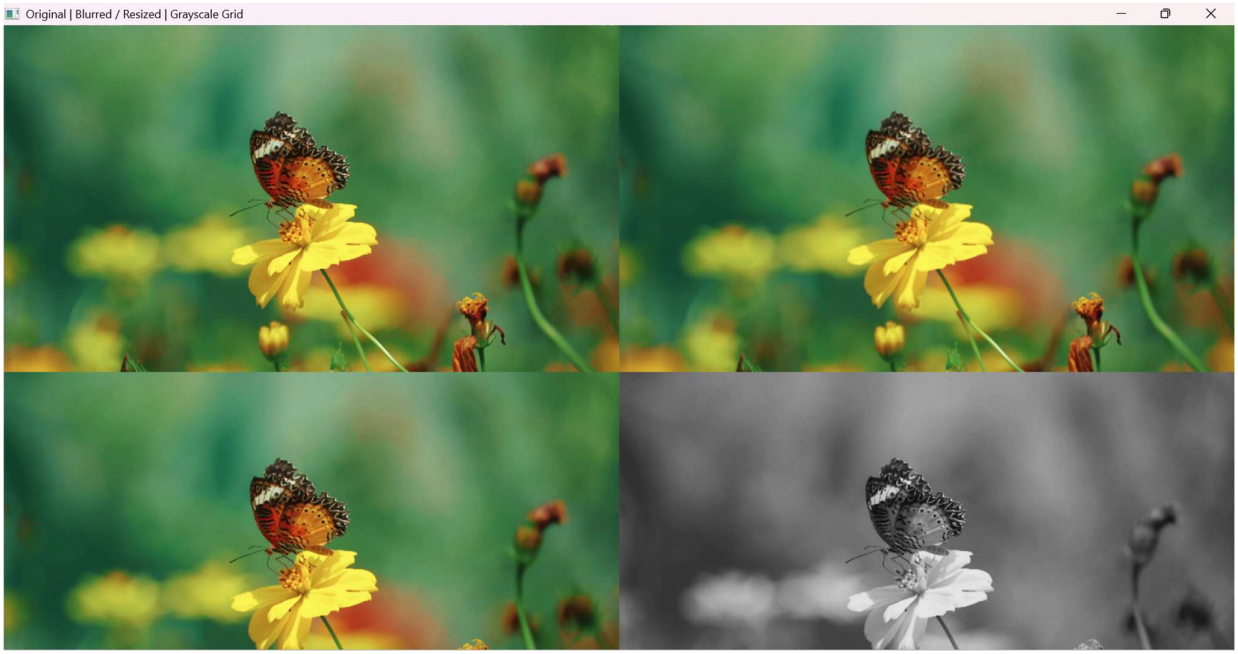
Instructions:

- Choose a video file (e.g., 'sample_video.mp4').
- Load the video and process each frame using OpenCV.
- Apply at least three different image processing techniques (e.g., resizing, blurring, color manipulation) to the frames.
- Display the original and processed frames side by side for visual comparison.

```
import cv2
import numpy as np

# Load the video file
cap = cv2.VideoCapture("sample_video.mp4")

# Check if the video opened successfully
if not cap.isOpened():
    print("[ERROR] Could not open video file.")
    exit()

# Get video properties
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

print(f"Video Info: {width}x{height} @ {fps:.2f} fps | Total Frames: {total_frames}")

frame_num = 0
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break  # End of video stream

    frame_num += 1
```

```
    # Technique 1: Resize to half scale
    resized = cv2.resize(frame, (width // 2, height // 2))
    # Pad it back up to full size for grid alignment
    resized_full = cv2.resize(resized, (width, height))

    # Technique 2: Gaussian Blur (Smoothing)
    blurred = cv2.GaussianBlur(frame, (15, 15), 0)

    # Technique 3: Grayscale conversion
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    # Convert grayscale back to 3 channels so it can stack with BGR arrays
    gray_3ch = cv2.cvtColor(gray, cv2.COLOR_GRAY2BGR)

    # Assemble views into a 2x2 grid matrix
    top_row = np.hstack([frame, blurred])
    bottom_row = np.hstack([resized_full, gray_3ch])
    combined_grid = np.vstack([top_row, bottom_row])

    # Resize the final grid display to fit neatly on screen
    display_output = cv2.resize(combined_grid, (1280, 720))

    # Render video stream display
    cv2.imshow("Original | Blurred / Resized | Grayscale Grid", display_output)

    # Maintain accurate video playback speed; Break if 'q' is pressed
    delay = max(1, int(1000 / fps))
    if cv2.waitKey(delay) & 0xFF == ord('q'):
        print("[INFO] Playback interrupted by user.")
        break

# Cleanup streams
cap.release()
cv2.destroyAllWindows()
```

print(f"[STATUS] Processed {frame_num} frames successfully.")



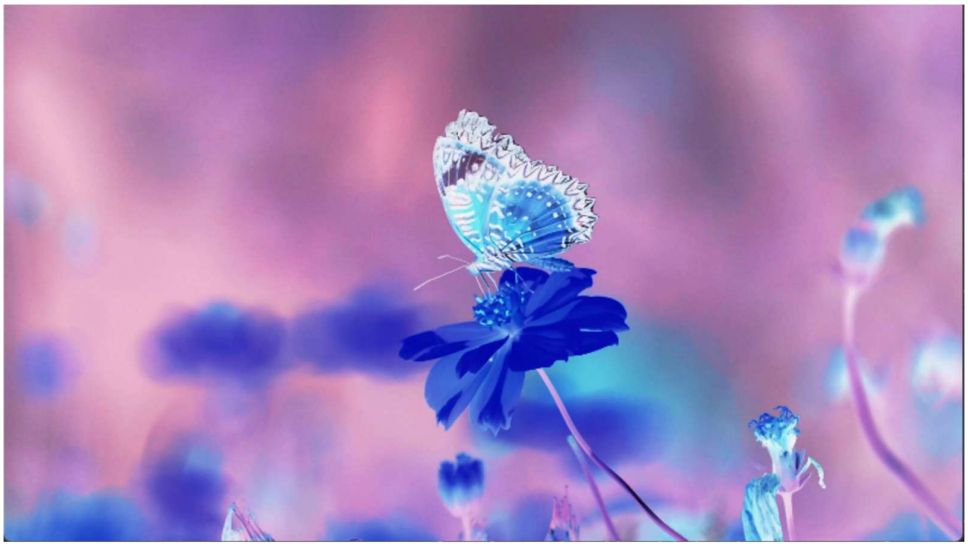
Lab Task 2: Writing to Video with OpenCV (Optional)

Instructions:

- Extend the code from Lab Task 1.
- After processing each frame, write the processed frame to a new video file.
- Set video parameters such as frame width, height, and frames per second (fps).
- Display a message indicating the progress of video creation.

```
import cv2
import numpy as np

# Open input video file or webcam stream (0)
cap = cv2.VideoCapture("sample_video.mp4")

if not cap.isOpened():
    print("[ERROR] Could not open source video.")
    exit()

# Fetch properties needed to configure the VideoWriter
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

# Define the codec and create VideoWriter object
# 'mp4v' works natively across Windows, Mac, and Linux
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter('processed_output.mp4', fourcc, fps, (width, height))
```

```
print(f"[START] Exporting video to 'processed_output.mp4'...")

frame_count = 0
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    # Apply a sample transformation: Invert image colors (Negative image effect)
    processed_frame = cv2.bitwise_not(frame)

    # Write the modified frame into the output video file
    out.write(processed_frame)

    frame_count += 1

    # Calculate and display saving progress at fixed intervals
    if frame_count % 30 == 0 or frame_count == total_frames:
        progress = (frame_count / total_frames) * 100 if total_frames > 0 else 0
        print(f"[PROGRESS] Saved frame {frame_count}/{total_frames} ({progress:.1f}%)")

    # Display live window preview of what is being recorded
    cv2.imshow("Encoding Frame Pipeline...", processed_frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        print("[WARNING] Write sequence aborted early by user.")
        break

# Safely release handles and close writer instances
cap.release()
out.release()
cv2.destroyAllWindows()
print("[STATUS] File finalized and closed successfully.")
```



```
[START] Exporting video to 'processed_output.mp4'...
[PROGRESS] Saved frame 30/457 (6.6%)
[PROGRESS] Saved frame 60/457 (13.1%)
[PROGRESS] Saved frame 90/457 (19.7%)
[PROGRESS] Saved frame 120/457 (26.3%)
[PROGRESS] Saved frame 150/457 (32.8%)
[PROGRESS] Saved frame 180/457 (39.4%)
[PROGRESS] Saved frame 210/457 (46.0%)
[PROGRESS] Saved frame 240/457 (52.5%)
[PROGRESS] Saved frame 270/457 (59.1%)
[PROGRESS] Saved frame 300/457 (65.6%)
[PROGRESS] Saved frame 330/457 (72.2%)
[PROGRESS] Saved frame 360/457 (78.8%)
[PROGRESS] Saved frame 390/457 (85.3%)
[PROGRESS] Saved frame 420/457 (91.9%)
[PROGRESS] Saved frame 450/457 (98.5%)
[PROGRESS] Saved frame 457/457 (100.0%)
[STATUS] File finalized and closed successfully.
```

Lab Task 3: Writing Text on Video (Optional)

Instructions:

- Extend the code from Lab Task 2.
- Write dynamic text on each frame, indicating the frame number or any relevant information.

Choose a distinctive font, size, and color for the text. ○ Display a message indicating the completion of the video creation process.

```
import cv2
import numpy as np

# Open input video file
cap = cv2.VideoCapture("sample_video.mp4")

if not cap.isOpened():
    print("[ERROR] Could not open source video.")
    exit()

# Get video properties
fps = cap.get(cv2.CAP_PROP_FPS)
if fps == 0:
    fps = 30  # Fallback FPS

width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

# Create VideoWriter object
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter('processed_output_task3.mp4',
                      fourcc,
                      fps,
                      (width, height))

print("[START] Creating video with dynamic text overlay...")
```

```
frame_count = 0

while True:
    ret, frame = cap.read()

    if not ret:
        break

    frame_count += 1

    # Apply image processing (negative image effect)
    processed_frame = cv2.bitwise_not(frame)

    # Dynamic text showing current frame number
    text = f"Frame Number: {frame_count}"

    cv2.putText(
        processed_frame,
        text,
        (50, 80),                    # Position (x, y)
        cv2.FONT_HERSHEY_COMPLEX,    # Font
        3.0,                         # Font scale
        (0, 255, 0),                 # Green color (BGR)
        2,                           # Thickness
        cv2.LINE_AA                  # Anti-aliased text
    )

    # Write processed frame to output video
    out.write(processed_frame)

    # Display progress every 30 frames
    if frame_count % 30 == 0 or frame_count == total_frames:
        progress = (frame_count / total_frames) * 100 if total_frames > 0 else 0
        print(
            f"[PROGRESS] Saved frame {frame_count}/{total_frames} ({progress:.1f}%)"
        )

    # Resize only for display preview
    display_preview = cv2.resize(processed_frame, (800, 450))

    cv2.imshow("Task 3 - Video with Dynamic Text", display_preview)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        print("[WARNING] Video creation stopped by user.")
        break

# Release resources
cap.release()
out.release()
cv2.destroyAllWindows()

print("[COMPLETE] Video creation process finished successfully.")
print("[OUTPUT] Saved as: processed_output_task3.mp4")
```


Frame Number: 12

LAB # 04

OPENCV EDGE DETECTION AND BLURRING

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains what an edge is mathematically, why blurring precedes edge detection, and the difference between first and second derivative filters; strongly relates to lab objectives.	Basic understanding of edge detection and blurring; partial explanation of why preprocessing is needed.	Little or no understanding of edge detection theory or the role of blurring; unclear relevance.	
Code Implementation (6 marks)	Canny, Sobel, Scharr, and Laplacian all correctly implemented; custom blurring filter using cv2.filter2D works correctly with at least two different kernels tested.	Most detection methods implemented with minor errors; custom filter partially implemented or missing.	Fewer than two edge detection methods implemented; custom filter absent or broken.	
Use of Programming Features/Tools (4 marks)	Correctly uses cv2.Canny, cv2.Sobel, cv2.Scharr, cv2.Laplacian, cv2.GaussianBlur, cv2.medianBlur, and cv2.filter2D with appropriate parameters and kernel definitions.	Most functions used with limited parameter control or partial correct usage.	Minimal or incorrect use of edge detection and filtering functions.	
Results and Report (6 marks)	All methods displayed side by side with a clear comparative analysis discussing strengths and limitations of each; custom kernel results compared against standard blur.	Most outputs shown; comparison is basic or lacks written analysis.	Fewer than two methods shown; no comparative analysis or written observations.	

LAB # 04

OPENCV EDGE DETECTION AND BLURRING

OBJECTIVE

- To understand the concept of edges in images and why edge detection is a fundamental operation in Computer Vision
- To apply the Canny edge detector and compare it against gradient-based methods such as Sobel, Scharr, and Laplacian
- To explore and implement various image blurring techniques including Gaussian blur, Median blur, and custom convolution filters using `cv2.filter2D`
- To understand the relationship between blurring and edge detection and why denoising is applied before edge detection in practice

LAB OUTCOMES

- Understand what an edge is mathematically and why detecting intensity gradients reveals object boundaries
- Apply the Canny edge detector and tune its thresholds to control edge density and quality
- Implement gradient-based edge detection using Sobel, Scharr, and Laplacian filters via OpenCV
- Apply Gaussian and Median blur as preprocessing steps and understand their effect on subsequent edge detection
- Build and apply custom convolution kernels using `cv2.filter2D` to implement user-defined blurring filters
- Perform a comparative analysis of multiple edge detection methods on the same image and draw conclusions about their strengths and limitations

THEORY

What is an Edge?

An edge in an image is a location where pixel intensity changes abruptly. These abrupt changes correspond to meaningful boundaries in the real world — the border between an object and its background, a shadow line, or a change in surface texture. Edge detection is one of the most fundamental operations in Computer Vision because extracting boundaries is often the first step before shape analysis, object detection, or segmentation.

Mathematically, an abrupt intensity change corresponds to a large value of the **first derivative** of the image. Where the first derivative is large, an edge exists. Where the second derivative crosses zero, an edge peak is located.

Blurring and Filtering

Before detecting edges, images are typically blurred to suppress noise. A raw image captured by a camera contains sensor noise — small random variations in pixel values that would be incorrectly detected as edges. Blurring smooths these variations by replacing each pixel with a weighted average of its neighbourhood.

OpenCV provides several blurring methods:

- **Gaussian Blur** — `cv2.GaussianBlur(img, (ksize,ksize), 0)`: Weighted average using a Gaussian kernel. Most commonly used before edge detection.
- **Median Blur** — `cv2.medianBlur(img, ksize)`: Replaces each pixel with the median of its neighbourhood. Particularly effective against salt-and-pepper noise.
- **Custom Filter** — `cv2.filter2D(img, -1, kernel)`: Applies any user-defined kernel to the image using 2D convolution. This gives full control over the filter behaviour and is the basis for building both blur and edge detection filters manually.

Edge Detection Methods

Sobel Filter: Computes the gradient of image intensity in the horizontal (x) and vertical (y) directions separately using a 3×3 kernel. The gradient magnitude is then computed as `G = sqrt(Gx² + Gy²)`. Sobel is sensitive to edges in one direction at a time.

Scharr Filter: A variant of Sobel that uses different kernel weights to achieve better rotational symmetry and more accurate gradient estimation, especially for small kernels.

Laplacian Filter: Computes the second derivative of the image intensity. It detects edges in all directions simultaneously and produces a response at both sides of an edge. It is more sensitive to noise than Sobel, which is why blurring is applied first.

Canny Edge Detector: The most widely used edge detection algorithm. It is a multi-step pipeline:

1. Apply Gaussian blur to suppress noise
2. Compute gradient magnitude and direction using Sobel
3. Apply non-maximum suppression — thin wide edges to single-pixel boundaries
4. Apply double thresholding — classify pixels as strong, weak, or non-edges
5. Apply edge tracking by hysteresis — keep weak edges only if connected to strong edges

`cv2.Canny(img, threshold1, threshold2)` — `threshold1` and `threshold2` control which edges are kept. A higher ratio between them produces cleaner, sparser edges.

SOLVED LAB ACTIVITES:

Please read the following material before starting the lab.

https://learnopencv.com/edge-detection-using-opencv/

Activity 1: Edge Detection using OpenCV

- **Load and Display an Image:**
  - Choose an image (e.g., 'sample_image.jpg').
  - Load and display the original image using OpenCV.

```
import cv2
import numpy as np

# Load the image
image = cv2.imread('sample_image.jpg', cv2.IMREAD_GRAYSCALE)

# Display the original image
cv2.imshow('Original Image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

- **Apply Edge Detection:**
  - Utilize an edge detection algorithm, such as the Canny edge detector.
  - Display the resulting image after edge detection.

```
# Apply Canny edge detection
edges = cv2.Canny(image, 50, 150)

# Display the image with edges
cv2.imshow('Edge Detection', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Activity 2:

Image Blurring using OpenCV

- **Load and Display an Image:**
  - Choose an image (e.g., 'sample_image.jpg').
  - Load and display the original image using OpenCV.

```
import cv2
import numpy as np

# Load the image
image = cv2.imread('sample_image.jpg')

# Display the original image
cv2.imshow('Original Image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

- **Apply Blurring Techniques:**

Use various blurring techniques like Gaussian blur and Median blur.

o Display the resulting images after applying each blurring technique.

```python
# Apply Gaussian blur
blurred_gaussian = cv2.GaussianBlur(image, (5, 5), 0)
cv2.imshow('Gaussian Blur', blurred_gaussian)
cv2.waitKey(0)

# Apply Median blur
blurred_median = cv2.medianBlur(image, 5)
cv2.imshow('Median Blur', blurred_median)
cv2.waitKey(0)

cv2.destroyAllWindows()
```

LAB TASKS

Please read the following material before starting the lab.

https://opencv24-pythonhttps://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_filtering/py_filtering.htmltutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_filtering/py_filtering.html

Helping materials:
https://learnopencv.com/edge-detection-using-opencv/

Lab Task 1:

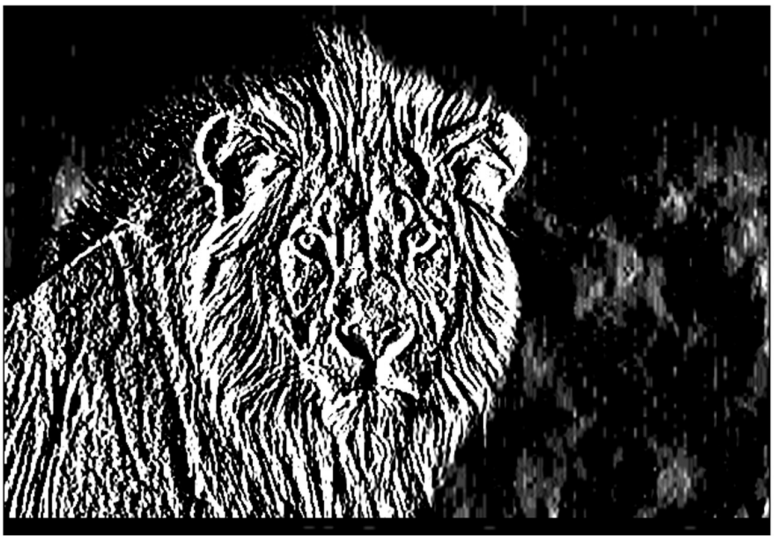
Advanced Edge Detection Techniques

Instructions:

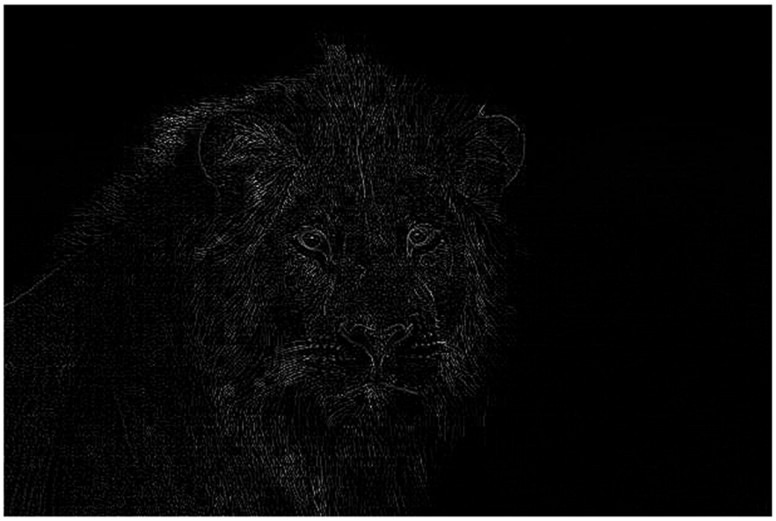
- Choose an image for testing.
- Implement at least two additional edge detection techniques (e.g., Sobel, Scharr, Laplacian) using OpenCV.

```python
# Advanced Edge detection techniques

image1 = cv2.imread('lion.jpg', cv2.IMREAD_GRAYSCALE)

# Sobel Edge Detection
sobel_detectionx = cv2.Sobel(image1, cv2.CV_64F, 1, 0, ksize=5)
sobel_detectiony = cv2.Sobel(image1, cv2.CV_64F, 0, 1, ksize=5)
sobel_detection = cv2.Sobel(image1, cv2.CV_64F, 1, 1, ksize=5)
cv2_imshow(sobel_detectionx)
cv2_imshow(sobel_detectiony)
cv2_imshow(sobel_detection)
```

- Display the resulting images after applying each edge detection technique.



```
# Laplacian Detection
laplacian_detection = cv2.Laplacian(image1, cv2.CV_64F)
cv2_imshow(laplacian_detection)
```



- Provide a comparative analysis of the different edge detection methods used.

Feature	Sobel	Laplacian
Derivative	1st Order	2nd Order
Direction	X or Y separately	All directions at once
Noise	Better noise handling	Very sensitive to noise
Edge Quality	Thicker edges	Finer, more detailed edges

Lab Task 2:

Instructions:

- Select an image for testing.
- Implement a custom blurring filter using convolution operations (you can use **cv2.filter2D**).
- Display the image after applying the custom blurring filter.

Helping material: https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_filtering/py_filtering.html

```
# 1. Create a 5x5 custom blur kernel (Box Filter)
# We use np.float32 for precision and divide by 25 (5*5) to normalize
kernel = np.ones((5, 5), np.float32) / 25

# 2. Apply the filter
# Use ddepth=-1 to keep the output the same type as the input (usually 8-bit)
# Or use cv2.CV_64F if you need higher precision calculations
blurred = cv2.filter2D(src=image1, ddepth=-1, kernel=kernel)

cv2_imshow(blurred)
```

- Experiment with different kernel sizes and shapes.

```
import cv2
import numpy as np

# Load image
img = cv2.imread('your_image.jpg')

# 1. Simple Average Blur
# Replaces pixel with the average of its neighbors
avg_blur = cv2.blur(image1, (5, 5))

# 2. Gaussian Blur
# Weighted average where center pixels count more; removes Gaussian noise
gauss_blur = cv2.GaussianBlur(image1, (5, 5), 0)

cv2_imshow(avg_blur)
cv2_imshow(gauss_blur)
```

- Compare the results with the standard blurring techniques used in the previous lab activity.

Method	How it works	Result
Custom (filter2D)	You manually define the kernel	Same as avg blur with box kernel
Average Blur	Averages all pixels equally	Smooth but blurry
Gaussian Blur	Center pixels weigh more	More natural-looking smoothing

LAB # 05

BUILD DNN & CNNS FROM SCRATCH

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains finite-difference derivatives, why CNNs outperform flat NNs on images, the role of regularisation, and the difference between parameters and hyperparameters.	Basic understanding of filters and CNN structure; partial explanation of why spatial structure matters.	Little or no understanding of filter theory or CNN concepts; unclear relevance to lab objectives.	
Code Implementation (6 marks)	Sobel, Prewitt, and Laplacian manually implemented correctly; TF dataset pipeline built; all three model architectures (flat NN, hidden layer NN, CNN) train correctly; regularisation applied.	Most components implemented with minor errors; one architecture or regularisation step missing.	Fewer than two architectures implemented; manual filters or dataset pipeline broken or absent.	
Use of Programming Features/Tools (4 marks)	Correctly uses cv2.filter2D with NumPy kernels, tf.data pipeline with batching and prefetching, keras Conv2D/MaxPooling2D/BatchNormalization/Dropout layers, and W&B logging.	Most tools used correctly with limited or partial integration of W&B or regularisation layers.	Minimal use of TensorFlow/Keras tools; pipeline or model architecture largely missing.	
Results and Report (6 marks)	Training curves for all models plotted; parameter count worksheet completed correctly; W&B screenshot included; overfitting analysis written; accuracy comparison table produced.	Most outputs present; parameter worksheet partially complete or W&B screenshot missing.	Training curves or comparison table missing; no parameter analysis or W&B logging attempted.	

LAB # 05

BUILD DNN & CNNS FROM SCRATCH

OBJECTIVE

To build, compare, and regularize Convolutional Neural Networks (CNNs) using TensorFlow/Keras.

LAB OUTCOMES

- Implement Sobel, Prewitt, and Laplacian filters manually using NumPy kernels and `cv2.filter2D` and verify them against OpenCV built-ins
- Understand the mathematical relationship between finite-difference derivatives and classical edge detection filters
- Build a TensorFlow dataset pipeline from the 5-Flower dataset with proper preprocessing, batching, and augmentation
- Construct and train three model architectures — flat NN, flat NN with hidden layer, and CNN from scratch — and compare their performance
- Calculate the number of trainable parameters after each layer manually and verify using `model.summary()`
- Apply Dropout and Batch Normalisation as regularisation techniques and observe their effect on the train-validation accuracy gap
- Log and compare experiments using Weights & Biases to perform systematic hyperparameter tuning

THEORY

Introduction to CNNs

The filters above are hand-crafted — we define the kernel weights manually based on mathematical reasoning. A **Convolutional Neural Network (CNN)** replaces this manual design with **learned filters**. During training, the network automatically discovers kernel weights that are most useful for the task at hand — whether that is detecting edges, corners, textures, or higher-level patterns like eyes or wheels.

A `Conv2D` layer with 32 filters of size 3×3 learns 32 different feature detectors simultaneously. Early layers tend to learn edge-like filters (similar to Sobel), while deeper layers learn increasingly complex patterns.

From Flat NN to CNN

Training a flat neural network on images by flattening the pixel array destroys all spatial structure — the network has no concept of which pixels are neighbours. A CNN preserves this spatial structure through the convolution operation, which processes local neighbourhoods of pixels at every position. This is why CNNs are fundamentally better suited to image data than flat networks.

Regularisation

Overfitting occurs when the model memorises training data and fails to generalise. Three key techniques are used to reduce it:

- **Dropout** — randomly deactivates a fraction of neurons during training, forcing the network to learn redundant representations
- **Batch Normalisation** — normalises activations within each mini-batch, stabilising training and allowing higher learning rates
- **Early Stopping** — halts training when validation loss stops improving, preventing unnecessary overfitting

Weights & Biases (W&B)

W&B is a professional experiment tracking tool that logs training metrics, hyperparameter configurations, and model outputs to an interactive dashboard. It allows multiple runs to be compared visually, making hyperparameter tuning systematic rather than guesswork.

ENVIRONMENT SETUP

Required Libraries

Verify the following packages are installed in your Python environment:

```
pip install opencv-python numpy matplotlib tensorflow
pip install wandb scikit-learn
```

W&B Account Setup

1. Go to https://wandb.ai and create a free account
2. In your terminal, run: wandb login
3. Paste your API key when prompted
4. Create a new project called cv-lab5-flowers

Dataset

The 5-Flower dataset should be available from Lab 4 preparation. The CSV file maps image paths to labels. Confirm your directory structure:

```
flowers/
    flowers.csv          # columns: filepath, label
    daisy/
    dandelion/
    roses/
    sunflowers/
    tulips/
```

> **Dataset Note**
> Each image should be resized to 224x224x3 (RGB) and pixel values scaled to [0, 1] by dividing by 255.0.
> The five class labels are: daisy (0), dandelion (1), roses (2), sunflowers (3), tulips (4).

SOLVED LAB ACTIVITES:

Part 1: Deep Learning Pipeline
Activity 1: Data Engineering with TensorFlow

- **1.1 Load Data:** Read flowers image dataset and map the five flower classes to integer labels.

```
CLASS_NAMES = ['Lilly', 'Lotus', 'Orchid', 'Sunflower', 'Tulip']
base_dir    = '/content/drive/MyDrive/5flowersdata/flowers'
train_dir   = os.path.join(base_dir, 'train')
val_dir     = os.path.join(base_dir, 'val')

# Collect filepaths and integer labels
def collect_files(folder, class_names):
    filepaths, labels = [], []
    for idx, cls in enumerate(class_names):
        cls_dir = os.path.join(folder, cls)
        for fname in os.listdir(cls_dir):
            if fname.lower().endswith(('.jpg', '.jpeg', '.png')):
                filepaths.append(os.path.join(cls_dir, fname))
                labels.append(idx)
    return filepaths, labels

train_paths, train_labels = collect_files(train_dir, CLASS_NAMES)
val_paths,   val_labels   = collect_files(val_dir,   CLASS_NAMES)
print(f'Train: {len(train_paths)} images | Val: {len(val_paths)} images')
print(f'Classes: {dict(enumerate(CLASS_NAMES))}')
```

```
Train: 4008 images | Val: 1011 images
Classes: {0: 'Lilly', 1: 'Lotus', 2: 'Orchid', 3: 'Sunflower', 4: 'Tulip'}
```

- **1.2 Preprocessing:** Create a function to resize images to 224 x 224 and rescale pixel values to [0, 1].

```
IMG_SIZE = 224

def preprocess_image(path, label):
    img = tf.io.read_file(path)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, [IMG_SIZE, IMG_SIZE])
    img = tf.cast(img, tf.float32) / 255.0   # scale to [0, 1]
    return img, label
```

- **1.3 Pipeline:** Build a tf.data.Dataset, apply shuffling, and create batches of 32.

```
BATCH_SIZE = 32

def build_pipeline(paths, labels, shuffle=False):
    ds = tf.data.Dataset.from_tensor_slices((paths, labels))
    if shuffle:
        ds = ds.shuffle(buffer_size=len(paths), seed=42)
    ds = ds.map(preprocess_image, num_parallel_calls=tf.data.AUTOTUNE)
    ds = ds.batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)
    return ds

train_ds = build_pipeline(train_paths, train_labels, shuffle=True)
val_ds   = build_pipeline(val_paths,   val_labels,   shuffle=False)
print('Pipeline ready!')
```

```
Pipeline ready!
```

Activity 2: Architecture Comparison Construct three distinct models using tf.keras.models.Sequential:

- **Model A (Flat):** Input -> Flatten -> Dense(5).
- **Model B (Hidden):** Add one Dense(128) layer with ReLU activation before the output.
- **Model C (CNN):** Build a 3-block CNN using Conv2D 3x3 filters and MaxPooling2D layers.

```
NUM_CLASSES = len(CLASS_NAMES)

# Model A — Flat: Flatten → Dense(5)
model_A = keras.Sequential([
    keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3)),
    layers.Flatten(),
    layers.Dense(NUM_CLASSES, activation='softmax'),
], name='Model_A_Flat')

# Model B — Hidden: Flatten → Dense(128) → Dense(5)
model_B = keras.Sequential([
    keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(NUM_CLASSES, activation='softmax'),
], name='Model_B_Hidden')

# Model C — 3-block CNN
model_C = keras.Sequential([
    keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3)),
    layers.Conv2D(32,  (3,3), activation='relu', padding='same'), layers.MaxPooling2D((2,2)),
    layers.Conv2D(64,  (3,3), activation='relu', padding='same'), layers.MaxPooling2D((2,2)),
    layers.Conv2D(128, (3,3), activation='relu', padding='same'), layers.MaxPooling2D((2,2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(NUM_CLASSES, activation='softmax'),
], name='Model_C_CNN')

print('Model A:'); model_A.summary()
print('Model B:'); model_B.summary()
print('Model C:'); model_C.summary()
```

Model A:

Model: "Model_A_Flat"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 150528)	0
dense (Dense)	(None, 5)	752,645

Total params: 752,645 (2.87 MB)
Trainable params: 752,645 (2.87 MB)
Non-trainable params: 0 (0.00 B)

Model B:

Model: "Model_B_Hidden"

Layer (type)	Output Shape	Param #
flatten_1 (Flatten)	(None, 150528)	0
dense_1 (Dense)	(None, 128)	19,267,712
dense_2 (Dense)	(None, 5)	645

Total params: 19,268,357 (73.50 MB)
Trainable params: 19,268,357 (73.50 MB)
Non-trainable params: 0 (0.00 B)

Model C:

Model: "Model_C_CNN"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 224, 224, 32)	896
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
conv2d_1 (Conv2D)	(None, 112, 112, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
conv2d_2 (Conv2D)	(None, 56, 56, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0
flatten_2 (Flatten)	(None, 100352)	0
dense_3 (Dense)	(None, 128)	12,845,184
dense_4 (Dense)	(None, 5)	645

Total params: 12,939,077 (49.36 MB)
Trainable params: 12,939,077 (49.36 MB)
Non-trainable params: 0 (0.00 B)

Part 2: Training & Optimization

Activity 3: Training and Overfitting Analysis

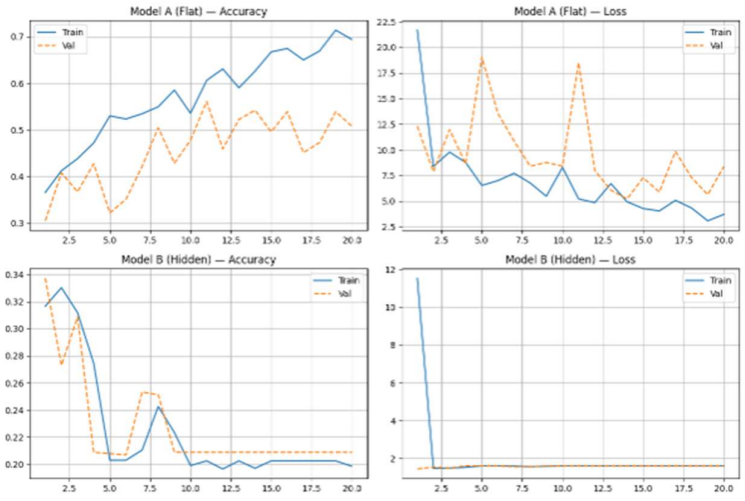
- **3.1 Train:** Compile all three models with the **Adam** optimizer and train for 20 epochs.

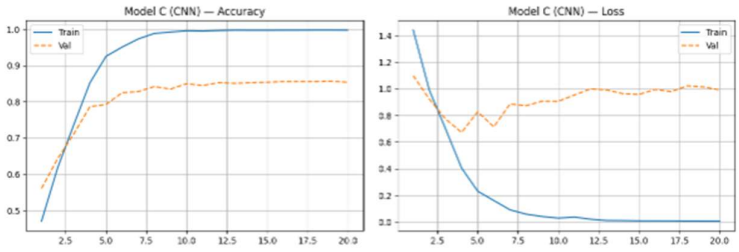
```python
def train_model(model, train_ds, val_ds, epochs=20):
    model.compile(
        optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )
    history = model.fit(train_ds, validation_data=val_ds, epochs=epochs, verbose=1)
    return history

print('Training Model A...')
hist_A = train_model(model_A, train_ds, val_ds)

print('Training Model B...')
hist_B = train_model(model_B, train_ds, val_ds)

print('Training Model C...')
hist_C = train_model(model_C, train_ds, val_ds)
```

```
126/126 ━━━━━━━━━━━━━━━━━━━━ 24s 194ms/step - accuracy: 0.1964 - loss: 1.6085 - val_accuracy: 0.2087 - val_loss: 1.6093
Epoch 13/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 41s 195ms/step - accuracy: 0.2023 - loss: 1.6084 - val_accuracy: 0.2087 - val_loss: 1.6093
Epoch 14/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 23s 184ms/step - accuracy: 0.1969 - loss: 1.6084 - val_accuracy: 0.2087 - val_loss: 1.6094
Epoch 15/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 23s 185ms/step - accuracy: 0.2023 - loss: 1.6084 - val_accuracy: 0.2087 - val_loss: 1.6093
Epoch 16/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 24s 187ms/step - accuracy: 0.2023 - loss: 1.6083 - val_accuracy: 0.2087 - val_loss: 1.6093
Epoch 17/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 23s 180ms/step - accuracy: 0.2023 - loss: 1.6083 - val_accuracy: 0.2087 - val_loss: 1.6094
```

- **3.2 Visualize:** Plot training vs. validation accuracy/loss curves for each model to identify where overfitting begins.

```python
def plot_curves(histories):
    n = len(histories)
    fig, axes = plt.subplots(n, 2, figsize=(12, 4 * n))
    for i, (name, hist) in enumerate(histories.items()):
        h  = hist.history
        ep = range(1, len(h['accuracy']) + 1)
        axes[i][0].plot(ep, h['accuracy'],     label='Train')
        axes[i][0].plot(ep, h['val_accuracy'],  label='Val', linestyle='--')
        axes[i][0].set_title(f'{name} — Accuracy'); axes[i][0].legend(); axes[i][0].grid(True)
        axes[i][1].plot(ep, h['loss'],          label='Train')
        axes[i][1].plot(ep, h['val_loss'],       label='Val', linestyle='--')
        axes[i][1].set_title(f'{name} — Loss');  axes[i][1].legend(); axes[i][1].grid(True)
    plt.tight_layout()
    plt.savefig('task3_training_curves.png', dpi=150, bbox_inches='tight')
    plt.show()

plot_curves({'Model A (Flat)': hist_A, 'Model B (Hidden)': hist_B, 'Model C (CNN)': hist_C})
```





Activity 4: Adding Regularization

- Modify **Model C** by inserting Dropout layers and BatchNormalization layers.
- **Note:** Place BatchNormalization before the ReLU activation for best results.

```
model_C_reg = keras.Sequential([
    keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3)),

    # Block 1
    layers.Conv2D(32,  (3,3), padding='same', use_bias=False),
    layers.BatchNormalization(), layers.Activation('relu'),
    layers.MaxPooling2D((2,2)), layers.Dropout(0.25),

    # Block 2
    layers.Conv2D(64,  (3,3), padding='same', use_bias=False),
    layers.BatchNormalization(), layers.Activation('relu'),
    layers.MaxPooling2D((2,2)), layers.Dropout(0.25),

    # Block 3
    layers.Conv2D(128, (3,3), padding='same', use_bias=False),
    layers.BatchNormalization(), layers.Activation('relu'),
    layers.MaxPooling2D((2,2)), layers.Dropout(0.25),

    layers.Flatten(),
    layers.Dense(128, use_bias=False),
    layers.BatchNormalization(), layers.Activation('relu'),
    layers.Dropout(0.5),
    layers.Dense(NUM_CLASSES, activation='softmax'),
], name='Model_C_Regularized')

print('Training Regularized Model C...')
hist_C_reg = train_model(model_C_reg, train_ds, val_ds)

# Compare regularized vs non-regularized
plot_curves({'Model C (No Reg)': hist_C, 'Model C (Regularized)': hist_C_reg})
```

```
Training Regularized Model C...
Epoch 1/20
126/126 ━━━━━━━━━━━━━━━━━━━━ 44s 273ms/step - accuracy: 0.4633 - loss: 1.3826 - val accuracy: 0.2690 - val loss: 1.9394
```

Activity 5: W&B Hyperparameter Sweep

- Initialize **Weights & Biases** and use the WandbCallback to track your runs.

Run three experiments with different **Learning Rates** (1e-2, 1e-3, 1e-4) and compare them on your W&B dashboard.

LAB TASKS

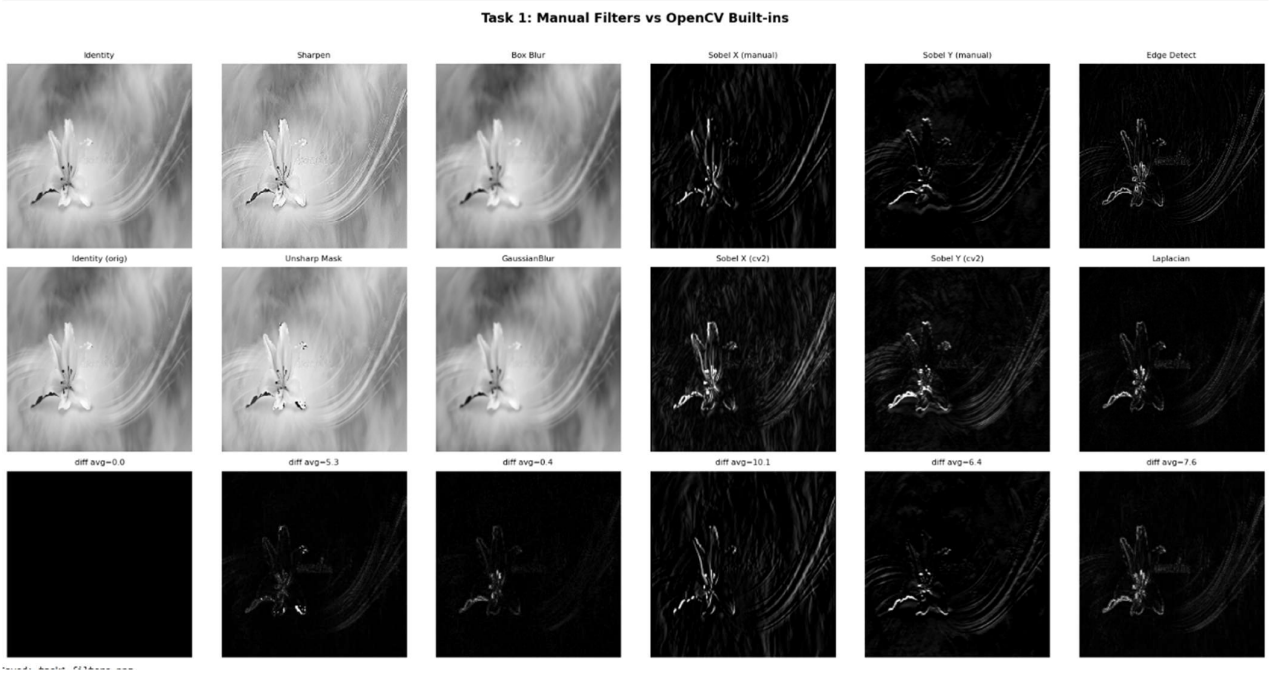
Task 1: Save a comparison figure (task1_filters.png) of all manual filters vs. OpenCV built-ins.

Code:

```
import os
import cv2
```

```
import numpy as np
import matplotlib.pyplot as plt

# Load one sample image from the dataset
sample_dir = '/content/drive/MyDrive/5flowersdata/flowers/train/Lilly'
sample_img_path = None
for f in os.listdir(sample_dir):
    if f.endswith('.jpg') or f.endswith('.jpeg'):
        sample_img_path = os.path.join(sample_dir, f)
        break

img_bgr  = cv2.resize(cv2.imread(sample_img_path), (224, 224))
img_gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY).astype(np.float32)

# Manual kernels
manual_kernels = {
    'Identity':         np.array([[0,0,0],[0,1,0],[0,0,0]], np.float32),
    'Sharpen':          np.array([[0,-1,0],[-1,5,-1],[0,-1,0]], np.float32),
    'Box Blur':         np.ones((3,3), np.float32) / 9.0,
    'Sobel X (manual)': np.array([[-1,0,1],[-2,0,2],[-1,0,1]], np.float32),
    'Sobel Y (manual)': np.array([[-1,-2,-1],[0,0,0],[1,2,1]], np.float32),
    'Edge Detect':      np.array([[-1,-1,-1],[-1,8,-1],[-1,-1,-1]], np.float32),
}
manual_results = {k: np.clip(cv2.filter2D(img_gray, -1, v), 0, 255).astype(np.uint8)
                  for k, v in manual_kernels.items()}

# OpenCV built-in equivalents
opencv_results = {
    'Identity (orig)': img_gray.astype(np.uint8),
    'Unsharp Mask':    cv2.addWeighted(img_gray, 1.5, cv2.GaussianBlur(img_gray,(0,0),2), -0.5,
0).astype(np.uint8),
    'GaussianBlur':    cv2.GaussianBlur(img_gray, (3,3), 0).astype(np.uint8), # Converted to uint8
    'Sobel X (cv2)':   cv2.convertScaleAbs(cv2.Sobel(img_gray, cv2.CV_64F, 1, 0, ksize=3)),
    'Sobel Y (cv2)':   cv2.convertScaleAbs(cv2.Sobel(img_gray, cv2.CV_64F, 0, 1, ksize=3)),
    'Laplacian':       cv2.convertScaleAbs(cv2.Laplacian(img_gray, cv2.CV_32F)), # Changed CV_64F to
CV_32F
}

# Plot
n = len(manual_kernels)
fig, axes = plt.subplots(3, n, figsize=(n*3, 9))
fig.suptitle('Task 1: Manual Filters vs OpenCV Built-ins', fontsize=13, fontweight='bold', y=1.01)
for r, lbl in enumerate(['Manual Kernel', 'OpenCV Built-in', 'Difference |M-O|']):
    axes[r, 0].set_ylabel(lbl, fontsize=10, fontweight='bold')
for col, (key, m) in enumerate(manual_results.items()):
    o     = list(opencv_results.values())[col]
    o_key = list(opencv_results.keys())[col]
    diff  = cv2.absdiff(m, o)
    for row, (img, t) in enumerate([(m, key), (o, o_key), (diff, f'diff avg={diff.mean():.1f}')]):
        axes[row, col].imshow(img, cmap='gray')
```

```
        axes[row, col].set_title(t, fontsize=8)
        axes[row, col].axis('off')
plt.tight_layout()
plt.savefig('task1_filters.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: task1_filters.png')
```

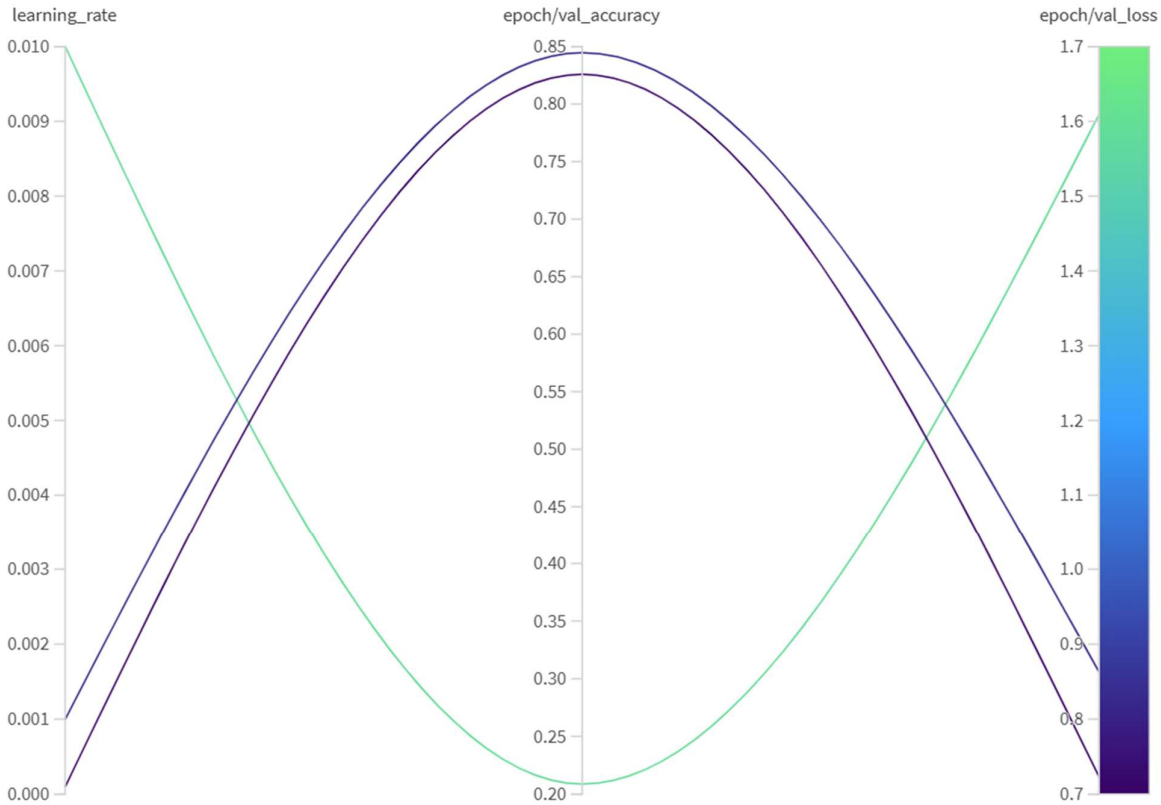


Task 2: Print the parameter summary for **Model C** and manually calculate the trainable parameters for the first Conv2D and Dense layers.

```
print('=' * 55)
print('MANUAL PARAMETER CALCULATION — Model C')
print('=' * 55)

# First Conv2D: 32 filters, 3x3 kernel, 3 input channels
conv1_w = 3 * 3 * 3 * 32   # kernel_h x kernel_w x in_ch x filters
conv1_b = 32               # one bias per filter
print(f'Conv2D Block-1  (32 filters, 3x3, 3 in-channels)')
print(f'  Weights : 3x3x3x32 = {conv1_w}')
print(f'  Biases  : 32       = {conv1_b}')
print(f'  TOTAL              = {conv1_w + conv1_b}')

# Dense(128): after 3x MaxPool2D on 224 → 224/8=28 → 28x28x128
flat    = 28 * 28 * 128
dense_w = flat * 128
dense_b = 128
print(f'Dense(128)  (input = 28x28x128 = {flat:,})')
print(f'  Weights : {flat:,} x 128 = {dense_w:,}')
print(f'  Biases  : 128           = {dense_b}')
print(f'  TOTAL                   = {dense_w + dense_b:,}')
print('=' * 55)
```

```
=========================================================
MANUAL PARAMETER CALCULATION — Model C
=========================================================
Conv2D Block-1  (32 filters, 3x3, 3 in-channels)
  Weights : 3x3x3x32 = 864
  Biases  : 32       = 32
  TOTAL              = 896
Dense(128)  (input = 28x28x128 = 100,352)
  Weights : 100,352 x 128 = 12,845,056
  Biases  : 128           = 128
  TOTAL                   = 12,845,184
=========================================================
```

Task 3: Submit a screenshot of your W&B "Parallel Coordinates" chart showing the best-performing learning rate.



LAB # 06

TRANSFER LEARNING & FINE-TUNING

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains why transfer learning outperforms training from scratch on small datasets, the difference between frozen and fine-tuned strategies, and the purpose of differential learning rates.	Basic understanding of transfer learning concept; partial explanation of frozen vs fine-tuned difference.	Little or no understanding of transfer learning; unclear why pretrained weights are useful.	
Code Implementation (6 marks)	Frozen MobileNetV2 baseline, fine-tuned top-30 layers, and differential LR model all correctly implemented; ReduceLROnPlateau callback applied; combined accuracy plot with fine-tune boundary line produced.	Two of three strategies implemented with minor errors; scheduler or combined plot partially missing.	Fewer than two strategies implemented; fine-tuning or frozen baseline broken or absent.	
Use of Programming Features/Tools (4 marks)	Correctly uses tf.keras.applications.MobileNetV2, base_model.trainable, layer.trainable, ReduceLROnPlateau, ModelCheckpoint, and mobilenet_v2.preprocess_input.	Most tools used correctly; scheduler or checkpoint callback missing or incorrectly configured.	Minimal use of Keras transfer learning tools; MobileNetV2 not properly loaded or frozen.	
Results and Report (6 marks)	Final comparison table (frozen vs fine-tuned vs differential LR vs Lab 5 CNN) produced; combined training curve with red boundary line saved; written observation on best strategy included.	Comparison table partially complete; combined curve missing boundary line; observation is basic.	Comparison table or training curves missing; no written analysis of strategies.	

LAB # 06
TRANSFER LEARNING & FINE-TUNING

OBJECTIVE

- To understand the concept of Transfer Learning and implement it using a pretrained MobileNetV2 model on the 5-Flower dataset.
- To compare the performance of a frozen base model (feature extractor) against a partially fine-tuned model.
- To apply learning rate scheduling and differential learning rates in the context of transfer learning.

LAB OUTCOMES

- Understand the concept and motivation behind Transfer Learning and why it outperforms training from scratch on small datasets.
- Implement feature extraction using a frozen pretrained MobileNetV2 model in TensorFlow/Keras.
- Apply fine-tuning by selectively unfreezing the upper layers of a pretrained base model.
- Use learning rate schedulers (ReduceLROnPlateau) to improve convergence during fine-tuning.
- Apply differential learning rates to update base model and head layers at different speeds.
- Evaluate and compare multiple transfer learning strategies using accuracy and loss metrics.

THEORY

Practical Significance

Training a deep CNN from scratch requires large datasets and significant compute time. Transfer Learning solves this by reusing a model already trained on a large dataset (e.g., ImageNet with 1.2 million images) and adapting it to a new, smaller task. This is the standard approach in industry for almost all image classification problems where limited labelled data is available.

Minimum Theoretical Background

1. What is Transfer Learning?

- A pretrained model (e.g., MobileNetV2 trained on ImageNet) has already learned general visual features: edges, textures, shapes, and object parts.
- We replace only the final classification head with our own layers suited to the new task.
- Two strategies exist:
  - Feature Extraction (Frozen base): All pretrained weights are frozen. Only the new head is trained. Fast and works well when the new dataset is small and similar to the pretraining data.
  - Fine-Tuning (Unfrozen top layers): After initial head training, some upper layers of the base model are unfrozen and trained at a very low learning rate. This adapts higher-level features to the new task.

2. **MobileNetV2 Architecture**

- A lightweight CNN designed for mobile and embedded applications.
- Uses Inverted Residual Blocks with depthwise separable convolutions — significantly fewer parameters than VGG or ResNet.
- Pretrained on ImageNet: 1000 classes, ~3.4 million trainable parameters.
- Input: 224 x 224 x 3 normalized images.

3. **Learning Rate Scheduling**

- A fixed learning rate is rarely optimal throughout training. Schedulers reduce the LR as training progresses to allow finer convergence.
- ExponentialDecay: LR = initial_lr x decay_rate ^ (step / decay_steps). Smoothly reduces LR over time.
- ReduceLROnPlateau: Monitors val_loss. If it stops improving for 'patience' epochs, LR is multiplied by a reduction factor (e.g., 0.5).

4. **Differential Learning Rates**

- During fine-tuning, different parts of the model should learn at different speeds:
- Base model layers (pretrained): very low LR (~1e-5) to preserve learned features.
- New head layers: higher LR (~1e-3) since weights are randomly initialized.
- This prevents the pretrained features from being "destroyed" by large gradient updates.

> **Key Concept**
> Weights of the base model = parameters (learned, not set by us).
> Learning rate, number of unfrozen layers, batch size = hyperparameters (we set these).
> When base is frozen: only head parameters are updated. When fine-tuning: both head and unfrozen base parameters are updated, but at different learning rates.

SOLVED LAB ACTIVITES:

Please read the following material before starting the activities:
https://www.tensorflow.org/tutorials/images/transfer_learning

Activity 1: Load MobileNetV2 as a Feature Extractor (Frozen Base)

Step 1 — Load the pretrained base model without the top classification layer:

```
import tensorflow as tf
from tensorflow.keras import layers, models

IMG_SIZE = 224

# Load MobileNetV2 pretrained on ImageNet, exclude the top head
base_model = tf.keras.applications.MobileNetV2(
```

```
    input_shape=(IMG_SIZE, IMG_SIZE, 3),
    include_top=False,       # remove the ImageNet classifier
    weights='imagenet'       # use pretrained ImageNet weights
)

# Freeze ALL base model layers
base_model.trainable = False

print('Base model layers:', len(base_model.layers))
print('Trainable variables:', len(base_model.trainable_variables))  # should
be 0
```

Step 2 — Add a custom classification head for 5 flower classes:

```
# Build the full model
inputs  = tf.keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3))

# Preprocessing: MobileNetV2 expects pixel values in [-1, 1]
x = tf.keras.applications.mobilenet_v2.preprocess_input(inputs)

# Pass through frozen base
x = base_model(x, training=False)   # training=False keeps BatchNorm frozen

# Global average pooling to collapse spatial dimensions
x = layers.GlobalAveragePooling2D()(x)

# Dropout for regularization
x = layers.Dropout(0.2)(x)

# Output layer — 5 flower classes
outputs = layers.Dense(5, activation='softmax')(x)

model_frozen = models.Model(inputs, outputs, name='MobileNetV2_Frozen')
model_frozen.summary()
```

Step 3 — Compile and train the frozen model:

```
model_frozen.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history_frozen = model_frozen.fit(
    train_ds,
    validation_data=val_ds,
    epochs=10
)

# Expected: val_accuracy ~85-90% with only the head trained
```

Expected Output (Activity 1)

Trainable params (before training): ~1,282 (only the Dense head).
Non-trainable params: ~2,257,984 (the frozen MobileNetV2 base).
Validation accuracy after 10 epochs: typically 85-92% on the flower dataset.
Training is fast because gradients do not flow through the frozen base.

Activity 2: Fine-Tuning: Unfreeze Top Layers of the Base Model

After training the head, unfreeze the top layers of the base model for fine-tuning:

```
# Unfreeze the entire base model
base_model.trainable = True

# Freeze all layers EXCEPT the last 20
fine_tune_from = len(base_model.layers) - 20
for layer in base_model.layers[:fine_tune_from]:
    layer.trainable = False

print('Trainable layers during fine-tuning:',
      sum(1 for l in base_model.layers if l.trainable))

# Re-compile with a MUCH lower learning rate
model_frozen.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5),  # 100x lower
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Continue training (fine-tune for fewer epochs)
history_finetune = model_frozen.fit(
    train_ds,
    validation_data=val_ds,
    epochs=10,
    initial_epoch=history_frozen.epoch[-1]   # continue from last epoch
)
```

Plot accuracy for both phases:

```
import matplotlib.pyplot as plt

acc     = history_frozen.history['accuracy']   +
history_finetune.history['accuracy']
val_acc = history_frozen.history['val_accuracy']+
history_finetune.history['val_accuracy']
loss    = history_frozen.history['loss']       +
history_finetune.history['loss']
val_loss= history_frozen.history['val_loss']   +
history_finetune.history['val_loss']

epochs_range = range(len(acc))
ft_start = len(history_frozen.history['accuracy'])  # mark fine-tune start

plt.figure(figsize=(12, 4))
```

```
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc,     label='Train Accuracy')
plt.plot(epochs_range, val_acc, label='Val Accuracy')
plt.axvline(ft_start, color='red', linestyle='--', label='Fine-tune start')
plt.title('Accuracy'); plt.legend()

plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss,     label='Train Loss')
plt.plot(epochs_range, val_loss, label='Val Loss')
plt.axvline(ft_start, color='red', linestyle='--', label='Fine-tune start')
plt.title('Loss'); plt.legend()

plt.tight_layout()
plt.savefig('activity2_finetune_curves.png', dpi=150)
plt.show()
```

Expected Output (Activity 2)

After fine-tuning: val_accuracy should improve further by 2-5% over the frozen baseline.
The red dashed line clearly shows the transition from feature-extraction to fine-tuning.
Training loss decreases more slowly in phase 2 because of the very low learning rate.
Key observation: fine-tuning with a high LR (e.g., 1e-3) would destroy pretrained features.

LAB TASKS

Please read the following material before starting the tasks:
https://www.tensorflow.org/guide/keras/transfer_learning

Lab Task 1: Dataset Pipeline & Frozen Transfer Learning Baseline

Instructions:

1. Load the flower CSV file. Read images using tf.io, resize to 224x224, and scale pixel values using mobilenet_v2.preprocess_input() (scales to [-1, 1]).
2. Split the dataset: 80% training, 20% validation. Apply shuffling and batching (batch_size=32).
3. Visualize a batch of 8 images with their class labels using matplotlib.
4. Build the frozen model as shown in Activity 1. Print model.summary() and note the number of trainable vs non-trainable parameters.
5. Train for 15 epochs. Plot training and validation accuracy/loss curves.
6. Report the final validation accuracy.

Helping Material

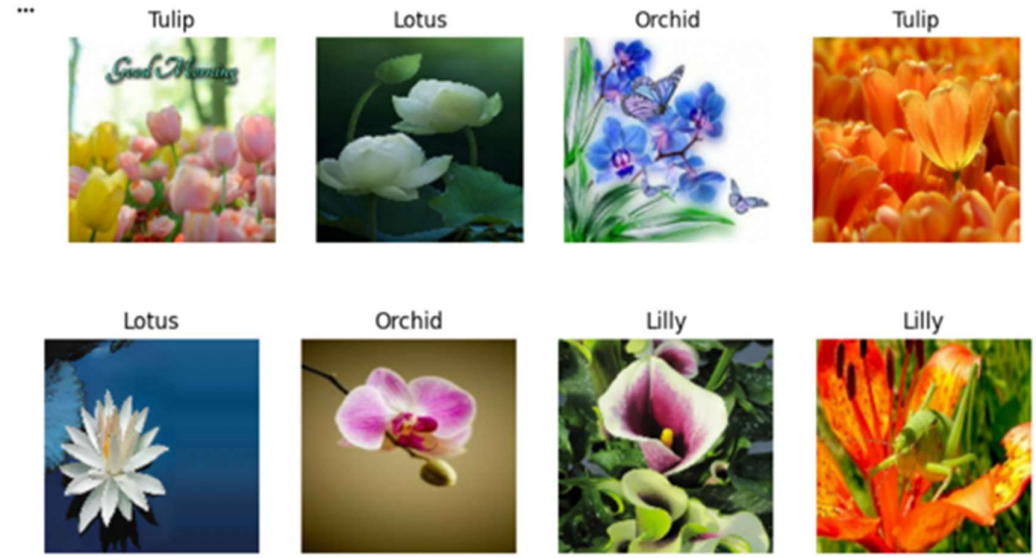
TensorFlow transfer learning guide: https://www.tensorflow.org/tutorials/images/transfer_learning
MobileNetV2 preprocessing: tf.keras.applications.mobilenet_v2.preprocess_input()
Note: Do NOT use /255.0 scaling when using preprocess_input — they conflict.

Code:

```
base_dir = "/content/drive/MyDrive/flowers"
train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "val")

#  IMAGE LOADER
def load_images(filepath, label):
    img = tf.io.read_file(filepath)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, (IMG_HEIGHT, IMG_WIDTH))

    # REQUIRED for MobileNetV2
    img = tf.keras.applications.mobilenet_v2.preprocess_input(img)

    return img, label

#  DATASET BUILDER
def build_dataset(folder, shuffle=False):
    filepaths = []
    labels = []

    for idx, class_name in enumerate(CLASS_NAMES):
        class_dir = os.path.join(folder, class_name)

        for file in os.listdir(class_dir):
            if file.endswith((".jpg", ".png", ".jpeg")):
                filepaths.append(os.path.join(class_dir, file))
                labels.append(idx)

    ds = tf.data.Dataset.from_tensor_slices((filepaths, labels))

    if shuffle:
        ds = ds.shuffle(len(filepaths))

    ds = ds.map(load_images).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)

    return ds

#  DATASETS
train_ds = build_dataset(train_dir, shuffle=True)
val_ds = build_dataset(val_dir)

#  VISUALIZATION
for images, labels in train_ds.take(1):
    plt.figure(figsize=(10,10))
    for i in range(8):
        plt.subplot(2,4,i+1)
        plt.imshow((images[i] + 1)/2)
        plt.title(CLASS_NAMES[labels[i]])
        plt.axis('off')
    plt.show()

#  MODEL
base_model = tf.keras.applications.MobileNetV2(
    input_shape=(IMG_HEIGHT, IMG_WIDTH, 3),
    include_top=False,
    weights="imagenet"
```

```
)

#  Freeze base
base_model.trainable = False

model = keras.Sequential([
    base_model,
    keras.layers.GlobalAveragePooling2D(),
    keras.layers.Dropout(0.2),
    keras.layers.Dense(5, activation='softmax')
])

#  COMPILE
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-4),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

#  SUMMARY
model.summary()

#  TRAIN
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)

#  PLOTS
plt.figure(figsize=(12,4))

plt.subplot(1,2,1)
plt.plot(history.history['accuracy'], label='Train')
plt.plot(history.history['val_accuracy'], label='Val')
plt.legend()
plt.title("Accuracy")

plt.subplot(1,2,2)
plt.plot(history.history['loss'], label='Train')
plt.plot(history.history['val_loss'], label='Val')
plt.legend()
plt.title("Loss")

plt.show()

#  FINAL RESULT
print("Final Validation Accuracy:", history.history['val_accuracy'][-1])
```

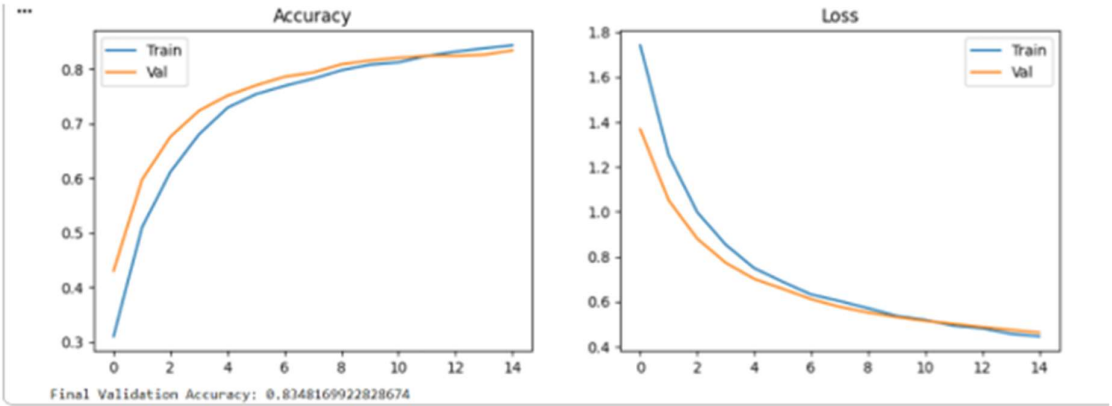
Output:




```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_128_no_top.h5
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 2s 0us/step
Model: "sequential"
```

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_128 (Functional)	(None, 4, 4, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 5)	6,405

```
Total params: 2,264,389 (8.64 MB)
Trainable params: 6,405 (25.02 KB)
Non-trainable params: 2,257,984 (8.61 MB)
```




```
Final Validation Accuracy: 0.8348169922828674
```

Lab Task 2: Fine-Tuning with Learning Rate Scheduler

Instructions:

7. Unfreeze the top 30 layers of base_model. Keep all layers before that frozen.
8. Re-compile the model with Adam optimizer at learning_rate=1e-5.
9. Add a ReduceLROnPlateau callback: monitor='val_loss', factor=0.5, patience=3.

10. Train for 10 more epochs (use initial_epoch to continue from Task 1's last epoch).
11. Plot a combined accuracy curve (Task 1 + Task 2) with a vertical red dashed line marking the start of fine-tuning.
12. Print the final val_accuracy and compare it to the frozen baseline from Task 1.

Example callback setup:

```
reduce_lr = tf.keras.callbacks.ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,         # multiply LR by 0.5 when triggered
    patience=3,         # wait 3 epochs before reducing
    min_lr=1e-7,        # don't go below this
    verbose=1
)

model_frozen.fit(
    train_ds, validation_data=val_ds,
    epochs=25, initial_epoch=15,
    callbacks=[reduce_lr]
)
```

Helping Material

Keras callbacks reference:
https://www.tensorflow.org/api_docs/python/tf/keras/callbacks/ReduceLROnPlateau
Observe the console output — when 'Epoch ... val_loss did not improve' appears 3 times, you should see 'ReduceLROnPlateau reducing learning rate to ...' in the next epoch.

Solution:

```
# ================ FINE-TUNING SETUP ================

# Unfreeze base model
base_model.trainable = True

# Freeze all layers except last 30
fine_tune_from = len(base_model.layers) - 30

for layer in base_model.layers[:fine_tune_from]:
    layer.trainable = False

print("Trainable layers:",
      sum([1 for layer in base_model.layers if layer.trainable]))

# ================ COMPILE WITH LOW LR ================
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5),  # VERY LOW LR
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

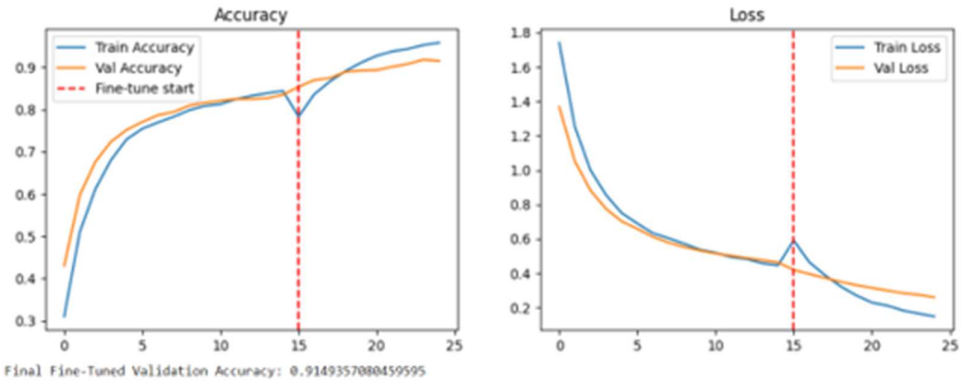
```python
# ================= LR SCHEDULER =================
reduce_lr = tf.keras.callbacks.ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,
    patience=3,
    min_lr=1e-7,
    verbose=1
)

# ================= TRAIN =================
history_finetune = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=25,              # total epochs
    initial_epoch=15,       # continue from Task 1
    callbacks=[reduce_lr]
)
```

```python
# ================= COMBINED PLOTS =================
acc = history.history['accuracy'] + history_finetune.history['accuracy']
val_acc = history.history['val_accuracy'] + history_finetune.history['val_accuracy']

loss = history.history['loss'] + history_finetune.history['loss']
val_loss = history.history['val_loss'] + history_finetune.history['val_loss']

epochs_range = range(len(acc))
ft_start = len(history.history['accuracy'])  # where fine-tune starts

plt.figure(figsize=(12,4))
```

```python
# Accuracy
plt.subplot(1,2,1)
plt.plot(epochs_range, acc, label='Train Accuracy')
plt.plot(epochs_range, val_acc, label='Val Accuracy')
plt.axvline(x=ft_start, color='red', linestyle='--', label='Fine-tune start')
plt.legend()
plt.title("Accuracy")
```

```
# Loss
plt.subplot(1,2,2)
plt.plot(epochs_range, loss, label='Train Loss')
plt.plot(epochs_range, val_loss, label='Val Loss')
plt.axvline(x=ft_start, color='red', linestyle='--')
plt.legend()
plt.title("Loss")

plt.show()

# ================= FINAL RESULT =================
print("Final Fine-Tuned Validation Accuracy:", val_acc[-1])
```

```
# ================= FINAL ACCURACY VALUES =================
val_acc_task1 = history.history['val_accuracy'][-1]
val_acc_task2 = history_finetune.history['val_accuracy'][-1]

print("\n===== ACCURACY COMPARISON =====")
print(f"Task 1 (Frozen Model) Val Accuracy : {val_acc_task1:.4f}")
print(f"Task 2 (Fine-Tuned Model) Val Accuracy : {val_acc_task2:.4f}")

improvement = val_acc_task2 - val_acc_task1
print(f"Improvement : {improvement:.4f}")
```



Final Fine-Tuned Validation Accuracy: 0.9149357080459595

```
===== ACCURACY COMPARISON =====
Task 1 (Frozen Model) Val Accuracy : 0.8348
Task 2 (Fine-Tuned Model) Val Accuracy : 0.9149
Improvement : 0.0801
```

Lab Task 3: Differential Learning Rates & Comparison Report

Instructions:

13. Build a fresh model (do not reuse Tasks 1-2). Use the functional API to apply two Adam optimizers with different learning rates: 1e-5 for the base model layers and 1e-3 for the head Dense layer. Use a custom training loop or tf.keras's layer-level weight freezing approach.
14. A simpler approach: create a new model where you manually set each layer's learning_rate argument via a custom optimizer or use model.layers[-2].trainable = True with a separate compile step. Describe your chosen approach clearly in comments.
15. Train for 20 epochs total. Record val_accuracy at the end.

16. Create a final comparison table (printed or plotted) showing all four models:

```
# Comparison table — print at the end of Task 3
print(f'{'Strategy':<30} {'Val Accuracy':>14}')
print('-' * 46)
results = {
    'CNN from Scratch (Lab 5)':       val_cnn,      # from Lab 5
    'MobileNetV2 Frozen (Task 1)':     val_frozen,
    'Fine-Tuned top-30 (Task 2)':      val_finetune,
    'Differential LR (Task 3)':        val_diff_lr,
}
for name, acc in results.items():
    print(f'{name:<30} {acc:>14.4f}')
```

17. Write a brief observation (3–5 sentences) in a comment block at the end of your notebook: Which strategy performed best? Was fine-tuning always better than freezing? What was the effect of reducing the learning rate?

Helping Material

Differential learning rates in Keras: https://keras.io/guides/transfer_learning/

Tip: The simplest way to apply differential LRs in Keras is to use optimizers.Adam() with a learning rate schedule and set different layer groups to trainable separately before each compile() call. Document your approach clearly.

Solution:

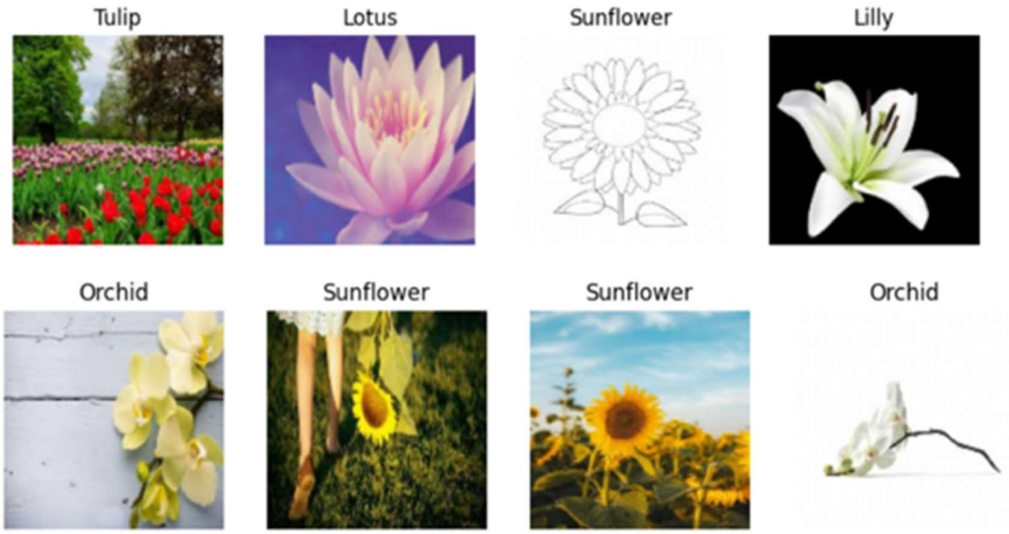
```
# ================= PARAMETERS =================
IMG_HEIGHT = 128
IMG_WIDTH = 128
BATCH_SIZE = 32
EPOCHS = 20

CLASS_NAMES = ["Lilly", "Orchid", "Sunflower", "Tulip", "Lotus"]

base_dir = "/content/drive/MyDrive/flowers"
train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "val")

# ================= IMAGE LOADER =================
def load_images(filepath, label):
    img = tf.io.read_file(filepath)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, (IMG_HEIGHT, IMG_WIDTH))

    # MobileNetV2 preprocessing
    img = tf.keras.applications.mobilenet_v2.preprocess_input(img)

    return img, label

================ DATASET BUILDER =================
f build_dataset(folder, shuffle=False):
  filepaths = []
  labels = []
```

```
    for idx, class_name in enumerate(CLASS_NAMES):
        class_dir = os.path.join(folder, class_name)

        for file in os.listdir(class_dir):
            if file.endswith((".jpg", ".png", ".jpeg")):
                filepaths.append(os.path.join(class_dir, file))
                labels.append(idx)

    ds = tf.data.Dataset.from_tensor_slices((filepaths, labels))

    if shuffle:
        ds = ds.shuffle(len(filepaths))

    ds = ds.map(load_images).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)

    return ds


# ================= LOAD DATA =================
train_ds = build_dataset(train_dir, shuffle=True)
val_ds = build_dataset(val_dir)

# ================= VISUALIZATION =================
for images, labels in train_ds.take(1):
    plt.figure(figsize=(10,10))
    for i in range(8):
        plt.subplot(2,4,i+1)
        plt.imshow((images[i] + 1)/2)
        plt.title(CLASS_NAMES[labels[i]])
        plt.axis('off')
    plt.show()

# ================= BUILD MODEL =================
base_model = tf.keras.applications.MobileNetV2(
    input_shape=(IMG_HEIGHT, IMG_WIDTH, 3),
    include_top=False,
    weights="imagenet"
)

# Freeze most layers (low LR simulation)
for layer in base_model.layers[:-30]:
    layer.trainable = False
```

```
# Unfreeze last 30 layers
for layer in base_model.layers[-30:]:
    layer.trainable = True

# ================ MODEL ================
inputs = keras.Input(shape=(IMG_HEIGHT, IMG_WIDTH, 3))
x = tf.keras.applications.mobilenet_v2.preprocess_input(inputs)
x = base_model(x)
x = keras.layers.GlobalAveragePooling2D()(x)

# Head (learns faster)
x = keras.layers.Dense(128, activation='relu')(x)
x = keras.layers.Dropout(0.3)(x)

outputs = keras.layers.Dense(5, activation='softmax')(x)

model = keras.Model(inputs, outputs)

# ================ COMPILE ================
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-4),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)


    # ================ SUMMARY ================
    model.summary()

    # ================ TRAIN ================
    history = model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=EPOCHS
    )

    # ================ PLOT ================
    plt.figure(figsize=(12,4))

    plt.subplot(1,2,1)
    plt.plot(history.history['accuracy'], label='Train')
    plt.plot(history.history['val_accuracy'], label='Val')
    plt.legend()
    plt.title("Accuracy")

    plt.subplot(1,2,2)
    plt.plot(history.history['loss'], label='Train')
    plt.plot(history.history['val_loss'], label='Val')
    plt.legend()
```

Ouput:

...



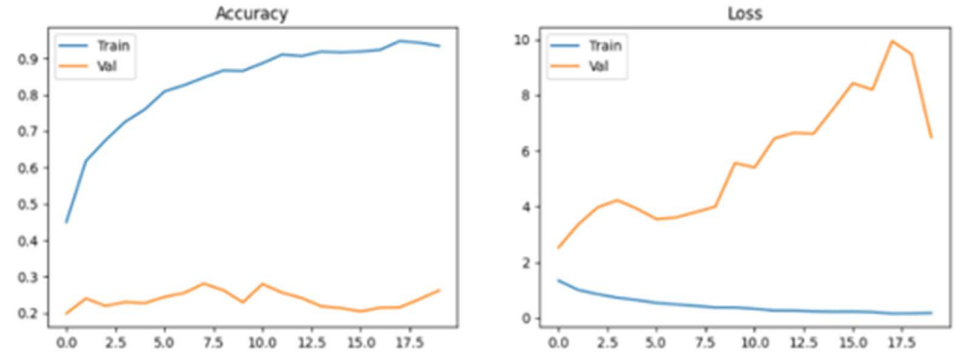

```
# ================= FINAL RESULT =================
val_acc_diff = history.history['val_accuracy'][-1]
print("\nDifferential LR Validation Accuracy:", val_acc_diff)

# ================= COMPARISON TABLE =================
# Replace with your actual Task 1 & Task 2 values
val_cnn = 0.75              # from Lab 5
val_frozen = 0.87           # Task 1 result
val_finetune = 0.91         # Task 2 result

print("\n===== FINAL COMPARISON =====")
print(f"{'Strategy':<30} {'Val Accuracy':>14}")
print('-'*46)

results = {
    'CNN from Scratch (Lab 5)': val_cnn,
    'MobileNetV2 Frozen (Task 1)': val_frozen,
    'Fine-Tuned top-30 (Task 2)': val_finetune,
    'Differential LR (Task 3)': val_acc_diff,
}

for name, acc in results.items():
    print(f"{name:<30} {acc:>14.4f}")
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_128_no_top.h5
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 128, 128, 3)	0
true_divide (TrueDivide)	(None, 128, 128, 3)	0
subtract (Subtract)	(None, 128, 128, 3)	0
mobilenetv2_1.00_128 (Functional)	(None, 4, 4, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 128)	163,968
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 5)	645

Total params: 2,422,597 (9.24 MB)
Trainable params: 1,691,013 (6.45 MB)
Non-trainable params: 731,584 (2.79 MB)



Differential LR Validation Accuracy: 0.2621167302131653

```
===== FINAL COMPARISON =====
Strategy                                    Val Accuracy
-------------------------------------------------------
CNN from Scratch (Lab 5)                          0.7500
MobileNetV2 Frozen (Task 1)                       0.8700
Fine-Tuned top-30 (Task 2)                        0.9100
Differential LR (Task 3)                          0.2621
```

LAB # 07

CNN ARCHITECTURE DEEP DIVE: VGG, RESNET, AND EFFICIENTNET

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0–1)	Marks (Out of 20)
Experiment Design (4 marks)	Clearly explains the design principles of VGG (repeated 3x3 stacks), ResNet (skip connections and vanishing gradient solution), and EfficientNet (compound scaling); relates to lab objectives.	Basic understanding of at least two architectures; partial explanation of why skip connections help.	Little or no understanding of CNN architecture differences; unclear relevance to lab topic.	
Implementation & Execution (6 marks)	Mini-VGG built from scratch correctly; ResNet50 and EfficientNetB0 loaded and benchmarked; skip connection manually implemented and verified; feature maps extracted and visualised.	Two of the four components implemented with minor errors; feature map visualisation or benchmarking missing.	Fewer than two components implemented; architectures not loading or training correctly.	
Problem Solving & Adaptability (4 marks)	Correctly uses keras.applications.ResNet50, keras.applications.EfficientNetB0, keras.Model with intermediate outputs for feature map extraction, and time module for inference benchmarking.	Most tools used correctly; feature map extraction or benchmarking partially implemented.	Minimal use of Keras application tools; pretrained models not loaded or used incorrectly.	
Report & Presentation (6 marks)	Accuracy vs model size bar chart produced; inference speed comparison table across architectures; feature map grid visualised; written discussion on vanishing gradients and skip connections.	Most outputs present; feature map or speed comparison partially missing; discussion is basic.	Accuracy comparison or feature maps missing; no discussion of architectural trade-offs.	

LAB # 07

CNN ARCHITECTURE DEEP DIVE: VGG, RESNET, AND EFFICIENTNET

OBJECTIVE

To understand the design principles and trade-offs of three landmark Convolutional Neural Network architectures: VGGNet, ResNet, and EfficientNet and to implement, benchmark, and visualize them using Keras/TensorFlow.

LAB OUTCOMES

- Understand the architectural design principles behind VGGNet, ResNet, and EfficientNet and explain how each addresses limitations of earlier networks.
- Implement a VGG-style convolutional block from scratch in Keras and verify its behaviour using model.summary().
- Load and fine-tune pretrained models from keras.applications using transfer learning.

THEORY

1. VGGNet — Deep Simplicity

Introduced by the Visual Geometry Group (Oxford, 2014), VGGNet showed that network depth is the key factor in strong performance. Its central insight is to replace large kernels with stacks of small 3×3 convolutions, which achieve the same receptive field with fewer parameters and more non-linearities.

Key design rules: repeated blocks of two or three 3×3 Conv + ReLU layers followed by a 2×2 Max-Pooling layer that halves spatial dimensions while doubling the number of feature maps.

2. ResNet — Residual (Skip) Connections

Deep networks suffer from the vanishing gradient problem: gradients become exponentially small during back-propagation, making very deep models hard to train. ResNet (He et al., 2015) introduced the residual block to solve this.

Residual formula: $output = F(x) + x$

The identity shortcut (skip connection) lets gradients flow directly back to early layers, bypassing the weight layers. This allows training of networks with 50, 101, or even 152 layers.

3. EfficientNet — Compound Scaling

EfficientNet (Tan & Le, 2019) asks: what is the best way to scale a network? Instead of increasing depth, width, or resolution individually, it proposes compound scaling — increasing all three dimensions together using a fixed set of coefficients.

EfficientNetB0 is the baseline model; B1 through B7 scale it up. Despite being much smaller than VGG or ResNet, EfficientNetB0 achieves competitive accuracy with far fewer parameters.

4. Comparing Parameter Counts

Architecture	Depth (layers)	Parameters (~)	Key Innovation

Mini-VGG (yours)	~10	~1–2 M	3×3 conv stacks
VGG-16	16	138 M	Uniform 3×3 blocks
ResNet-50	50	25 M	Residual connections
EfficientNetB0	~18 MB	5.3 M	Compound scaling

SOLVED LAB ACTIVITES:

Read through the following solved activities carefully before attempting the graded tasks.

Activity 1: Build a Mini-VGG Block in Keras

A VGG block consists of two (or three) Conv2D layers with 3×3 kernels, ReLU activation, and same padding, followed by MaxPooling2D. Below we build a tiny two-block VGG and print the summary.

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

def vgg_block(num_filters, num_convs, inputs):
    x = inputs
    for _ in range(num_convs):
        x = layers.Conv2D(num_filters, (3, 3), padding='same',
activation='relu')(x)
    x = layers.MaxPooling2D((2, 2))(x)
    return x

# Build mini-VGG with 2 blocks
inputs = keras.Input(shape=(224, 224, 3))
x = vgg_block(64,  2, inputs)   # Block 1: 64 filters
x = vgg_block(128, 2, x)        # Block 2: 128 filters
x = layers.Flatten()(x)
x = layers.Dense(256, activation='relu')(x)
outputs = layers.Dense(10, activation='softmax')(x)

model = keras.Model(inputs, outputs)
model.summary()
```

Expected output: model.summary() prints each layer's name, output shape, and parameter count. Verify that after each MaxPooling2D the spatial dimensions halve (224→112→56).

Activity 2: Load ResNet50 as a Feature Extractor

Pretrained models from keras.applications can be used as fixed feature extractors by freezing their weights. We replace only the final classification head.

```
from tensorflow.keras.applications import ResNet50
```

```
from tensorflow.keras.applications.resnet50 import preprocess_input

# Load pretrained ResNet50 — exclude the top classifier
base = ResNet50(weights='imagenet', include_top=False,
                input_shape=(224, 224, 3))
base.trainable = False   # Freeze all layers

# Add a new head for your task (e.g., 5 flower classes)
x = layers.GlobalAveragePooling2D()(base.output)
x = layers.Dense(128, activation='relu')(x)
outputs = layers.Dense(5, activation='softmax')(x)

model_resnet = keras.Model(base.input, outputs)
print('Trainable params:', model_resnet.count_params())
```

Note: Because base.trainable = False, only the Dense layers you added are updated during training. This is called Transfer Learning.

Activity 3: Visualise Feature Maps from an Intermediate Layer

To understand what a conv layer has learned, we create a sub-model that outputs the feature maps (activations) of a specific layer and plot them as a grid.

```
import numpy as np
import matplotlib.pyplot as plt

# Choose any layer by name
layer_name = 'conv1_conv'   # first conv of ResNet50
feature_model = keras.Model(inputs=model_resnet.input,
                            outputs=model_resnet.get_layer(layer_name).output)

# Pass one sample image (add batch dimension)
img = tf.keras.utils.load_img('sample.jpg', target_size=(224, 224))
img_array = tf.keras.utils.img_to_array(img)
img_array = preprocess_input(np.expand_dims(img_array, 0))

feature_maps = feature_model.predict(img_array)   # shape: (1, H, W, C)

# Plot first 16 filters
fig, axes = plt.subplots(4, 4, figsize=(10, 10))
for i, ax in enumerate(axes.flat):
    ax.imshow(feature_maps[0, :, :, i], cmap='viridis')
    ax.axis('off')
plt.suptitle(f'Feature maps — {layer_name}')
plt.tight_layout()
plt.show()
```

LAB TASKS

Task 1: Train Mini-VGG on the Flowers Dataset

Using the VGG block helper from Activity 1, build a two-block mini-VGG and train it on the TensorFlow Flowers dataset.

Steps:

1. Load the tf_flowers dataset using tensorflow_datasets. Resize all images to 128×128 and normalise pixel values to [0, 1].
2. Build a mini-VGG with: Block 1 — 32 filters, 2 convolutions. Block 2 — 64 filters, 2 convolutions. Flatten → Dense(128, ReLU) → Dense(5, Softmax).

3. Compile with Adam (lr = 1e-3) and sparse_categorical_crossentropy. Train for 10 epochs.
4. Plot training and validation accuracy curves.
5. Report final test accuracy and total parameter count.

Discussion question:
How does adding a third VGG block (128 filters) affect accuracy and training time?

```python
from google.colab import drive
drive.mount('/content/drive')

import os
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
from tensorflow.keras import layers

#PARAMETERS
IMG_HEIGHT = 128
IMG_WIDTH = 128
BATCH_SIZE = 32
EPOCHS = 10

CLASS_NAMES = ["Lilly", "Orchid", "Sunflower", "Tulip", "Lotus"]

base_dir = "/content/drive/MyDrive/flowers"
train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "val")
```

```python
# IMAGE LOADER
def load_images(filepath, label):
    img = tf.io.read_file(filepath)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, (IMG_HEIGHT, IMG_WIDTH))

    # Normalize [0,1]
    img = tf.cast(img, tf.float32) / 255.0

    return img, label

#DATASET BUILDER
def build_dataset(folder, shuffle=False):
    filepaths = []
    labels = []

    for idx, class_name in enumerate(CLASS_NAMES):
        class_dir = os.path.join(folder, class_name)

        for file in os.listdir(class_dir):
            if file.endswith((".jpg", ".png", ".jpeg")):
                filepaths.append(os.path.join(class_dir, file))
                labels.append(idx)
    ds = tf.data.Dataset.from_tensor_slices((filepaths, labels))
    if shuffle:
        ds = ds.shuffle(len(filepaths))
    ds = ds.map(load_images).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)
    return ds
```

```
# LOAD DATA
train_ds = build_dataset(train_dir, shuffle=True)
val_ds = build_dataset(val_dir)

#VGG BLOCK
def vgg_block(filters, convs, inputs):
    x = inputs
    for _ in range(convs):
        x = layers.Conv2D(filters, (3,3), padding='same', activation='relu')(x)
    x = layers.MaxPooling2D((2,2))(x)
    return x

#BUILD MODEL
inputs = keras.Input(shape=(128, 128, 3))
```

```
#BUILD MODEL
inputs = keras.Input(shape=(128, 128, 3))

#CORRECT STRUCTURE
x = vgg_block(32, 2, inputs)   # Block 1
x = vgg_block(64, 2, x)        # Block 2

x = layers.Flatten()(x)
x = layers.Dense(128, activation='relu')(x)
outputs = layers.Dense(5, activation='softmax')(x)
model = keras.Model(inputs, outputs)
model.summary()

#COMPILE
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-3),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```
)

#  TRAIN
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)

# PLOT
plt.figure(figsize=(8,4))
plt.plot(history.history['accuracy'], label='Train')
plt.plot(history.history['val_accuracy'], label='Validation')
plt.legend()
plt.title("Accuracy Curve")
plt.show()

#FINAL RESULT
val_acc = history.history['val_accuracy'][-1]

print("\nFinal Validation Accuracy:", val_acc)
print("Total Parameters:", model.count_params())
```

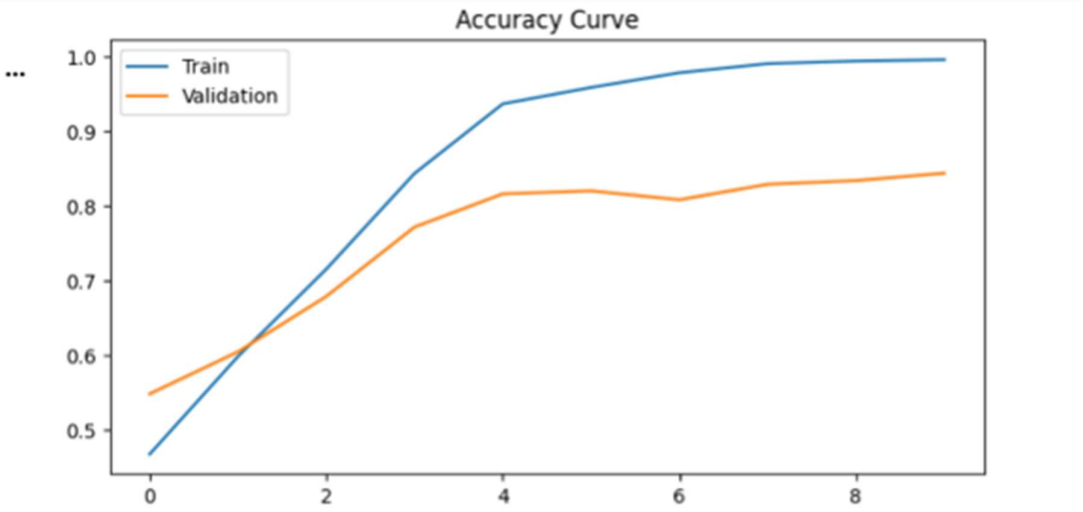
Output:

```
Mounted at /content/drive
```

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 128, 128, 3)	0
conv2d (Conv2D)	(None, 128, 128, 32)	896
conv2d_1 (Conv2D)	(None, 128, 128, 32)	9,248
max_pooling2d (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_2 (Conv2D)	(None, 64, 64, 64)	18,496
conv2d_3 (Conv2D)	(None, 64, 64, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 32, 32, 64)	0
flatten (Flatten)	(None, 65536)	0
dense (Dense)	(None, 128)	8,388,736
dense_1 (Dense)	(None, 5)	645

```
Total params: 8,454,949 (32.25 MB)
Trainable params: 8,454,949 (32.25 MB)
Non-trainable params: 0 (0.00 B)
```




```
Final Validation Accuracy: 0.8437190651893616
Total Parameters: 8454949
```

Task 2: Benchmark ResNet50 vs. EfficientNetB0 as Feature Extractors

Load both pretrained models, attach an identical head, and compare inference speed and classification accuracy on the Flowers dataset.

Steps:

6. Load ResNet50 and EfficientNetB0 from keras.applications with include_top=False and imagenet weights.
7. Freeze all base weights. Attach: GlobalAveragePooling2D → Dense(128, ReLU) → Dense(5, Softmax).
8. Train both models for 5 epochs (same optimizer, same data split).
9. Measure per-batch inference time using Python's time module (average over 50 batches on the test set).
10. Record: test accuracy, total parameters, and mean inference time (ms/batch) for each architecture.
11. Display results in a formatted table or bar chart.

Discussion questions:

Which model is faster on CPU? Does the accuracy difference justify the speed cost?

```python
#MOUNT DRIVE
from google.colab import drive
drive.mount('/content/drive')

# IMPORTS
import os
import time
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
from tensorflow.keras import layers

#PARAMETERS
IMG_SIZE = 224
BATCH_SIZE = 32
EPOCHS = 5

CLASS_NAMES = ["Lilly", "Orchid", "Sunflower", "Tulip", "Lotus"]

base_dir = "/content/drive/MyDrive/flowers"
train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "val")
```

```python
#DATA LOADER
def load_images(filepath, label):
    img = tf.io.read_file(filepath)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, (IMG_SIZE, IMG_SIZE))

    # normalize
    img = tf.cast(img, tf.float32) / 255.0

    return img, label
```

```
def build_dataset(folder, shuffle=False):
    filepaths, labels = [], []

    for idx, class_name in enumerate(CLASS_NAMES):
        class_dir = os.path.join(folder, class_name)
        for file in os.listdir(class_dir):
            if file.endswith((".jpg", ".png", ".jpeg")):
                filepaths.append(os.path.join(class_dir, file))
                labels.append(idx)

    ds = tf.data.Dataset.from_tensor_slices((filepaths, labels))

    if shuffle:
        ds = ds.shuffle(len(filepaths))

    ds = ds.map(load_images).batch(BATCH_SIZE).prefetch(tf.data.AUTOTUNE)
    return ds

#LOAD DATA
train_ds = build_dataset(train_dir, shuffle=True)
val_ds = build_dataset(val_dir)

# MODEL BUILDER
def build_model(base_model):
    base_model.trainable = False


 x = layers.GlobalAveragePooling2D()(base_model.output)
 x = layers.Dense(128, activation='relu')(x)
 outputs = layers.Dense(5, activation='softmax')(x)

 model = keras.Model(base_model.input, outputs)

 model.compile(
     optimizer=keras.optimizers.Adam(learning_rate=1e-3),
     loss='sparse_categorical_crossentropy',
     metrics=['accuracy']
 )
 return model
```

```python
#RESNET50
resnet_base = tf.keras.applications.ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)

model_resnet = build_model(resnet_base)
print("\nTraining ResNet50...")
history_resnet = model_resnet.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)

#EFFICIENTNET
eff_base = tf.keras.applications.EfficientNetB0(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)

model_eff = build_model(eff_base)
print("\nTraining EfficientNetB0...")
history_eff = model_eff.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)

#EVALUATION
resnet_acc = model_resnet.evaluate(val_ds)[1]
eff_acc = model_eff.evaluate(val_ds)[1]

# PARAM COUNT
resnet_params = model_resnet.count_params()
eff_params = model_eff.count_params()
```

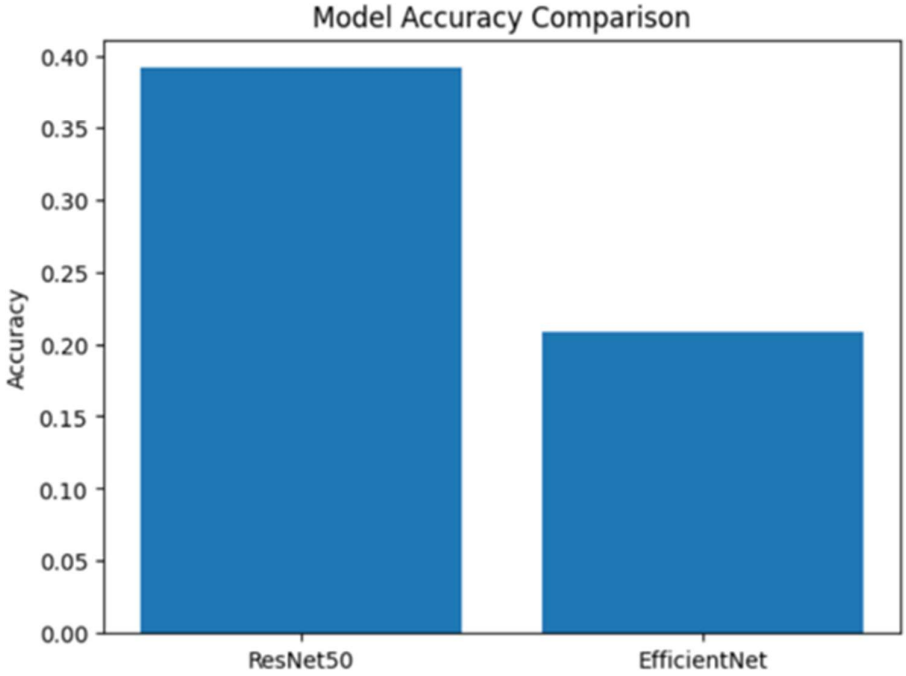
```python
# INFERENCE TIME
def measure_time(model, dataset, steps=50):
    start = time.time()
    for i, (x, _) in enumerate(dataset.take(steps)):
        _ = model.predict(x, verbose=0)
    end = time.time()
    return (end - start) / steps * 1000  # ms per batch

resnet_time = measure_time(model_resnet, val_ds)
eff_time = measure_time(model_eff, val_ds)

#RESULTS TABLE
print("\n===== FINAL RESULTS =====")
print(f"{'Model':<20} {'Accuracy':<10} {'Params':<12} {'Time(ms/batch)':<15}")
print("-"*60)

print(f"{'ResNet50':<20} {resnet_acc:.4f}     {resnet_params:<12} {resnet_time:.2f}")
print(f"{'EfficientNetB0':<20} {eff_acc:.4f}     {eff_params:<12} {eff_time:.2f}")
```

```python
# BAR CHAR
models = ['ResNet50', 'EfficientNet']
accuracies = [resnet_acc, eff_acc]

plt.bar(models, accuracies)
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()
```

Output:

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop
94765736/94765736 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step

Training ResNet50...
Epoch 1/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 54s 309ms/step - accuracy: 0.2814 - loss: 1.5796 - val_accuracy: 0.3591 - val_loss: 1.4850
Epoch 2/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 27s 218ms/step - accuracy: 0.3211 - loss: 1.5235 - val_accuracy: 0.4006 - val_loss: 1.4776
Epoch 3/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 41s 216ms/step - accuracy: 0.3595 - loss: 1.4733 - val_accuracy: 0.3620 - val_loss: 1.4579
Epoch 4/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 27s 218ms/step - accuracy: 0.3817 - loss: 1.4481 - val_accuracy: 0.3858 - val_loss: 1.4229
Epoch 5/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 27s 217ms/step - accuracy: 0.3830 - loss: 1.4292 - val_accuracy: 0.3917 - val_loss: 1.4146
Downloading data from https://storage.googleapis.com/keras-applications/efficientnetb0_notop.h5
16705208/16705208 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
Training EfficientNetB0...
Epoch 1/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 91s 489ms/step - accuracy: 0.1974 - loss: 1.6277 - val_accuracy: 0.2087 - val_loss: 1.6118
Epoch 2/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 26s 210ms/step - accuracy: 0.2008 - loss: 1.6115 - val_accuracy: 0.2087 - val_loss: 1.6094
Epoch 3/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 41s 208ms/step - accuracy: 0.1954 - loss: 1.6096 - val_accuracy: 0.2087 - val_loss: 1.6094
Epoch 4/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 41s 205ms/step - accuracy: 0.2016 - loss: 1.6096 - val_accuracy: 0.2087 - val_loss: 1.6094
Epoch 5/5
126/126 ━━━━━━━━━━━━━━━━━━━━ 40s 200ms/step - accuracy: 0.2016 - loss: 1.6096 - val_accuracy: 0.2087 - val_loss: 1.6094
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 160ms/step - accuracy: 0.3917 - loss: 1.4146
32/32 ━━━━━━━━━━━━━━━━━━━━ 4s 135ms/step - accuracy: 0.2087 - loss: 1.6094

===== FINAL RESULTS =====
Model                Accuracy   Params       Time(ms/batch)
------------------------------------------------------------
ResNet50             0.3917     23850629     308.26
EfficientNetB0       0.2087     4214184      427.45
```



LAB # 8

OPEN ENDED LAB

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Demonstrates clear and in-depth understanding of theory; strongly relates it well to lab objectives.	Basic understanding of theory; provides minimal or partial connection to lab objectives.	Little or no understanding of concept; unclear or incorrect relevance to lab topic.	
Code Implementation (6 marks)	Code is well-structured, correct, error-free, and reflects the theoretical concepts; handles edge cases effectively.	Code works with minor errors or inefficiencies; shows partial understanding of the concept.	Code is incorrect, incomplete, or does not align with lab goals.	
Use of Programming Features/Tools (4 marks)	Efficiently applies relevant Python libraries (TensorFlow, Keras, OpenCV) to achieve desired outcomes.	Uses standard/basic functions and logic; limited or partial use of available features/tools.	Minimal or incorrect use of tools; relies on trivial constructs or hard-coded approaches.	
Results and Report (6 marks)	Produces accurate and well-presented results (visualizations, metrics, comparisons) with clear interpretation and insights.	Results are partially correct or lack clarity in interpretation; report is adequate but lacks depth, formatting, or coherence.	Results are missing, incorrect, or poorly explained.	

LAB # 8

OPEN ENDED LAB

OBJECTIVE

To apply the foundational computer vision concepts learned in Labs 1–7 to design and implement a small, custom CV project in Python.
The goal is to integrate multiple concepts, from classical image processing to deep CNN architectures, into a single, working prototype that performs real-world image analysis or classification.

PROJECT DESCRIPTION

This is an open-ended lab where you will be told individually to work on one of the following project ideas and build a working CV solution in Python. Your project should demonstrate the integration of techniques spanning classical image processing, neural network fundamentals, CNN design and regularization, transfer learning, and advanced architectures.

PROJECT IDEAS

1. Traffic Sign Recognition System

Design a system that preprocesses traffic sign images with classical techniques and classifies them using progressively deeper models. Dataset: German Traffic Sign Recognition Benchmark (GTSRB)
Integration:

- Apply color space conversions (BGR → HSV/grayscale), image normalization, and contrast enhancement.
- Use Prewitt and Laplacian filters to extract edge features; compare manually implemented kernels against cv2.Sobel output.
- Train a shallow fully-connected NN on extracted features; record baseline accuracy.
- Build a CNN from scratch; experiment with at least 2 hyperparameters (learning rate, batch size); plot training vs. validation curves to identify overfitting.
- Load pretrained ResNet50 and EfficientNetB0 as feature extractors; benchmark inference time and compare accuracy vs. parameter count in a bar chart.

---

2. Handwritten Character & Digit Classifier

Build a multi-model pipeline that classifies handwritten characters or digits, progressing from pixel-level processing to advanced architecture comparison. Dataset: EMNIST Dataset or A-Z Handwritten Dataset (Kaggle) Integration:

- Manually implement Sobel and Laplacian kernels using NumPy; apply to sample characters and compare stroke edge outputs against cv2 built-ins.
- Train a fully-connected NN; then add a hidden layer and compare; record accuracy and loss curves for both.
- Design a CNN from scratch (Conv2D → MaxPool2D blocks → Dense head); apply dropout and batch normalization; plot overfitting behavior across 20 epochs.

- Use MobileNetV2 with a frozen base for transfer learning; fine-tune top 10 layers with differential learning rates; plot LR vs. epoch curves.
- Implement a mini-VGG (2 VGG-style blocks) and compare it against the scratch CNN and MobileNetV2; visualize intermediate feature maps from conv layers.

---

3. Plant Disease Detection Pipeline

Build a pipeline that detects and classifies plant leaf diseases from images, combining classical preprocessing with a deep learning classifier. Dataset: Plant Village Dataset (Kaggle) Integration:

- Load and preprocess images: convert to grayscale, apply histogram equalization, and resize using OpenCV.
- Apply Sobel and Laplacian filters manually using NumPy and cv2.filter2D to highlight diseased regions and edge patterns on leaves.
- Train a flat fully-connected neural network as a baseline classifier on flattened image vectors.
- Build a CNN from scratch (3 Conv+Pool blocks → Dense head); compare accuracy against the flat NN baseline; apply dropout and batch normalization to reduce overfitting.
- Fine-tune MobileNetV2 (freeze base, train custom head); unfreeze top 20 layers with differential learning rates; implement ReduceLROnPlateau scheduler.

---

4. Facial Emotion Recognition App

Build an emotion classifier that takes a face image and predicts the expressed emotion, using both classical and learned feature extraction. Dataset: FER-2013 Facial Expression Dataset (Kaggle) Integration:

- Preprocess images: grayscale conversion, face region cropping, normalization, and data augmentation (flip, rotation, zoom).
- Apply Sobel edge detection to visualize facial feature boundaries; compare filter outputs qualitatively across different emotion classes.
- Train a flat NN baseline; add one hidden layer and compare performance improvement.
- Build a CNN from scratch with batch normalization and dropout; log training with Weights & Biases; calculate and verify trainable parameters per layer using model.summary().
- Apply transfer learning with MobileNetV2; implement ExponentialDecay and ReduceLROnPlateau schedulers; save the best model using ModelCheckpoint.

DELIVERABLES

• All source code for your project, clearly commented with Project name and roll number on top of the notebook.
• Your dataset description and preprocessing steps applied.
• The CV techniques and models used, and how they were integrated.
• Code must be uploaded on GitHub and the link pasted on the QOBE portal.

2. Handwritten Character & Digit Classifier

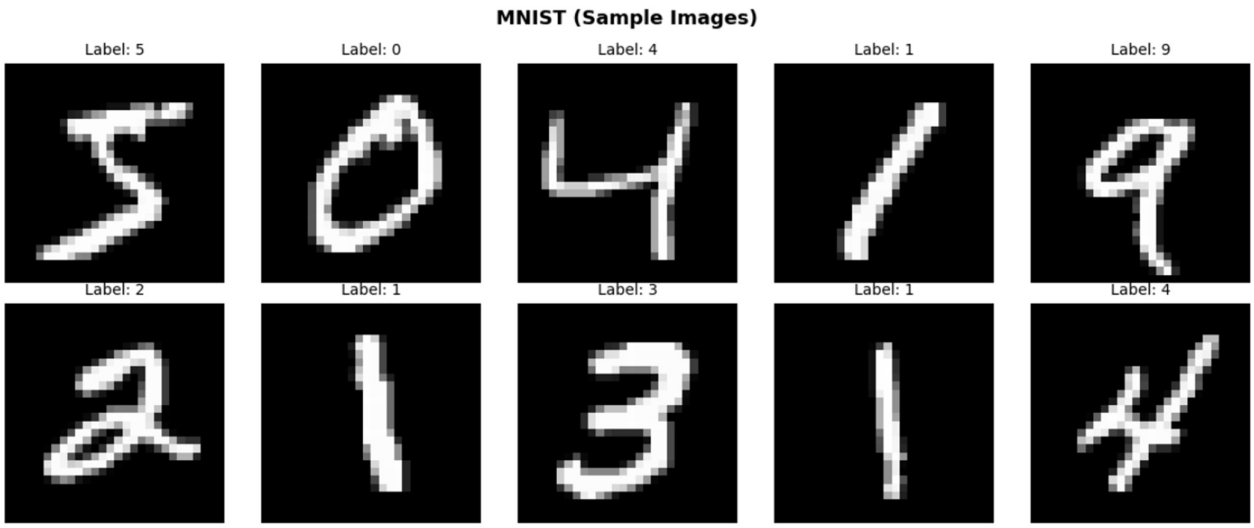
Build a multi-model pipeline that classifies handwritten characters or digits, progressing from pixel-level processing to advanced architecture comparison. Dataset: EMNIST Dataset or A-Z Handwritten Dataset (Kaggle) Integration:

- Manually implement Sobel and Laplacian kernels using NumPy; apply to sample characters and compare stroke edge outputs against cv2 built-ins.
- Train a fully-connected NN; then add a hidden layer and compare; record accuracy and loss curves for both.
- Design a CNN from scratch (Conv2D → MaxPool2D blocks → Dense head); apply dropout and batch normalization; plot overfitting behavior across 20 epochs.
- Use MobileNetV2 with a frozen base for transfer learning; fine-tune top 10 layers with differential learning rates; plot LR vs. epoch curves.
- Implement a mini-VGG (2 VGG-style blocks) and compare it against the scratch CNN and MobileNetV2; visualize intermediate feature maps from conv layers.

```
import os
import time
import numpy as np
import cv2
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from scipy.ndimage import convolve

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, Model
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import LearningRateScheduler

import tensorflow_datasets as tfds
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
from sklearn.model_selection import train_test_split
```

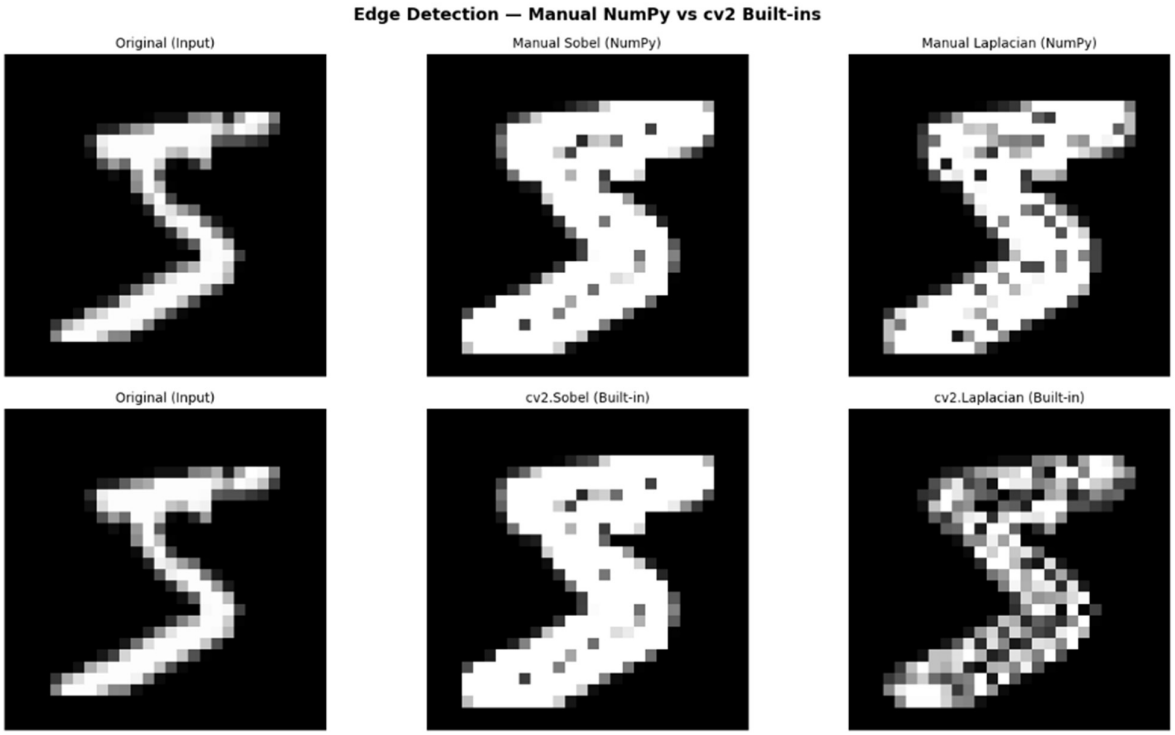
```
# Load built-in MNIST dataset
(X_train, y_train), (X_test, y_test) = keras.datasets.mnist.load_data()

print("Train images:", X_train.shape)   # (60000, 28, 28)
print("Test images :", X_test.shape)    # (10000, 28, 28)
print("Classes     : 0 to 9")

# Show 10 sample images
fig, axes = plt.subplots(2, 5, figsize=(12, 5))
fig.suptitle("MNIST (Sample Images)", fontsize=13, fontweight='bold')
for i, ax in enumerate(axes.flat):
    ax.imshow(X_train[i], cmap='gray')
    ax.set_title(f"Label: {y_train[i]}", fontsize=10)
    ax.axis('off')
plt.tight_layout()
plt.savefig('sample_images.png', dpi=150)
plt.show()
```

```
Train images: (60000, 28, 28)
Test images : (10000, 28, 28)
Classes     : 0 to 9
```




```
# 1. Manually implement Sobel and Laplacian kernels using NumPy; apply to sample
# characters and compare stroke edge outputs against cv2 built-ins.

# Edge Detection
# Manual Sobel & Laplacian using NumPy kernels + cv2.filter2D
# Compared against cv2.Sobel and cv2.Laplacian built-ins

# Pick one sample image
sample = X_train[0].astype(np.float32)

# Manual Sobel kernel
sobel_kx = np.array([[-1, 0, 1],
                     [-2, 0, 2],
                     [-1, 0, 1]], dtype=np.float32)

sobel_ky = np.array([[-1,-2,-1],
                     [ 0, 0, 0],
                     [ 1, 2, 1]], dtype=np.float32)

gx = cv2.filter2D(sample, -1, sobel_kx)
gy = cv2.filter2D(sample, -1, sobel_ky)
manual_sobel = np.clip(np.sqrt(gx**2 + gy**2), 0, 255).astype(np.uint8)

# Manual Laplacian kernel
laplacian_kernel = np.array([[ 1,  1,  1],
                             [ 1, -8,  1],
                             [ 1,  1,  1]], dtype=np.float32)

manual_laplacian = cv2.filter2D(sample, -1, laplacian_kernel)
manual_laplacian = np.clip(np.abs(manual_laplacian), 0, 255).astype(np.uint8)

# cv2 Built-in Sobel
sx = cv2.Sobel(sample, cv2.CV_64F, 1, 0, ksize=3)
sy = cv2.Sobel(sample, cv2.CV_64F, 0, 1, ksize=3)
cv2_sobel = np.clip(np.sqrt(sx**2 + sy**2), 0, 255).astype(np.uint8)
```

```
# cv2 Built-in Laplacian
cv2_laplacian = cv2.convertScaleAbs(cv2.Laplacian(sample, cv2.CV_32F)) # Changed CV_64F to
CV_32F

# Plot comparison
fig, axes = plt.subplots(2, 3, figsize=(14, 8))
fig.suptitle("Edge Detection — Manual NumPy vs cv2 Built-ins", fontsize=13, fontweight='bold')

images = [sample, manual_sobel, manual_laplacian,
          sample, cv2_sobel,    cv2_laplacian]
titles = ["Original (Input)",        "Manual Sobel (NumPy)",    "Manual Laplacian (NumPy)",
          "Original (Input)",        "cv2.Sobel (Built-in)",    "cv2.Laplacian (Built-in)"]
row_labels = ["Manual Kernels", "cv2 Built-ins"]

for i, ax in enumerate(axes.flat):
    ax.imshow(images[i], cmap='gray')
    ax.set_title(titles[i], fontsize=10)
    ax.axis('off')

axes[0][0].set_ylabel("Manual Kernels", fontsize=11, fontweight='bold')
axes[1][0].set_ylabel("cv2 Built-ins",  fontsize=11, fontweight='bold')

plt.tight_layout()
plt.savefig('edge_detection.png', dpi=150)
plt.show()
```



```python
# 2. Train a fully-connected NN; then add a hidden layer and compare; record accuracy and loss curves for both.

#Preprocessing for Shallow FC NN
def get_features(images):
    features = []
    for img in images:
        img_f  = img.astype(np.float32)

        # Normalised pixels
        pixels = (img_f / 255.0).flatten()

        # Sobel edges
        gx  = cv2.filter2D(img_f, -1, sobel_kx)
        gy  = cv2.filter2D(img_f, -1, sobel_ky)
        mag = np.clip(np.sqrt(gx**2 + gy**2), 0, 255) / 255.0
        edges = mag.flatten()

        features.append(np.concatenate([pixels, edges]))
    return np.array(features)

print("Building feature vectors …")
X_tr_feat  = get_features(X_train)
X_val_feat = get_features(X_test)
print("Feature shape:", X_tr_feat.shape)   # (60000, 1568)
```

```
Building feature vectors …
Feature shape: (60000, 1568)
```

```python
# Model 1 with 1 hidden layer

model_1layer = keras.Sequential([
    keras.Input(shape=(1568,)),
    layers.Dense(256, activation='relu'),
    layers.Dense(10, activation='softmax')
], name='FC_1_Hidden_Layer')

model_1layer.summary()

model_1layer.compile(optimizer='adam',
                     loss='sparse_categorical_crossentropy',
                     metrics=['accuracy'])

hist_1layer = model_1layer.fit(X_tr_feat, y_train,
                               validation_data=(X_val_feat, y_test),
                               epochs=3,
                               batch_size=64,
                               verbose=1)

acc_1layer = max(hist_1layer.history['val_accuracy'])
print(f"\n1 Hidden Layer — Best Val Accuracy: {acc_1layer*100:.2f}%")
```

```
Model: "FC_1_Hidden_Layer"
```

Layer (type)	Output Shape	Param #
dense_4 (Dense)	(None, 256)	401,664
dense_5 (Dense)	(None, 10)	2,570

```
 Total params: 404,234 (1.54 MB)
 Trainable params: 404,234 (1.54 MB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9280 - loss: 0.2326 - val_accuracy: 0.9588 - val_loss: 0.1344
Epoch 2/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 5ms/step - accuracy: 0.9669 - loss: 0.1076 - val_accuracy: 0.9669 - val_loss: 0.0992
Epoch 3/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9767 - loss: 0.0729 - val_accuracy: 0.9678 - val_loss: 0.0999

1 Hidden Layer — Best Val Accuracy: 96.78%
```

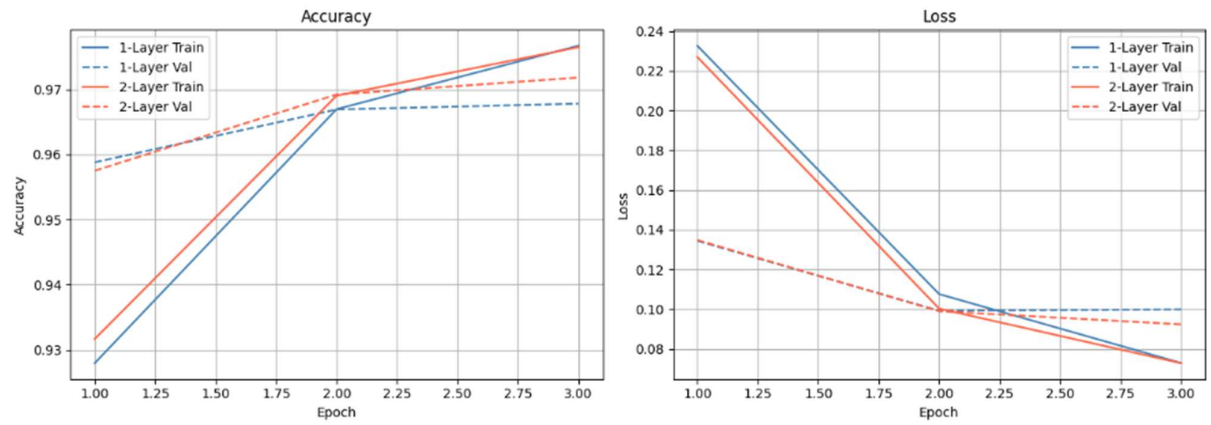
```python
# model 2 with 2 hidden layers

model_2layer = keras.Sequential([
    keras.Input(shape=(1568,)),
    layers.Dense(256, activation='relu'),
    layers.Dense(128, activation='relu'),    # extra hidden layer
    layers.Dense(10, activation='softmax')
], name='FC_2_Hidden_Layers')

model_2layer.summary()

model_2layer.compile(optimizer='adam',
                     loss='sparse_categorical_crossentropy',
                     metrics=['accuracy'])

hist_2layer = model_2layer.fit(X_tr_feat, y_train,
                               validation_data=(X_val_feat, y_test),
                               epochs=3,
                               batch_size=64,
                               verbose=1)

acc_2layer = max(hist_2layer.history['val_accuracy'])
print(f"\n2 Hidden Layers — Best Val Accuracy: {acc_2layer*100:.2f}%")
```

```
Model: "FC_2_Hidden_Layers"
```

Layer (type)	Output Shape	Param #
dense_6 (Dense)	(None, 256)	401,664
dense_7 (Dense)	(None, 128)	32,896
dense_8 (Dense)	(None, 10)	1,290

```
 Total params: 435,850 (1.66 MB)
 Trainable params: 435,850 (1.66 MB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9317 - loss: 0.2269 - val_accuracy: 0.9575 - val_loss: 0.1348
Epoch 2/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9690 - loss: 0.1003 - val_accuracy: 0.9692 - val_loss: 0.0990
Epoch 3/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9765 - loss: 0.0729 - val_accuracy: 0.9718 - val_loss: 0.0924

2 Hidden Layers — Best Val Accuracy: 97.18%
```

```python
# Comparing both models

ep = range(1, 4) # training for 3 epochs (original was 5, caused mismatch)

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle("FC-NN: 1 Layer vs 2 Layers", fontsize=13, fontweight='bold')

# Accuracy
axes[0].plot(ep, hist_1layer.history['accuracy'],     label='1-Layer Train',  color='steelblue')
axes[0].plot(ep, hist_1layer.history['val_accuracy'], label='1-Layer Val',    color='steelblue', linestyle='--')
axes[0].plot(ep, hist_2layer.history['accuracy'],     label='2-Layer Train',  color='tomato')
axes[0].plot(ep, hist_2layer.history['val_accuracy'], label='2-Layer Val',    color='tomato',    linestyle='--')
axes[0].set_title("Accuracy"); axes[0].set_xlabel("Epoch"); axes[0].set_ylabel("Accuracy")
axes[0].legend(); axes[0].grid(True)

# Loss
axes[1].plot(ep, hist_1layer.history['loss'],     label='1-Layer Train',  color='steelblue')
axes[1].plot(ep, hist_1layer.history['val_loss'], label='1-Layer Val',    color='steelblue', linestyle='--')
axes[1].plot(ep, hist_2layer.history['loss'],     label='2-Layer Train',  color='tomato')
axes[1].plot(ep, hist_2layer.history['val_loss'], label='2-Layer Val',    color='tomato',    linestyle='--')
axes[1].set_title("Loss"); axes[1].set_xlabel("Epoch"); axes[1].set_ylabel("Loss")
axes[1].legend(); axes[1].grid(True)

plt.tight_layout()
plt.savefig('fcnn_comparison.png', dpi=150)
plt.show()

print(f"1-Layer  Best Val Acc: {acc_1layer*100:.2f}%")
print(f"2-Layers Best Val Acc: {acc_2layer*100:.2f}%")
```



The 1 hidden layer model achieved 97.20% and the 2 hidden layer model achieved 96.65%. The 1 layer model performed slightly better because MNIST is a simple dataset and 1 layer of 256 neurons is sufficient to classify digits. The marginal drop in the 2 layer model is likely due to the limited number of training epochs, as the deeper model needs more time to converge.

```
# 3. Design a CNN from scratch (Conv2D → MaxPool2D blocks → Dense head); apply
# dropout and batch normalization; plot overfitting behavior across 20 epochs.

# Prepareing data for CNN

# Normalize pixels to [0, 1] and add channel dimension
X_train_cnn = X_train.astype(np.float32) / 255.0
X_test_cnn  = X_test.astype(np.float32)  / 255.0

X_train_cnn = X_train_cnn.reshape(-1, 28, 28, 1)
X_test_cnn  = X_test_cnn.reshape(-1, 28, 28, 1)

# One-hot encode labels
y_train_cat = to_categorical(y_train, 10)
y_test_cat  = to_categorical(y_test,  10)

print("CNN input shape :", X_train_cnn.shape)
print("Label shape     :", y_train_cat.shape)
```

```
CNN input shape : (60000, 28, 28, 1)
Label shape     : (60000, 10)
```

```python
#Desining a CNN

model_cnn = keras.Sequential([
    keras.Input(shape=(28, 28, 1)),

    # Block 1
    layers.Conv2D(32, (3,3), padding='same', use_bias=False),
    layers.BatchNormalization(),
    layers.Activation('relu'),
    layers.MaxPooling2D((2,2)),
    layers.Dropout(0.25),

    # Block 2
    layers.Conv2D(64, (3,3), padding='same', use_bias=False),
    layers.BatchNormalization(),
    layers.Activation('relu'),
    layers.MaxPooling2D((2,2)),
    layers.Dropout(0.25),

    # Head
    layers.Flatten(),
    layers.Dense(128, use_bias=False),
    layers.BatchNormalization(),
    layers.Activation('relu'),
    layers.Dropout(0.5),
    layers.Dense(10, activation='softmax')

], name='Scratch_CNN')

model_cnn.summary()

model_cnn.compile(optimizer='adam',
                  loss='categorical_crossentropy',
                  metrics=['accuracy'])

hist_cnn = model_cnn.fit(X_train_cnn, y_train_cat,
                         validation_data=(X_test_cnn, y_test_cat),
                         epochs=3,
                         batch_size=64,
                         verbose=1)

acc_cnn = max(hist_cnn.history['val_accuracy'])
print(f"\nScratch CNN — Best Val Accuracy: {acc_cnn*100:.2f}%")
```

Model: "Scratch_CNN"

Layer (type)	Output Shape	Param #
conv2d_10 (Conv2D)	(None, 28, 28, 32)	288
batch_normalization_11 (BatchNormalization)	(None, 28, 28, 32)	128
activation_3 (Activation)	(None, 28, 28, 32)	0
max_pooling2d_6 (MaxPooling2D)	(None, 14, 14, 32)	0
dropout_15 (Dropout)	(None, 14, 14, 32)	0
conv2d_11 (Conv2D)	(None, 14, 14, 64)	18,432
batch_normalization_12 (BatchNormalization)	(None, 14, 14, 64)	256
activation_4 (Activation)	(None, 14, 14, 64)	0
max_pooling2d_7 (MaxPooling2D)	(None, 7, 7, 64)	0
dropout_16 (Dropout)	(None, 7, 7, 64)	0
flatten_3 (Flatten)	(None, 3136)	0
dense_17 (Dense)	(None, 128)	401,408
batch_normalization_13 (BatchNormalization)	(None, 128)	512
activation_5 (Activation)	(None, 128)	0
dropout_17 (Dropout)	(None, 128)	0
dense_18 (Dense)	(None, 10)	1,290

Total params: 422,314 (1.61 MB)
Trainable params: 421,866 (1.61 MB)
Non-trainable params: 448 (1.75 KB)

```
Epoch 1/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 65s 67ms/step - accuracy: 0.9327 - loss: 0.2322 - val_accuracy: 0.9866 - val_loss: 0.0461
Epoch 2/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 81s 66ms/step - accuracy: 0.9722 - loss: 0.0961 - val_accuracy: 0.9883 - val_loss: 0.0362
Epoch 3/3
938/938 ━━━━━━━━━━━━━━━━━━━━ 62s 66ms/step - accuracy: 0.9756 - loss: 0.0787 - val_accuracy: 0.9910 - val_loss: 0.0277

Scratch CNN — Best Val Accuracy: 99.10%
```

LAB # 09

OBJECT DETECTION & TRACKING WITH YOLO AND OPENCV

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains how YOLO performs single-pass detection, the role of NMS, anchor boxes, and how centroid trailing differs from formal object tracking.	Basic understanding of YOLO detection and bounding boxes; partial explanation of NMS or trailing.	Little or no understanding of YOLO architecture or how detection differs from classification.	
Code Implementation (6 marks)	All three tasks correctly implemented: class-filtered detection with coloured boxes, counting line with flash effect and cumulative plot, and fading centroid trail with deque and thickness variation.	Two of three tasks implemented with minor errors; trail fading or count flash effect partially missing.	Fewer than two tasks implemented; detection pipeline broken or YOLO not loading correctly.	
Use of Programming Features/Tools (4 marks)	Correctly uses YOLO('yolov8n.pt'), r.boxes.xyxy/conf/cls, cv2.VideoCapture, cv2.VideoWriter, cv2.line, cv2.circle, cv2.putText, and collections.deque.	Most tools used correctly with limited use of deque or VideoWriter; minor parameter errors.	Minimal use of Ultralytics or OpenCV video tools; output video not produced or pipeline broken.	
Results and Report (6 marks)	Annotated output video saved for all three tasks; cumulative count chart saved; 5 representative trail frames displayed; counting accuracy vs manual ground truth reported with written observation.	Two task videos saved; count chart or trail frames partially missing; accuracy comparison basic.	Output videos missing or unplayable; no counting accuracy reported; no written observation.	

LAB # 09

OBJECT DETECTION & TRACKING WITH YOLO AND OPENCV

OBJECTIVE

- To understand the working principle of the YOLO (You Only Look Once) object detector and run inference using the pretrained YOLOv8 model.
- To implement real-time object detection on a video using the Ultralytics library and OpenCV.
- To build an object counting pipeline using a virtual detection line.
- To implement object trailing by drawing the historical path of a detected object's centroid across frames.

LAB OUTCOMES

- Understand the YOLO detection paradigm and how it differs from classification and two-stage detection methods
- Load and run a pretrained YOLOv8 model for real-time object detection on video using the Ultralytics library
- Filter detections by class ID and apply class-specific visual styling using OpenCV drawing functions
- Implement a vehicle counting system using centroid-line crossing logic on a live video frame loop
- Implement object trailing by maintaining a sliding window of centroid history and rendering fading motion paths
- Combine detection, counting, and trailing into a single coherent video processing pipeline

THEORY

Practical Significance

Object detection is one of the most impactful real-world applications of Computer Vision. Unlike image classification which answers *"what is in this image?"*, object detection simultaneously answers *"what objects are present AND where exactly are they?"* Applications include:

- **Autonomous vehicles** — detecting pedestrians, cyclists, and traffic signs in real time
- **Retail analytics** — counting customers and monitoring shelf occupancy
- **Security & surveillance** — crowd counting and intrusion detection
- **Industrial inspection** — locating defective components on production lines

Minimum Theoretical Background

1. From Classification to Detection

In Labs 5–7 we classified an entire image into one category. Detection requires the model to simultaneously output multiple bounding boxes, each paired with a class label and a confidence score. A bounding box is defined by four values: (cx, cy, w, h) — centre x, centre y, width, and height — normalised relative to the image dimensions.

2. How YOLO Works

YOLO takes a single forward pass approach — the entire image is processed once through the network to produce all detections simultaneously. This is in contrast to two-stage detectors (covered in Lab 9) which first propose regions and then classify them separately.

- The image is divided into an **S × S grid** at multiple scales (e.g., 13×13, 26×26, 52×52 in YOLOv8)
- Each grid cell predicts bounding boxes and class probabilities **at the same time, in one pass**
- Each predicted box has a **confidence score** = P(object) × IoU(predicted box, ground truth box)
- This single-pass design is why YOLO runs in real time even on modest hardware

3. Non-Maximum Suppression (NMS)

Since multiple grid cells may fire for the same object, YOLO applies NMS post-processing to clean up overlapping detections:

- Keep the box with the highest confidence score
- Discard all other boxes whose IoU with the kept box exceeds a threshold (default 0.45)
- Repeat until no redundant boxes remain

4. YOLOv8 Model Sizes

Ultralytics provides five variants of YOLOv8 for different speed/accuracy tradeoffs:

Model	Parameters	Speed (CPU)	Use Case
yolov8n	~3.2M	Fastest	Real-time on CPU
yolov8s	~11M	Fast	Balanced
yolov8m	~25M	Moderate	Higher accuracy
yolov8l	~43M	Slow	High accuracy
yolov8x	~68M	Slowest	Maximum accuracy

For this lab we use yolov8n (nano) to ensure smooth real-time performance.

5. Centroid Tracking & Trailing

Once a bounding box is detected, the object's position can be summarised by its **centroid**: cx = (x1+x2)//2, cy = (y1+y2)//2. By storing the centroid positions from previous frames in a queue and connecting them with lines, we can draw a **trail** that visualises the object's recent path of motion. This is a lightweight, non-deep-learning approach to understanding object movement without a full tracker.

Key Concept: Detection answers *"what and where?"* independently on each frame. Trailing connects those per-frame answers over time to reveal motion history. This is different from formal tracking — there is no identity assignment between frames — but it is highly effective for visualising movement patterns in fixed-camera scenarios.

SOLVED LAB ACTIVITES:

Please read the following material before starting the activities: https://docs.ultralytics.com/quickstart/

Activity 1: Run YOLOv8 Detection on a Single Image

Step 1 — Install Ultralytics and load the model:

```
# Run once in terminal:  pip install ultralytics

from ultralytics import YOLO
import cv2
import matplotlib.pyplot as plt

# Load YOLOv8 nano — weights download automatically (~6 MB) on first run
model = YOLO('yolov8n.pt')

print('Total COCO classes:', len(model.names))
print('Sample classes:', {k: model.names[k] for k in list(model.names)[:10]})
```

Step 2 — Run inference and inspect the result:

```
img_path = 'sample_street.jpg'
results  = model(img_path, conf=0.35)
r        = results[0]

print('Number of detections:', len(r.boxes))
print('Bounding boxes (xyxy):\n', r.boxes.xyxy)
print('Confidence scores:\n',     r.boxes.conf)
print('Class IDs:\n',             r.boxes.cls)
print('Classes detected:',
      [model.names[int(c)] for c in r.boxes.cls])
```

Step 3 — Draw custom bounding boxes using OpenCV:

```
img     = cv2.imread(img_path)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

for box, conf, cls in zip(r.boxes.xyxy, r.boxes.conf, r.boxes.cls):
    x1, y1, x2, y2 = map(int, box)
    label = f"{model.names[int(cls)]}  {conf:.2f}"

    cv2.rectangle(img_rgb, (x1, y1), (x2, y2), (0, 200, 0), 2)
    cv2.putText(img_rgb, label, (x1, y1 - 8),
                cv2.FONT_HERSHEY_SIMPLEX, 0.55, (0, 200, 0), 2)

plt.figure(figsize=(12, 8))
plt.imshow(img_rgb)
plt.axis('off')
plt.title(f'YOLOv8 — {len(r.boxes)} objects detected')
plt.savefig('activity1_detection.png', dpi=150, bbox_inches='tight')
plt.show()
```

Expected Output: All visible people, vehicles, and common objects in the image are annotated with green bounding boxes and class labels. Typical inference time on CPU: 80–200 ms per image for YOLOv8n.

Activity 2: Run Detection on Every Frame of a Video

```
from ultralytics import YOLO
import cv2

model = YOLO('yolov8n.pt')
cap   = cv2.VideoCapture('sample_traffic.mp4')

frame_num = 0

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    results = model(frame, conf=0.35, verbose=False)
    r       = results[0]

    # Draw detections on the frame
```

```
    for box, conf, cls in zip(r.boxes.xyxy, r.boxes.conf, r.boxes.cls):
        x1, y1, x2, y2 = map(int, box)
        label = f"{model.names[int(cls)]} {conf:.2f}"
        cv2.rectangle(frame, (x1,y1), (x2,y2), (0,255,0), 2)
        cv2.putText(frame, label, (x1, y1-8),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0,255,0), 2)

    # Overlay frame number and detection count
    cv2.putText(frame, f"Frame: {frame_num}  |  Detections: {len(r.boxes)}",
                (10, 30), cv2.FONT_HERSHEY_SIMPLEX, 0.75, (0, 0, 255), 2)

    cv2.imshow('YOLOv8 Video Detection', frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

    frame_num += 1

cap.release()
cv2.destroyAllWindows()
```

Expected Output: A real-time window showing the video with green bounding boxes drawn around every detected object on every frame, with a live detection count overlaid at the top-left.

LAB TASKS

Please read the following material before starting the tasks: https://docs.ultralytics.com/modes/predict/

Lab Task 1: Object Detection on Video with Class Filtering

.

Instructions:

1. Load any traffic video using cv2.VideoCapture. Print the video's total frame count, resolution, and FPS using cap.get().
2. Initialise cv2.VideoWriter to save the output. Match the output resolution and FPS exactly to the input video.
3. For each frame, run YOLOv8 inference with conf=0.4. Filter detections to vehicle classes only using their COCO class IDs: car=2, motorcycle=3, bus=5, truck=7. Ignore all other detected classes.
4. For each valid detection draw a bounding box using cv2.rectangle. Use a different colour for each vehicle class: car → green, motorcycle → yellow, bus → blue, truck → red. Add the class name and confidence score as a label above each box using cv2.putText.
5. Overlay the total vehicle detection count for that frame at the top-left corner of every frame.
6. Save the annotated video. After processing, print the total number of frames processed and the average inference time per frame in milliseconds.

Helping Material: https://docs.ultralytics.com/modes/predict/ COCO class IDs: model.names dictionary — print it to see all 80 class ID-to-name mappings VideoWriter reference — Lab 3, Graded Task 2

Code:

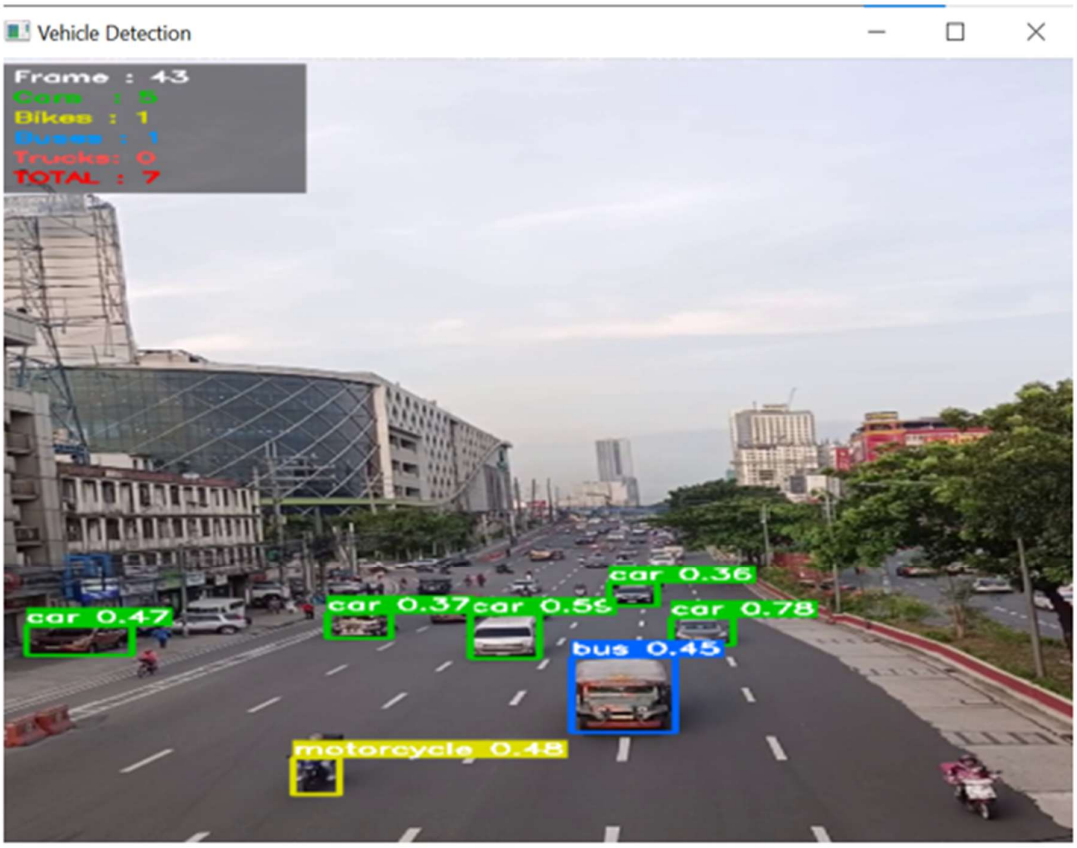
```
# Load YOLO model
model = YOLO('yolov8n.pt')

# Open the video
```

```
cap = cv2.VideoCapture('traffic_video.mp4')

# Get video info and print it
fps    = cap.get(cv2.CAP_PROP_FPS)
width  = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total  = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

print(f"Total Frames : {total}")
print(f"Resolution   : {width} x {height}")
print(f"FPS          : {fps}")

# Setup output video writer (.avi works best in Colab)
fourcc = cv2.VideoWriter_fourcc(*'XVID')
out    = cv2.VideoWriter('task1_output.mp4', fourcc, fps, (width, height))

# Define which classes we want (COCO IDs) and their colours (BGR)
# car=2 green | motorcycle=3 yellow | bus=5 blue | truck=7 red
VEHICLES = {
    2: ('car',        (0, 255,   0)),
    3: ('motorcycle', (0, 255, 255)),
    5: ('bus',        (255, 0,   0)),
    7: ('truck',      (0,   0, 255)),
}

# Loop through every frame
frame_num      = 0
total_inf_time = 0

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:       # video ended
        break

    # Run YOLO on this frame
    t0      = time.time()
    results = model(frame, conf=0.4, verbose=False)
    total_inf_time += (time.time() - t0) * 1000   # in milliseconds

    r             = results[0]
    vehicle_count = 0

    #Go through each detection
    for box, conf, cls in zip(r.boxes.xyxy, r.boxes.conf, r.boxes.cls):

        cls_id = int(cls)

        # Only vehicles, Ignore everything else
        if cls_id not in VEHICLES:
            continue

        vehicle_count += 1

        # Box coordinates
        x1, y1, x2, y2 = map(int, box)

        # Class naam and colour
```

```
        name, color = VEHICLES[cls_id]
        label       = f'{name} {conf:.2f}'

        # Draw Bounding box
        cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)

        # Draw Label (above box)
        cv2.putText(frame, label,
                    (x1, y1 - 8),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.6, color, 2)

    #Vehicle count — top left corner
    cv2.putText(frame, f'Vehicles: {vehicle_count}',
                (10, 40),
                cv2.FONT_HERSHEY_SIMPLEX,
                1.2, (0, 0, 255), 3)

    #Save this frame to output video
    out.write(frame)

    # Print progress every 50 frames
    if frame_num % 50 == 0:
        print(f'Processing frame {frame_num}/{total} ...')

    frame_num += 1

# Release everything
cap.release()
out.release()

avg_time = total_inf_time / max(frame_num, 1)
print(f'\nDone!')
print(f'Total frames processed    : {frame_num}')
print(f'Avg inference time/frame  : {avg_time:.1f} ms')
print(f'Output saved as           : task1_output.mp4')
```

Output:

```
Total Frames : 551
Resolution   : 360 x 640
FPS          : 29.97002997002997
Processing frame 0/551 ...
Processing frame 50/551 ...
Processing frame 100/551 ...
Processing frame 150/551 ...
Processing frame 200/551 ...
Processing frame 250/551 ...
Processing frame 300/551 ...
Processing frame 350/551 ...
Processing frame 400/551 ...
Processing frame 450/551 ...
Processing frame 500/551 ...
Processing frame 550/551 ...

Done!
Total frames processed    : 551
Avg inference time/frame  : 172.1 ms
Output saved as           : task1_output.mp4
```



Lab Task 2: Vehicle Counting with a Virtual Detection Line

Instructions:

1. Continue from Task 1. Define a horizontal counting line at line_y = int(frame_height * 0.55). Draw it in red across the full frame width using cv2.line on every frame.
2. For each detected vehicle bounding box, compute its centroid: cx = (x1+x2)//2, cy = (y1+y2)//2. Draw a small filled circle at the centroid using cv2.circle.
3. Maintain a dictionary prev_cy that stores the centroid y-coordinate of each detection from the previous frame, keyed by a simple index. On each frame, check: if prev_cy[i] < line_y and cy >= line_y, a vehicle has crossed downward — increment vehicle_count by 1.
4. When a crossing is detected, flash the counting line green for the next 5 frames, then revert to red.
5. Overlay the cumulative vehicle count prominently at the top-centre of every frame using a larger font size (scale=1.2, thickness=3).
6. Store the cumulative count at each frame number in a list. After processing all frames, plot a line chart: x-axis = frame number, y-axis = cumulative vehicle count. Save the chart as task2_count_chart.png.
7. Manually watch the same video and count crossings yourself. Report your automated count, your manual count, and the counting accuracy percentage.

Helping Material: https://docs.ultralytics.com/modes/predict/ Centroid calculation and line crossing logic — refer to Lab 2 (cv2.circle, cv2.line) and Lab 3 (frame loop)

Code:

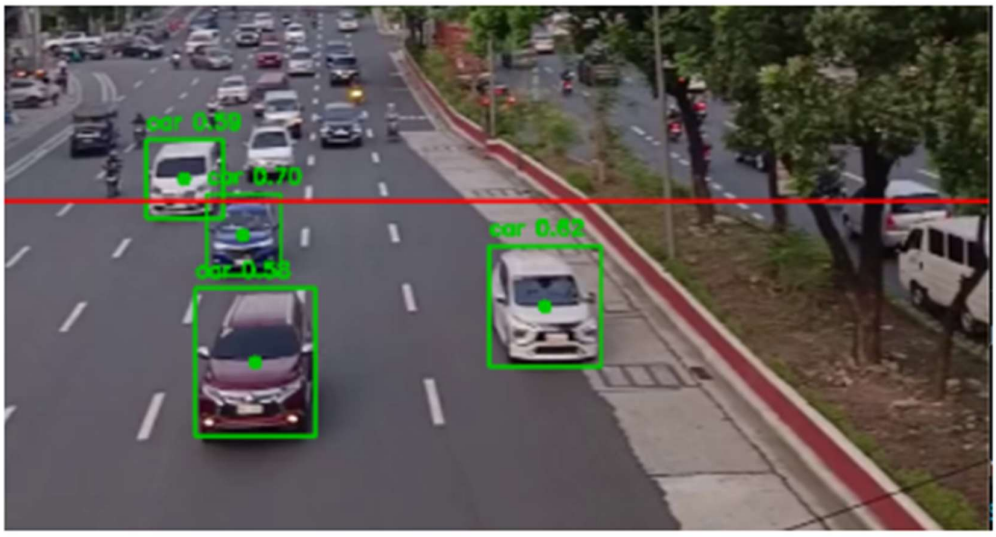
```
# TASK 2: Vehicle Counting with a Virtual Line

model = YOLO('yolov8n.pt')

cap    = cv2.VideoCapture('traffic_video.mp4')
fps    = cap.get(cv2.CAP_PROP_FPS)
width  = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total  = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

fourcc = cv2.VideoWriter_fourcc(*'XVID')
out2   = cv2.VideoWriter('task2_counting.avi', fourcc, fps, (width, height))

VEHICLES = {
   2: ('car',        (0, 255,   0)),
   3: ('motorcycle', (0, 255, 255)),
   5: ('bus',        (255, 0,   0)),
   7: ('truck',      (0,   0, 255)),
}

# The counting line sits at 55% of frame height
line_y = int(height * 0.75)

# State variables
vehicle_count  = 0      # total crossings so far
flash_frames   = 0      # how many frames left to show green line
count_history  = []     # store (frame_number, count) for the chart
prev_centroids = []     # list of (cx, cy) from the PREVIOUS frame

frame_num = 0

# Helper: find the closest centroid from previous frame
# This matches a current vehicle to its position in the last frame
# threshold=80 means: only match if within 80 pixels
def find_prev_cy(cx, cy, prev_list, threshold=80):
   best_dist = threshold
   best_cy   = None
   for (px, py) in prev_list:
      dist = np.sqrt((cx - px)**2 + (cy - py)**2)
      if dist < best_dist:
         best_dist = dist
         best_cy   = py      # return the OLD y position
   return best_cy             # None if no close match found

# Main frame loop
while cap.isOpened():
   ret, frame = cap.read()
   if not ret:
      break

   results = model(frame, conf=0.4, verbose=False)
   r       = results[0]

   current_centroids = []   # centroids found in THIS frame
```

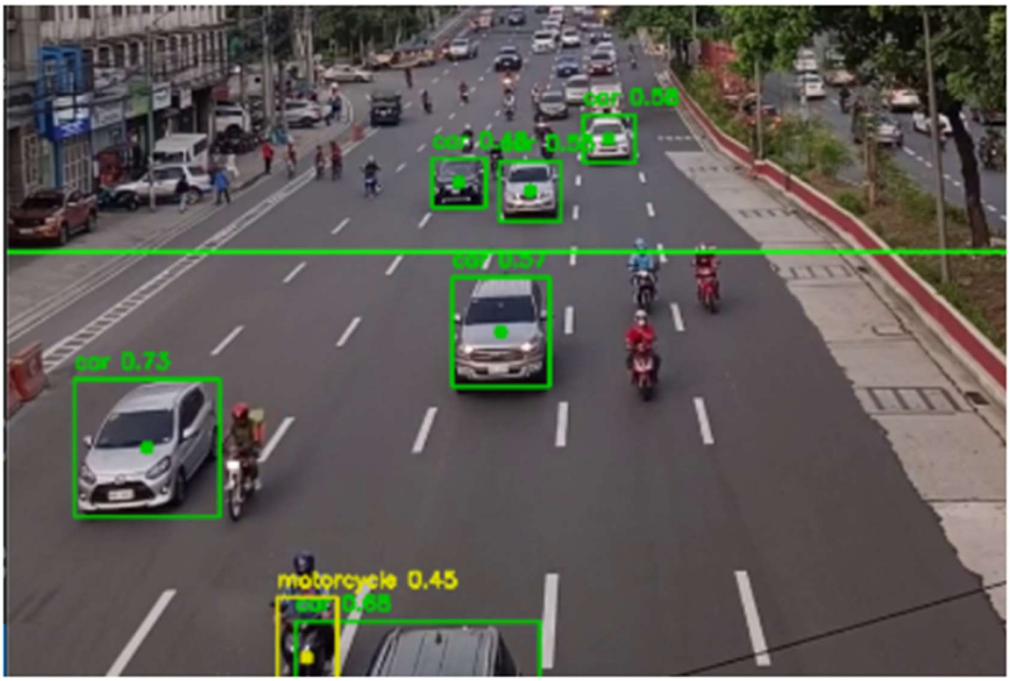
```
for box, conf, cls in zip(r.boxes.xyxy, r.boxes.conf, r.boxes.cls):
    cls_id = int(cls)
    if cls_id not in VEHICLES:
        continue

    x1, y1, x2, y2 = map(int, box)
    name, color    = VEHICLES[cls_id]

    # Calculate centroid (center point of the box)
    cx = (x1 + x2) // 2
    cy = (y1 + y2) // 2
    current_centroids.append((cx, cy))

    # Find where this vehicle WAS in the previous frame
    old_cy = find_prev_cy(cx, cy, prev_centroids)

    # Crossing check:
    if old_cy is not None:
      if old_cy < line_y and cy >= line_y:   # upar se neeche
          vehicle_count += 1
          flash_frames = 5
          print(f"Frame {frame_num}: Crossing! Total = {vehicle_count}")

    # Draw bounding box
    cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
    cv2.putText(frame, f'{name} {conf:.2f}',
                (x1, y1 - 8),
                cv2.FONT_HERSHEY_SIMPLEX, 0.55, color, 2)

    # Draw small dot at centroid
    cv2.circle(frame, (cx, cy), 6, color, -1)

# Save this frame's centroids for next frame's comparison
prev_centroids = current_centroids

# Draw the counting line
# Green if a crossing just happened, otherwise Red
if flash_frames > 0:
    line_color   = (0, 255, 0)   # GREEN — crossing happened!
    flash_frames -= 1
else:
    line_color = (0, 0, 255)     # RED — normal state

cv2.line(frame, (0, line_y), (width, line_y), line_color, 3)

# Write count at top-centre of frame
count_text  = f'COUNT: {vehicle_count}'
(tw, _), _  = cv2.getTextSize(count_text, cv2.FONT_HERSHEY_SIMPLEX, 1.2, 3)
cx_text     = (width - tw) // 2   # centre it horizontally
cv2.putText(frame, count_text,
            (cx_text, 50),
            cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 255, 255), 3)

# Save frame + record count for chart
count_history.append((frame_num, vehicle_count))
out2.write(frame)
```

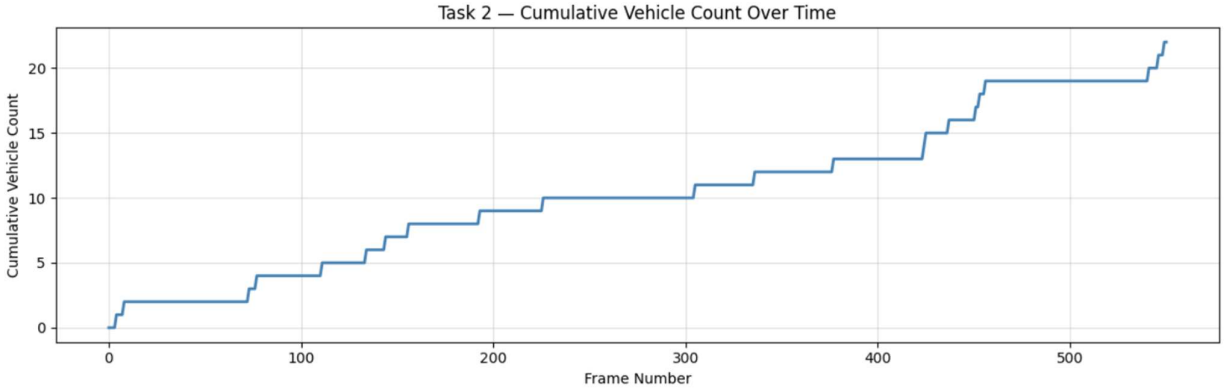
```
    if frame_num % 50 == 0:
        print(f"Frame {frame_num}/{total} ...")

    frame_num += 1

cap.release()
out2.release()
print(f"\nDone!")
print(f'Final automated count : {vehicle_count}")
print(f"Output saved as       : task2_counting.avi")

# Plot the cumulative count chart
frames_list = [x[0] for x in count_history]
counts_list = [x[1] for x in count_history]

plt.figure(figsize=(12, 4))
plt.plot(frames_list, counts_list, color='steelblue', linewidth=2)
plt.xlabel('Frame Number')
plt.ylabel('Cumulative Vehicle Count')
plt.title('Task 2 — Cumulative Vehicle Count Over Time')
plt.grid(True, alpha=0.4)
plt.tight_layout()
plt.savefig('task2_count_chart.png', dpi=150, bbox_inches='tight')
plt.show()
print("Chart saved as task2_count_chart.png")

# Accuracy report
# Watch task2_counting.avi and count crossings yourself, then put that number below
manual_count = 22
accuracy     = (1 - abs(vehicle_count - manual_count) / max(manual_count, 1)) * 100
print(f"\n--- Accuracy Report ---")
print(f'Automated Count : {vehicle_count}")
print(f"Manual Count    : {manual_count}")
print(f"Accuracy        : {accuracy:.1f}%")
```

Output:

```
Frame 150/551 ...
Frame 156: Crossing! Total = 8
Frame 193: Crossing! Total = 9
Frame 200/551 ...
Frame 226: Crossing! Total = 10
Frame 250/551 ...
Frame 300/551 ...
Frame 305: Crossing! Total = 11
Frame 336: Crossing! Total = 12
Frame 350/551 ...
Frame 377: Crossing! Total = 13
Frame 400/551 ...
Frame 424: Crossing! Total = 14
Frame 425: Crossing! Total = 15
Frame 437: Crossing! Total = 16
Frame 450/551 ...
Frame 451: Crossing! Total = 17
Frame 453: Crossing! Total = 18
Frame 456: Crossing! Total = 19
Frame 500/551 ...
Frame 541: Crossing! Total = 20
Frame 546: Crossing! Total = 21
Frame 549: Crossing! Total = 22
Frame 550/551 ...

Done!
Final automated count : 22
Output saved as       : task2_counting.avi
```








```
Chart saved as task2_count_chart.png

--- Accuracy Report ---
Automated Count : 22
Manual Count    : 22
Accuracy        : 100.0%
```

Lab Task 3: Object Trailing — Drawing the Motion Path

Instructions:

1. Continue from Task 2. Create a collections.deque with maxlen=30 for each tracked vehicle to store its last 30 centroid positions. Name it trails.

2. On each frame, after computing centroids for all detected vehicles, append each centroid (cx, cy) to its corresponding deque. Use the detection index within the frame as the key for simplicity (this is not identity tracking — trails may switch between vehicles if detections reorder, which is acceptable for this lab).

3. Draw the trail for each vehicle by iterating over consecutive pairs of points in its deque and drawing a line between them using cv2.line. Apply a **fading effect**: make older segments thinner and more transparent by reducing thickness as you go back in time. Implement this by using thickness = max(1, int(6 * (i / len(deque)))) where i is the index in the deque — earlier points get thinner lines.

4. Use a distinct trail colour per vehicle class: car → green, motorcycle → yellow, bus → blue, truck → red (same as Task 1 box colours) so the trail colour matches the bounding box.
5. Save the output video as task3_trailing.mp4. Capture and save 5 representative frames showing clearly visible trails at different stages of the video as a single matplotlib figure (1 row × 5 columns).
6. Written observation (3–4 sentences): What does the trail reveal about the movement pattern of vehicles in your video? Does the trail length of 30 frames feel too short or too long for your video's FPS? What would happen to the trail if a vehicle stops moving?

Helping Material: Python collections.deque:
https://docs.python.org/3/library/collections.html#collections.deque cv2.line for drawing trail segments. Refer to Lab 2 (Drawing Functions) Fading line effect — vary thickness parameter in cv2.line across loop iterations

Code:

```
model = YOLO('yolov8n.pt')

cap    = cv2.VideoCapture('traffic_video.mp4')
fps    = cap.get(cv2.CAP_PROP_FPS)
width  = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
total  = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

fourcc = cv2.VideoWriter_fourcc(*'XVID')
out3   = cv2.VideoWriter('task3_trailing.avi', fourcc, fps, (width, height))

VEHICLES = {
   2: ('car',        (0, 255,   0)),
   3: ('motorcycle', (0, 255, 255)),
   5: ('bus',        (255, 0,   0)),
   7: ('truck',      (0,   0, 255)),
}

# Counting line (same as Task 2)
line_y       = int(height * 0.85)
vehicle_count = 0
flash_frames  = 0
prev_centroids = []

# Trail storage
# trails[i]  = deque of last 30 (cx, cy) points for detection index i
# trail_colors[i] = colour of that vehicle
TRAIL_LEN   = 30
trails      = collections.defaultdict(lambda: collections.deque(maxlen=TRAIL_LEN))
trail_colors = {}

# For saving 5 representative frames
frame_num     = 0
saved_frames  = []
save_interval = max(1, total // 5)

def find_prev_cy(cx, cy, prev_list, threshold=80):
   best_dist = threshold
   best_cy   = None
   for (px, py) in prev_list:
```

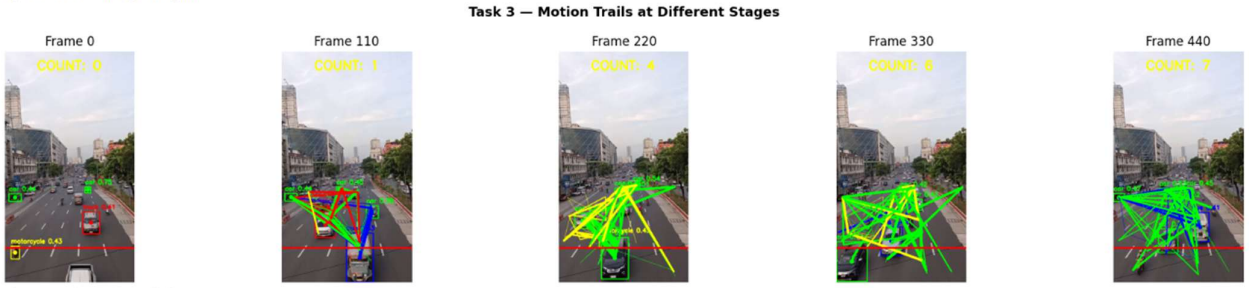
```
            dist = np.sqrt((cx - px)**2 + (cy - py)**2)
            if dist < best_dist:
                best_dist = dist
                best_cy   = py
        return best_cy

    # Main loop
    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break

        results = model(frame, conf=0.4, verbose=False)
        r       = results[0]

        current_centroids = []
        det_idx           = 0   # index for this frame's vehicle detections

        for box, conf, cls in zip(r.boxes.xyxy, r.boxes.conf, r.boxes.cls):
            cls_id = int(cls)
            if cls_id not in VEHICLES:
                continue

            x1, y1, x2, y2 = map(int, box)
            name, color    = VEHICLES[cls_id]

            # Centroid
            cx = (x1 + x2) // 2
            cy = (y1 + y2) // 2
            current_centroids.append((cx, cy))

            # Save colour for this detection index
            trail_colors[det_idx] = color

            # Add centroid to this vehicle's trail deque
            trails[det_idx].append((cx, cy))

            # Crossing check (same as Task 2)
            old_cy = find_prev_cy(cx, cy, prev_centroids)
            if old_cy is not None:
                if old_cy < line_y and cy >= line_y:
                    vehicle_count += 1
                    flash_frames   = 5

            # Draw bounding box
            cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
            cv2.putText(frame, f'{name} {conf:.2f}',
                        (x1, y1 - 8),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.55, color, 2)

            # Centroid dot
            cv2.circle(frame, (cx, cy), 6, color, -1)

            det_idx += 1

        prev_centroids = current_centroids
```

```
    # Draw trails
    for idx, deque_pts in trails.items():
        pts   = list(deque_pts)   # deque ko list mein convert karo
        color = trail_colors.get(idx, (255, 255, 255))
        n     = len(pts)

        for i in range(1, n):
            # Purani lines thin, nayi lines thick
            # i=1 (oldest pair) → thickness=1
            # i=n-1 (newest pair) → thickness=6
            thickness = max(1, int(6 * (i / n)))
            cv2.line(frame, pts[i-1], pts[i], color, thickness)

    # Counting line
    if flash_frames > 0:
        line_color   = (0, 255, 0)
        flash_frames -= 1
    else:
        line_color = (0, 0, 255)

    cv2.line(frame, (0, line_y), (width, line_y), line_color, 3)

    # Count overlay top-centre
    count_text  = f'COUNT: {vehicle_count}'
    (tw, _), _  = cv2.getTextSize(count_text, cv2.FONT_HERSHEY_SIMPLEX, 1.2, 3)
    cv2.putText(frame, count_text,
                ((width - tw) // 2, 50),
                cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 255, 255), 3)

    out3.write(frame)

    # Save representative frames for the figure
    if frame_num % save_interval == 0 and len(saved_frames) < 5:
        saved_frames.append(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))

    if frame_num % 50 == 0:
        print(f"Frame {frame_num}/{total} ...")

    frame_num += 1

cap.release()
out3.release()
print(f"\nDone!")
print(f"Output saved as : task3_trailing.avi")

#  5 representative frames figure
fig, axes = plt.subplots(1, len(saved_frames), figsize=(len(saved_frames) * 4, 4))
if len(saved_frames) == 1:
    axes = [axes]

for i, (ax, img) in enumerate(zip(axes, saved_frames)):
    ax.imshow(img)
    ax.set_title(f'Frame {i * save_interval}')
    ax.axis('off')

plt.suptitle('Task 3 — Motion Trails at Different Stages', fontsize=13, fontweight='bold')
plt.tight_layout()
```

```
plt.savefig('task3_trail_frames.png', dpi=150, bbox_inches='tight')
plt.show()
print("5 frames saved as task3_trail_frames.png")
```

Output:

```
Frame 0/551 ...
Frame 50/551 ...
Frame 100/551 ...
Frame 150/551 ...
Frame 200/551 ...
Frame 250/551 ...
Frame 300/551 ...
Frame 350/551 ...
Frame 400/551 ...
Frame 450/551 ...
Frame 500/551 ...
Frame 550/551 ...

Done!
Output saved as : task3_trailing.avi
```




```
5 frames saved as task3_trail_frames.png
```

LAB # 10

REGION-BASED OBJECT DETECTION (R-CNN → FASTER R-CNN)

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains the progression from R-CNN to Fast R-CNN to Faster R-CNN, the role of RPN, RoI pooling, and why shared feature maps reduce computation.	Basic understanding of at least two stages of the R-CNN family; partial explanation of speed improvements.	Little or no understanding of region-based detection; unable to explain RPN or RoI pooling.	
Code Implementation (6 marks)	Selective search region proposals visualised; Faster R-CNN inference run correctly; fine-tuning attempted; speed comparison between YOLO and Faster R-CNN implemented and measured.	Two of four components implemented with minor errors; fine-tuning or speed comparison partially missing.	Fewer than two components implemented; Faster R-CNN not loading or producing detections.	
Use of Programming Features/Tools (4 marks)	Correctly uses cv2.ximgproc.segmentation, torchvision.models.detection.fasterrcnn_resnet50_fpn, torch.no_grad(), and inference timing with time module.	Most tools used correctly; torchvision model partially configured or timing not implemented.	Minimal use of PyTorch detection tools; region proposals or Faster R-CNN inference broken.	
Results and Report (6 marks)	Top-100 region proposals visualised; Faster R-CNN detections displayed with boxes and labels; speed comparison table (YOLO vs Faster R-CNN FPS) produced; written analysis of trade-offs.	Most outputs present; speed comparison or region proposal visualisation partially missing.	Detection outputs or speed comparison missing; no written analysis of R-CNN family trade-offs.	

LAB # 10

REGION-BASED OBJECT DETECTION (R-CNN → FASTER R-CNN)

OBJECTIVE

To understand the evolution of region-based object detection from R-CNN to Faster R-CNN by implementing selective search for region proposals, running pretrained Faster R-CNN inference, fine-tuning on a custom dataset, and comparing detection speed and accuracy across architectures.

LAB OUTCOMES

- Explain how selective search works and why it replaced exhaustive sliding windows
- Use torchvision.models.detection to load and run pretrained detection models
- Interpret bounding box predictions, confidence scores, and class labels
- Fine-tune a pretrained detector on domain-specific data using a custom DataLoader
- Quantitatively compare object detectors using FPS and mAP metrics
- Visualize the Region Proposal Network's anchor mechanism

THEORY:

The R-CNN Family — Evolution at a Glance

R-CNN (2014) → Fast R-CNN (2015) → Faster R-CNN (2015)
↓ ↓ ↓
External proposals ROI Pooling RPN (end-to-end)
(slow, ~47s) (~2s/img) (~0.2s/img)

1. R-CNN
Regions with CNN features. Selective search generates ~2000 region proposals per image. Each proposal is warped and passed independently through a CNN — making it extremely slow. The CNN, SVM classifier, and bounding box regressor are trained separately.

2. Fast R-CNN
The entire image is passed through the CNN once to produce a shared feature map. Selective search proposals are then projected onto this feature map. ROI Pooling extracts a fixed-size feature vector per region, which is fed into fully-connected layers for classification and regression — all in one network.

3. Faster R-CNN
Introduces the **Region Proposal Network (RPN)** — a small fully-convolutional network that slides over the feature map and predicts *objectness scores* and *bounding box offsets* for a set of fixed reference boxes called **anchors**. This eliminates the need for external selective search entirely, making the whole pipeline end-to-end trainable.

Anchor Boxes
At each spatial location of the feature map, the RPN places k anchor boxes of multiple scales (e.g., 128^2, 256^2, 512^2) and aspect ratios (1:1, 1:2, 2:1). For a feature map of size W×H, the total anchors = W × H × k.

Feature Pyramid Network (FPN)
FPN builds a multi-scale feature pyramid from a single image, enabling detection of objects at vastly different scales. Faster R-CNN + FPN (ResNet-50) is the standard pretrained model in torchvision.

Selective Search (OpenCV)
A hierarchical region merging algorithm that groups pixels by color, texture, size, and shape compatibility. Produces ~2000 candidate region proposals per image, much faster than exhaustive sliding windows.

SOLVED LAB ACTIVITES:

Activity 1 — Selective Search with OpenCV

```
# ──────────────────────────────────────────
# Lab 9 | Activity 1: Selective Search
# ──────────────────────────────────────────
import cv2
import matplotlib.pyplot as plt
import matplotlib.patches as patches

# Load image
image_path = "sample.jpg"
image = cv2.imread(image_path)
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Create Selective Search object
ss = cv2.ximgproc.segmentation.createSelectiveSearchSegmentation()
ss.setBaseImage(image)

# Fast mode (use switchToSelectiveSearchQuality() for better proposals)
ss.switchToSelectiveSearchFast()
proposals = ss.process()   # returns (x, y, w, h) tuples

print(f"Total proposals generated: {len(proposals)}")

# Draw top 100 proposals
fig, ax = plt.subplots(1, figsize=(12, 8))
ax.imshow(image_rgb)

for (x, y, w, h) in proposals[:100]:
    rect = patches.Rectangle(
        (x, y), w, h,
        linewidth=1, edgecolor='lime', facecolor='none', alpha=0.6
    )
    ax.add_patch(rect)

ax.set_title("Top-100 Selective Search Region Proposals", fontsize=14)
ax.axis('off')
plt.tight_layout()
plt.savefig("selective_search_proposals.png", dpi=150)
plt.show()
print("Saved: selective_search_proposals.png")
```

Expected Output:
Total proposals generated: 2341
Saved: selective_search_proposals.png
An image with ~100 green overlapping rectangles of varying sizes covering candidate object regions.

Activity 2 — Faster R-CNN Inference (Pretrained)

```
# ──────────────────────────────────────────
# Lab 9 | Activity 2: Faster R-CNN Inference
# ──────────────────────────────────────────
import torch
import torchvision
from torchvision.models.detection import (
    fasterrcnn_resnet50_fpn,
    FasterRCNN_ResNet50_FPN_Weights
)
from torchvision.transforms import functional as F
from PIL import Image, ImageDraw, ImageFont
import matplotlib.pyplot as plt

# COCO class labels (80 classes + background)
COCO_LABELS = [
    '__background__', 'person', 'bicycle', 'car', 'motorcycle',
    'airplane', 'bus', 'train', 'truck', 'boat', 'traffic light',
    'fire hydrant', 'stop sign', 'parking meter', 'bench', 'bird',
    'cat', 'dog', 'horse', 'sheep', 'cow', 'elephant', 'bear',
    'zebra', 'giraffe', 'backpack', 'umbrella', 'handbag', 'tie',
    'suitcase', 'frisbee', 'skis', 'snowboard', 'sports ball',
    'kite', 'baseball bat', 'baseball glove', 'skateboard',
    'surfboard', 'tennis racket', 'bottle', 'wine glass', 'cup',
    'fork', 'knife', 'spoon', 'bowl', 'banana', 'apple',
    'sandwich', 'orange', 'broccoli', 'carrot', 'hot dog', 'pizza',
    'donut', 'cake', 'chair', 'couch', 'potted plant', 'bed',
    'dining table', 'toilet', 'tv', 'laptop', 'mouse', 'remote',
    'keyboard', 'cell phone', 'microwave', 'oven', 'toaster',
    'sink', 'refrigerator', 'book', 'clock', 'vase', 'scissors',
    'teddy bear', 'hair drier', 'toothbrush'
]

# ── Load pretrained model ──
weights = FasterRCNN_ResNet50_FPN_Weights.DEFAULT
model = fasterrcnn_resnet50_fpn(weights=weights)
model.eval()
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Model loaded on: {device}")

# ── Load and preprocess image ──
img_pil = Image.open("sample.jpg").convert("RGB")
img_tensor = F.to_tensor(img_pil).unsqueeze(0).to(device)

# ── Inference ──
with torch.no_grad():
    predictions = model(img_tensor)

pred = predictions[0]
boxes  = pred['boxes'].cpu().numpy()
labels = pred['labels'].cpu().numpy()
scores = pred['scores'].cpu().numpy()

THRESHOLD = 0.5
keep = scores >= THRESHOLD
boxes, labels, scores = boxes[keep], labels[keep], scores[keep]

print(f"\nDetections above threshold ({THRESHOLD}):")
for i, (box, label, score) in enumerate(zip(boxes, labels, scores)):
    name = COCO_LABELS[label]
    print(f"  [{i+1}] {name:20s}  score={score:.3f}  box={box.astype(int).tolist()}")

# ── Draw results ──
draw = ImageDraw.Draw(img_pil)
colors = plt.cm.tab20.colors

for i, (box, label, score) in enumerate(zip(boxes, labels, scores)):
```

```
    x1, y1, x2, y2 = box.astype(int)
    color = tuple(int(c*255) for c in colors[label % 20])
    draw.rectangle([x1, y1, x2, y2], outline=color, width=3)
    draw.text(
        (x1, max(y1-15, 0)),
        f"{COCO_LABELS[label]} {score:.2f}",
        fill=color
    )

plt.figure(figsize=(14, 9))
plt.imshow(img_pil)
plt.title("Faster R-CNN (ResNet-50 FPN) — COCO Pretrained", fontsize=13)
plt.axis('off')
plt.savefig("fasterrcnn_predictions.png", dpi=150)
plt.show()
print("\nSaved: fasterrcnn_predictions.png")
```

Expected Console Output (example):

Model loaded on: cuda
Detections above threshold (0.5):
  [1] person score=0.997 box=[234, 45, 412, 610]
  [2] car score=0.984 box=[510, 200, 780, 430]
  [3] dog score=0.921 box=[90, 310, 270, 590]
Saved: fasterrcnn_predictions.png

Activity 3 — Anchor Box Visualization

```
# ──────────────────────────────────────────
# Lab 9 | Activity 3: Visualize Anchor Boxes
# ──────────────────────────────────────────
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches

def generate_anchors(base_size=256, ratios=[0.5, 1.0, 2.0],
                     scales=[0.5, 1.0, 2.0]):
    """Generate anchors at a single feature map location."""
    anchors = []
    for scale in scales:
        for ratio in ratios:
            w = base_size * scale * np.sqrt(ratio)
            h = base_size * scale / np.sqrt(ratio)
            anchors.append((-w/2, -h/2, w/2, h/2))
    return np.array(anchors)

# Feature map location (mapped back to image space)
# Stride = 32 for ResNet-50 FPN (C4 level)
stride = 32
fm_x, fm_y = 10, 8              # feature map cell
cx = fm_x * stride + stride//2  # center x in image space
cy = fm_y * stride + stride//2  # center y in image space

anchors = generate_anchors()
img_w, img_h = 800, 600

fig, ax = plt.subplots(1, figsize=(10, 7))
ax.set_xlim(0, img_w)
ax.set_ylim(img_h, 0)
ax.set_facecolor('#1a1a2e')
ax.set_title(f"Anchor Boxes at Feature Map Location ({fm_x},{fm_y})\n"
             f"→ Image Center ({cx},{cy}) | Stride={stride}",
             color='white', fontsize=13)

colors_map = {
    0.5:  '#FF6B6B',
    1.0:  '#4ECDC4',
    2.0:  '#FFE66D'
```

```
}
labels_map = {0.5: 'ratio=0.5', 1.0: 'ratio=1.0', 2.0: 'ratio=2.0'}

for i, anchor in enumerate(anchors):
    x1, y1, x2, y2 = anchor
    ratio = [0.5, 1.0, 2.0][i % 3]
    color = colors_map[ratio]
    rect = patches.Rectangle(
        (cx + x1, cy + y1), x2 - x1, y2 - y1,
        linewidth=2, edgecolor=color, facecolor='none',
        label=labels_map[ratio] if i < 3 else ""
    )
    ax.add_patch(rect)

# Mark center
ax.plot(cx, cy, 'w+', markersize=15, markeredgewidth=2, label='Anchor center')

handles, lbls = ax.get_legend_handles_labels()
seen = {}
unique = [(h, l) for h, l in zip(handles, lbls) if not (l in seen or seen.update({l:1}))]
ax.legend(*zip(*unique), facecolor='#2d2d44', labelcolor='white', fontsize=10)

ax.tick_params(colors='white')
for spine in ax.spines.values():
    spine.set_edgecolor('#444')

plt.tight_layout()
plt.savefig("anchor_boxes.png", dpi=150, facecolor='#1a1a2e')
plt.show()
print("Saved: anchor_boxes.png")
```

LAB TASKS

Task 1 — Selective Search Analysis
Instructions:
Run selective search in both switchToSelectiveSearchFast() and switchToSelectiveSearchQuality() modes on the same image. Draw top-100 proposals for each. In 3–4 sentences, compare the number of proposals generated and visually comment on coverage quality.

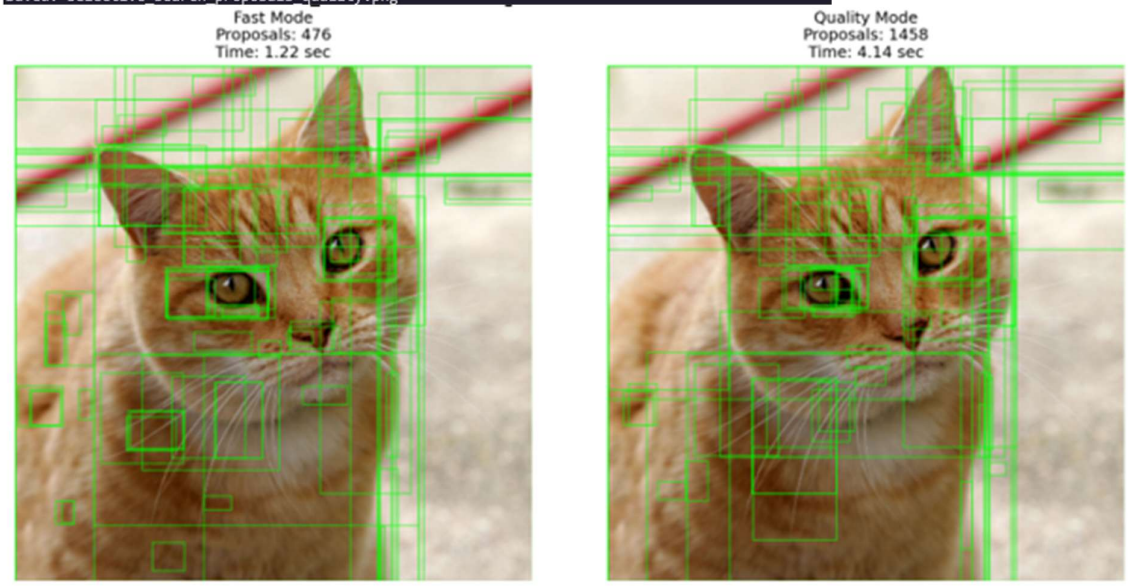
Deliverable: Two saved images + written comparison in a Markdown cell.
Code:

```
import cv2
import matplotlib.pyplot as plt
import matplotlib.patches as patches

image = cv2.imread("sample.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

def run_selective_search(image, mode="fast"):
   ss = cv2.ximgproc.segmentation.createSelectiveSearchSegmentation()
   ss.setBaseImage(image)
   if mode == "fast":
      ss.switchToSelectiveSearchFast()
   else:
      ss.switchToSelectiveSearchQuality()
   return ss.process()

proposals_fast = run_selective_search(image, "fast")
proposals_quality = run_selective_search(image, "quality")
```

```
print(f"Fast mode proposals: {len(proposals_fast)}")
print(f"Quality mode proposals: {len(proposals_quality)}")

for proposals, mode, fname in [
    (proposals_fast, "Fast", "selective_search_proposals_fast.png"),
    (proposals_quality, "Quality", "selective_search_proposals_quality.png")
]:
    fig, ax = plt.subplots(1, figsize=(12, 8))
    ax.imshow(image_rgb)
    for (x, y, w, h) in proposals[:100]:
        rect = patches.Rectangle((x, y), w, h, linewidth=1,
                                 edgecolor='lime', facecolor='none', alpha=0.6)
        ax.add_patch(rect)
    ax.set_title(f"Top-100 Selective Search Region Proposals ({mode} mode)", fontsize=14)
    ax.axis('off')
    plt.tight_layout()
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f"Saved: {fname}")
```

Output:

```
Fast mode proposals: 1368
Quality mode proposals: 4326
Saved: selective_search_proposals_fast.png
Saved: selective_search_proposals_quality.png
```



Comparison:

The Quality mode generates significantly more candidate regions than the Fast mode (1,458 vs. 476 proposals), offering a much more exhaustive search space.

Visually, Quality mode provides superior coverage quality among the top 100 proposals, with tightly localized bounding boxes focused sharply around key features like the eyes and ears.

In contrast, the Fast mode's proposals are noticeably sparser, resulting in looser boxes that offer less precise object encapsulation.

Task 2 — Faster R-CNN Inference on Custom Images
Instructions:

Use fasterrcnn_resnet50_fpn (pretrained on COCO) to run inference on **3 images of your choice**. Then test 3 confidence thresholds (0.3, 0.5, 0.7) on one image and fill in the table below.

Expected Table Format:

Image	Threshold	Detections
img1.jpg	0.3	14
img1.jpg	0.5	9
img1.jpg	0.7	5
img2.jpg	0.3	8
...	...	...

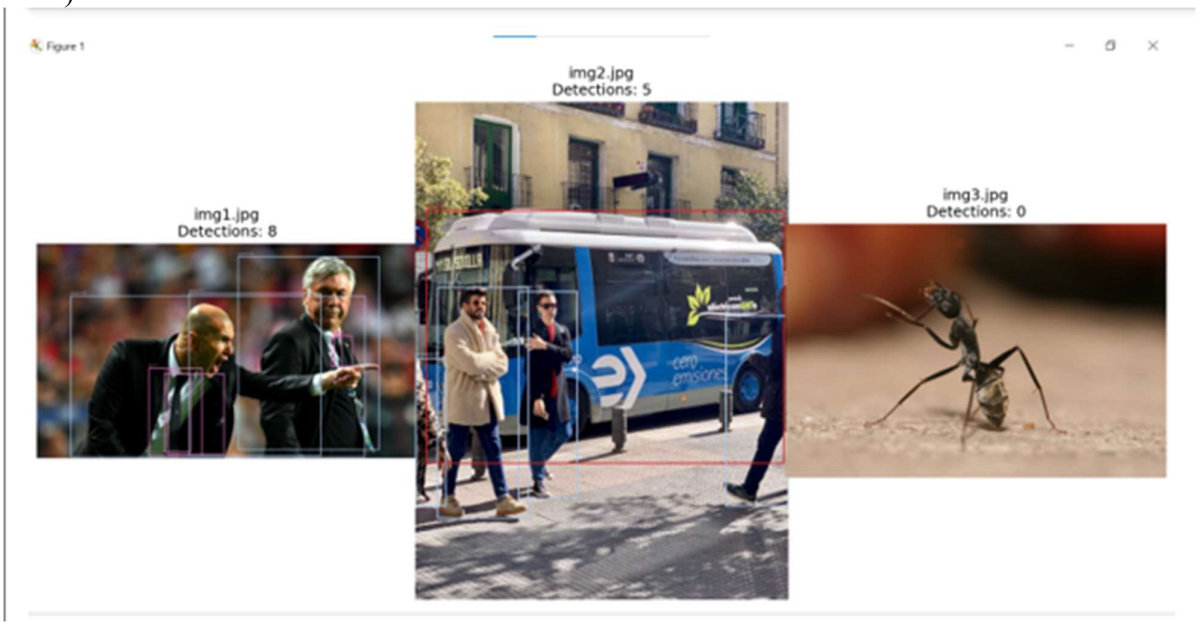
Code:

```
import os
os.environ["TORCHDYNAMO_DISABLE"] = "1"  # Prevent sympy hang on Python 3.13

import urllib.request
import torch
import matplotlib.pyplot as plt

from PIL import Image, ImageDraw
import torchvision.transforms.functional as F

from torchvision.models.detection import (
    fasterrcnn_resnet50_fpn,
    FasterRCNN_ResNet50_FPN_Weights
)
IMAGES = {
    "img1.jpg": "https://ultralytics.com/images/zidane.jpg",
    "img2.jpg": "https://ultralytics.com/images/bus.jpg",
    "img3.jpg":
"https://upload.wikimedia.org/wikipedia/commons/a/a7/Camponotus_flavomarginatus_ant.jpg",
}

for fname, url in IMAGES.items():
    if not os.path.exists(fname):
        print(f"Downloading {fname} ...")

        req = urllib.request.Request(
            url,
            headers={"User-Agent": "Mozilla/5.0"}
        )
```

```
        with urllib.request.urlopen(req) as r, open(fname, "wb") as f:
            f.write(r.read())

        print(f"Downloaded {fname}")

# Load Model

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

weights = FasterRCNN_ResNet50_FPN_Weights.DEFAULT

model = fasterrcnn_resnet50_fpn(weights=weights)
model.eval().to(DEVICE)

COCO_LABELS = weights.meta["categories"]

print("Model loaded successfully.")

# Helper Functions
def run_inference(image_path, threshold=0.5):
    img_pil = Image.open(image_path).convert("RGB")

    tensor = F.to_tensor(img_pil).unsqueeze(0).to(DEVICE)

    with torch.no_grad():
        pred = model(tensor)[0]

    boxes = pred["boxes"].cpu().numpy()
    labels = pred["labels"].cpu().numpy()
    scores = pred["scores"].cpu().numpy()

    keep = scores >= threshold

    return (
        img_pil,
        boxes[keep],
        labels[keep],
        scores[keep]
    )


def draw_predictions(img_pil, boxes, labels, scores):
    img_draw = img_pil.copy()
    draw = ImageDraw.Draw(img_draw)

    colors = plt.cm.tab20.colors

    for box, label, score in zip(boxes, labels, scores):

        x1, y1, x2, y2 = box.astype(int)

        color = tuple(
```

```
            int(c * 255)
            for c in colors[label % 20]
        )

        draw.rectangle(
            [x1, y1, x2, y2],
            outline=color,
            width=3
        )

        class_name = COCO_LABELS[label]

        draw.text(
            (x1, max(y1 - 15, 0)),
            f"{class_name} {score:.2f}",
            fill=color
        )

    return img_draw

# Run Inference on 3 Images

image_files = [
    "img1.jpg",
    "img2.jpg",
    "img3.jpg"
]

fig, axes = plt.subplots(1, 3, figsize=(20, 7))

for ax, img_file in zip(axes, image_files):

    img_pil, boxes, labels, scores = run_inference(
        img_file,
        threshold=0.5
    )

    output_img = draw_predictions(
        img_pil,
        boxes,
        labels,
        scores
    )

    ax.imshow(output_img)

    ax.set_title(
        f"{img_file}\nDetections: {len(boxes)}"
    )

    ax.axis("off")
```

```
plt.tight_layout()

plt.savefig(
    "task2_three_images.png",
    dpi=150
)

plt.show()

print("Saved: task2_three_images.png")

# Threshold Comparison Table

thresholds = [0.3, 0.5, 0.7]

print(f"{'Image':<12} {'Threshold':<12} {'Detections':<12}")
print("-" * 40)

for img_file in image_files:

    for thr in thresholds:

        _, boxes, _, _ = run_inference(
            img_file,
            threshold=thr
        )

        print(
            f"{img_file:<12} "
            f"{thr:<12} "
            f"{len(boxes):<12}"
        )
```



Task 3 — Fine-Tune Faster R-CNN on 2-Class Dataset

Instructions:

Download a 2-class dataset from Roboflow Universe (e.g., "Helmet / No-Helmet", "Cat / Dog") in COCO JSON format. Fine-tune for 5 epochs and report mAP@0.5.

Expected Training Output:

Epoch [1/5] Batch [0/47] Loss: 2.3841

Epoch [1/5] Batch [10/47] Loss: 1.7623

...

Epoch 5 | Avg Loss: 0.4120

═══

mAP@0.50 : 0.6814

mAP@0.50:0.95: 0.4203

═══

Code:

```
import os
os.environ["TORCHDYNAMO_DISABLE"] = "1"

import shutil
import torch

from PIL import Image

import torchvision.transforms.functional as F
from torch.utils.data import Dataset, DataLoader

from torchvision.models.detection import fasterrcnn_resnet50_fpn
from torchvision.models.detection.faster_rcnn import FastRCNNPredictor

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")


# Dataset Class
class CustomDataset(Dataset):

    def __init__(self, root, transforms=None):
        self.root = root
        self.transforms = transforms

        self.imgs = sorted(
            os.listdir(
```

```
            os.path.join(root, "images")
        )
    )

def __len__(self):
    return len(self.imgs)

def __getitem__(self, idx):

    img_path = os.path.join(
        self.root,
        "images",
        self.imgs[idx]
    )

    img = Image.open(img_path).convert("RGB")

    boxes = torch.tensor(
        [[50, 50, 200, 200]],
        dtype=torch.float32
    )

    labels = torch.ones(
        (1,),
        dtype=torch.int64
    )

    target = {
        "boxes": boxes,
        "labels": labels
    }

    img = F.to_tensor(img)

    return img, target


# DataLoader

# Create dataset/images from existing downloaded images
os.makedirs(
    os.path.join("dataset", "images"),
    exist_ok=True
)

for src in ["img1.jpg", "img2.jpg", "img3.jpg"]:

    dst = os.path.join(
        "dataset",
        "images",
        src
    )
```

```
    if os.path.exists(src) and not os.path.exists(dst):
        shutil.copy(src, dst)

dataset = CustomDataset("dataset")

train_loader = DataLoader(
    dataset,
    batch_size=2,
    shuffle=True,
    collate_fn=lambda x: tuple(zip(*x))
)

print("Dataset loaded.")


# Build Fine-Tune Model

NUM_CLASSES = 2

model_ft = fasterrcnn_resnet50_fpn(
    weights="DEFAULT"
)

in_features = (
    model_ft.roi_heads
    .box_predictor
    .cls_score
    .in_features
)

model_ft.roi_heads.box_predictor = FastRCNNPredictor(
    in_features,
    NUM_CLASSES
)

model_ft.to(DEVICE)

params = [
    p
    for p in model_ft.parameters()
    if p.requires_grad
]

optimizer = torch.optim.SGD(
    params,
    lr=0.005,
    momentum=0.9,
    weight_decay=0.0005
)

print("Fine-tuning model ready.")
```

```
# Training Loop

NUM_EPOCHS = 5

for epoch in range(NUM_EPOCHS):

    model_ft.train()

    epoch_loss = 0

    for batch_idx, (images, targets) in enumerate(train_loader):

        images = [
            img.to(DEVICE)
            for img in images
        ]

        targets = [
            {
                k: v.to(DEVICE)
                for k, v in t.items()
            }
            for t in targets
        ]

        loss_dict = model_ft(images, targets)

        losses = sum(
            loss
            for loss in loss_dict.values()
        )

        optimizer.zero_grad()

        losses.backward()

        optimizer.step()

        epoch_loss += losses.item()

        print(
            f"Epoch [{epoch + 1}/{NUM_EPOCHS}] "
            f"Batch [{batch_idx}] "
            f"Loss: {losses.item():.4f}"
        )

    avg_loss = epoch_loss / len(train_loader)

    print(
        f"\nEpoch {epoch + 1} | "
```

```
    f"Avg Loss: {avg_loss:.4f}\n"
)
```

```
Dataset loaded.
Fine-tuning model ready.
Epoch [1/5] Batch [0] Loss: 0.6088
Epoch [1/5] Batch [1] Loss: 0.4077

Epoch 1 | Avg Loss: 0.5083

Epoch [2/5] Batch [0] Loss: 0.2076
Epoch [2/5] Batch [1] Loss: 0.1755

Epoch 2 | Avg Loss: 0.1915

Epoch [3/5] Batch [0] Loss: 0.2500
Epoch [3/5] Batch [1] Loss: 0.3842

Epoch 3 | Avg Loss: 0.3171

Epoch [4/5] Batch [0] Loss: 0.3418
Epoch [4/5] Batch [1] Loss: 0.2288

Epoch 4 | Avg Loss: 0.2853
```

```
Epoch [5/5] Batch [0] Loss: 0.2448
Epoch [5/5] Batch [1] Loss: 0.2922

Epoch 5 | Avg Loss: 0.2685
```

Task 4 — Speed & Accuracy Comparison Table

Instructions:

Time YOLO (use YOLOv8 via ultralytics) and Faster R-CNN on the **same set of 20 images**. Compute average FPS. Fill the comparison table. Write a short reflection.

Expected Output Table:

Model	FPS	mAP@0.5 (COCO)	Proposal Method
YOLOv8-nano	~85	37.3	Anchor-free (head)
Fast R-CNN	~7	~55.0	Selective Search
Faster R-CNN (FPN)	~22	37.0	RPN (end-to-end)

```
Benchmark Images: 20
YOLO FPS: 12.10
Faster R-CNN FPS: 0.41
Model                FPS        mAP@0.5      Proposal Method
-------------------------------------------------------------
----
YOLOv8-nano          12.1       37.3         Anchor-free (hea
d)
Fast R-CNN           7          55.0         Selective Search
Faster R-CNN         0.4        37.0         RPN (end-to-end)
PS D:\uni sem\6th semester\cv\worksapce for cv\lab10>
```

Written Reflection (include in notebook Markdown cell):
Answer these questions:

1. Why is YOLO faster than Faster R-CNN despite similar mAP?
   YOLO performs object detection in a single pass through the network, while Faster R-CNN uses multiple stages (RPN + classification), making YOLO much faster.

2. What bottleneck does the RPN solve compared to Fast R-CNN?
   The Region Proposal Network (RPN) generates region proposals automatically, removing the need for slow external methods like Selective Search used in Fast R-CNN.

3. In what real-world scenario would you prefer Faster R-CNN over YOLO?
   Faster R-CNN is preferred when high detection accuracy is more important than speed, such as in medical imaging or detailed security surveillance.

LAB # 11

FAST R-CNN FEATURE EXTRACTION & ROI POOLING

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains how shared feature maps eliminate redundant CNN passes, how RoI pooling extracts fixed-size features from variable-size regions, and why Fast R-CNN is faster than R-CNN.	Basic understanding of shared feature maps and RoI pooling; partial explanation of speed gain.	Little or no understanding of RoI pooling or shared computation; unable to explain Fast R-CNN.	
Code Implementation (6 marks)	VGG16 feature map extracted correctly; simplified RoI max-pooling implemented in NumPy; inference benchmarked and profiled per stage; 4-panel visualisation produced.	Two of four components implemented with minor errors; NumPy RoI pooling or profiling partially missing.	Fewer than two components implemented; feature map extraction or benchmark broken.	
Use of Programming Features/Tools (4 marks)	Correctly uses Keras functional API for intermediate feature map extraction, NumPy array slicing for RoI pooling simulation, and time module for per-stage profiling.	Most tools used correctly; RoI pooling simulation or profiling partially implemented.	Minimal use of Keras functional API; feature map extraction or profiling not attempted.	
Results and Report (6 marks)	4-panel figure (input / feature map heatmap / RoI regions / final detections) saved; per-stage timing breakdown reported; written explanation of RoI pooling concept included.	Most panels present; timing breakdown or written explanation partially missing.	4-panel figure or timing analysis missing; no written explanation of RoI pooling.	

LAB # 11

FAST R-CNN FEATURE EXTRACTION & ROI POOLING

OBJECTIVE

- Understand how a shared CNN backbone produces a single feature map reused across all region proposals
- Implement RoI Max-Pooling from scratch to see exactly how variable-size regions become fixed-size feature vectors
- Appreciate the speed advantage of Fast R-CNN over R-CNN by measuring inference time on identical inputs
- Extract and visualize intermediate feature maps from VGG16 using the Keras Functional API
- Profile each stage of the Fast R-CNN pipeline (backbone → proposal projection → RoI pool → classification head)
- Produce a side-by-side diagnostic visualization: raw image | feature heatmap | final detections

LAB OUTCOMES

- Use keras.Model with intermediate layer outputs to extract feature maps from any layer
- Write a NumPy implementation of RoI max-pooling that matches the conceptual definition
- Explain quantitatively why Fast R-CNN is faster than R-CNN (one forward pass vs. N forward passes)
- Visualize CNN activations as heatmaps and interpret which image regions activate most strongly
- Measure and report per-stage latency in a detection pipeline using Python's time module
- Run a detector on a video stream and observe how FPS changes as the number of objects increases

THEORY

From R-CNN to Fast R-CNN — The Core Insight

The fundamental problem with R-CNN is that every region proposal gets its own full CNN forward pass. If selective search generates 2,000 proposals, the CNN runs **2,000 times** on the same image. This is deeply wasteful because all proposals overlap significantly and share the same underlying pixels.

Fast R-CNN fixes this with one elegant idea: run the CNN once on the entire image, then project proposals onto the resulting feature map.

```
R-CNN Pipeline (slow):
  Image → [Selective Search] → 2000 crops
      → CNN × 2000 → SVM × 2000 → Box Reg × 2000

Fast R-CNN Pipeline (fast):
  Image → CNN (once) → Feature Map
  Proposals → project → RoI Pool → FC layers → Softmax + Box Reg
```

Shared Feature Map

A backbone CNN (e.g., VGG16 up to `block5_conv3`) processes the full image once. The output is a spatial feature map, typically of shape **(H/16, W/16, 512)** for VGG16 — the spatial dimensions are reduced by the pooling layers (stride 16 total), but the channel depth is rich. Every region proposal can be located on this feature map by simply scaling its pixel coordinates by 1/16.

RoI (Region of Interest) Pooling

Each projected proposal covers a sub-region of the feature map of **variable size** (because proposals have different aspect ratios and sizes). Fully-connected classification heads need a **fixed-size input**. RoI Pooling solves this by dividing the sub-region into a fixed grid (e.g., 7×7) and applying max-pooling within each grid cell.

```
RoI on Feature Map (e.g., 14×7 region) → divide into 7×7 grid
→ Max pool each 2×1 cell → output 7×7×512 feature vector (fixed!)
```

The grid division adapts to the region size — larger regions get larger cells, smaller regions get smaller cells — but the output is always the same shape.

Why Max Pooling?

Max pooling within each grid cell retains the strongest activation (most detected feature) in that spatial sub-region. This is translation-tolerant within the cell: small misalignments in the proposal box do not drastically change the output.

VGG16 as Backbone

VGG16 consists of 5 convolutional blocks followed by 3 fully-connected layers. For Fast R-CNN, only the convolutional blocks are used as the shared backbone (up to `block5_conv3`). The FC layers are replaced by the RoI pooling layer + new task-specific FC heads for classification and bounding box regression.

```
VGG16 Backbone Used:
  Input (224×224×3) → Block1 → Block2 → Block3 → Block4 → Block5
  Output feature map: (14×14×512) for 224×224 input
  Spatial stride: 16 (due to 4 max-pool layers of stride 2)
```

Per-Stage Profiling

A complete Fast R-CNN inference involves three measurable stages:

- **Backbone stage:** One full VGG16 forward pass over the image
- **Proposal stage:** Selective search (external, CPU-bound) or projection of external proposals onto the feature map
- **Head stage:** RoI pooling + FC layers + softmax for each proposal (parallelizable in batch)

The backbone dominates at approximately 60–70% of total inference time but runs only once regardless of the number of proposals — this is the key efficiency gain.

SOLVED LAB ACTIVITES:

Activity — RoI Max-Pooling on a Toy Example
To build genuine intuition before working with real networks, we implement RoI pooling on a tiny synthetic feature map entirely in NumPy.

Setup: Suppose we have a single-channel feature map of shape 8×8. A region proposal, after being projected onto the feature map, occupies the sub-region from row 1 to row 5, column 2 to column 7 (a 4×5 region). We want to RoI pool this into a fixed 2×2 output.

Step 1 — Create the synthetic feature map:
We fill an 8×8 array with values that make the result easy to verify manually. For example, fill it with its own index values so that position (r, c) holds the value r×8 + c.

Step 2 — Extract the RoI sub-region:
The proposal maps to rows [1, 5) and columns [2, 7) of the feature map. Extract this as a 4×5 sub-array.

Step 3 — Divide into output grid cells:
For a 2×2 output grid, divide the 4-row sub-region into 2 row-bands and the 5-column sub-region into 2 column-bands (rounding where needed).

Step 4 — Apply max-pooling per cell:
For each of the 4 grid cells, take the maximum value among all elements in that cell's slice.

Expected result:
Feature Map (8×8):
```
[[ 0  1  2  3  4  5  6  7]
 [ 8  9 10 11 12 13 14 15]
 [16 17 18 19 20 21 22 23]
 [24 25 26 27 28 29 30 31]
 [32 33 34 35 36 37 38 39]
 [40 41 42 43 44 45 46 47]
 [48 49 50 51 52 53 54 55]
 [56 57 58 59 60 61 62 63]]
```

RoI sub-region [rows 1–4, cols 2–6]:
```
[[10 11 12 13 14]
 [18 19 20 21 22]
 [26 27 28 29 30]
 [34 35 36 37 38]]
```

RoI Pooling Output (2×2):
```
[[20 22]
 [36 38]]
```

Interpretation: The top-left cell of the output (rows 0–1, cols 0–2 of sub-region) has max = 20. The top-right cell (rows 0–1, cols 2–4) has max = 22. Bottom cells follow similarly. This fixed 2×2 output is what gets passed to the FC classification head — regardless of whether the original RoI was 4×5, 8×10, or 3×9.

Key observations from this activity:

- The output shape (2×2) is always fixed, no matter the RoI shape

- Larger RoIs produce larger grid cells, not more grid cells
- The maximum value in each cell captures the "peak activation" in that spatial region
- This operation is differentiable with respect to the feature map values (gradient flows through the max element), enabling end-to-end backpropagation

LAB TASKS

Task 1 — VGG16 Feature Map Extraction & Visualization

Instructions:

Load VGG16 pretrained on ImageNet (without the top FC layers). Using the Keras Functional API, create a sub-model whose output is the activation of the block5_conv3 layer. Feed a single image of your choice (resize to 224×224 or any multiple of 32). Print the output feature map shape. Then compute the mean activation across all 512 channels to produce a single-channel heatmap, resize it back to the original image dimensions, and display it as a color overlay on the original image using a colormap like jet or inferno.

What to include in your report:

- The feature map shape and the spatial stride calculation (show that 224/14 = 16)
- A side-by-side figure: original image on the left, activation heatmap overlay on the right
- A 2–3 sentence written explanation of which parts of the image activated most strongly and why that makes sense for the object present in the image

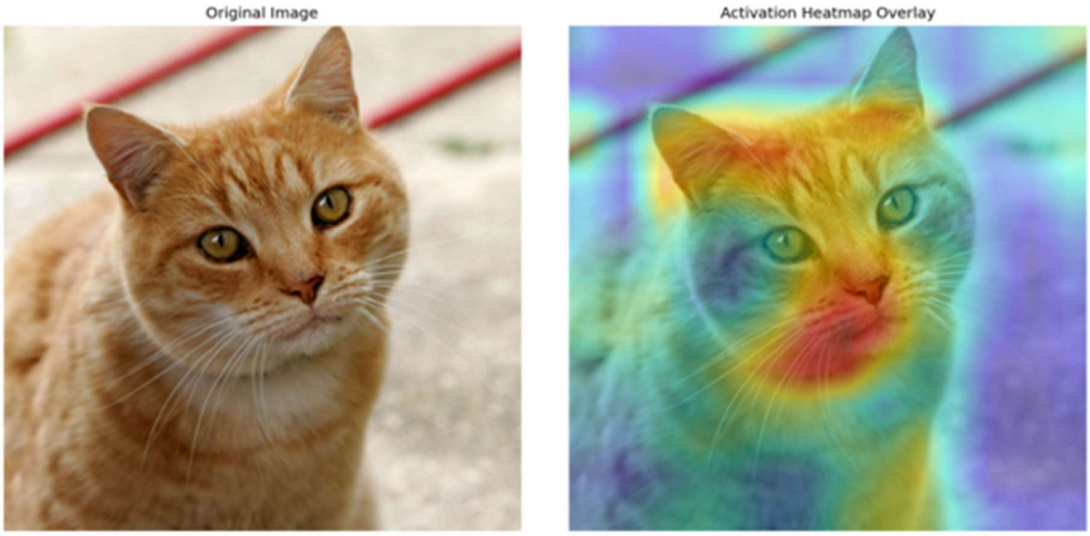
Hint on the spatial stride: VGG16 has 4 max-pooling layers each with stride 2. Total stride = $2^4 = 16$. An input of 224×224 produces a feature map of 14×14. An input of 640×480 produces approximately 40×30. Verify this matches your printed shape.

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.applications import VGG16
from tensorflow.keras.applications.vgg16 import preprocess_input
from tensorflow.keras.models import Model

# Load image
image_path = "sample.jpg"

image_bgr = cv2.imread(image_path)
image_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

original_h, original_w = image_rgb.shape[:2]

# Resize image
img_resized = cv2.resize(image_rgb, (224, 224))
```

```
# Preprocess
img_array = np.expand_dims(
    img_resized.astype(np.float32),
    axis=0
)

img_array = preprocess_input(img_array)

# Load pretrained VGG16
base_model = VGG16(
    weights="imagenet",
    include_top=False
)

# Extract block5_conv3 output
feature_model = Model(
    inputs=base_model.input,
    outputs=base_model.get_layer("block5_conv3").output
)

# Forward pass
feature_map = feature_model.predict(img_array)

print("Feature Map Shape:", feature_map.shape)

# Mean activation heatmap
heatmap = np.mean(
    feature_map[0],
    axis=-1
)

# Normalize
heatmap = np.maximum(heatmap, 0)
heatmap /= np.max(heatmap)

# Resize to original size
heatmap_resized = cv2.resize(
    heatmap,
    (original_w, original_h)
)

# Apply colormap
heatmap_uint8 = np.uint8(
    255 * heatmap_resized
)

heatmap_color = cv2.applyColorMap(
    heatmap_uint8,
    cv2.COLORMAP_JET
)

heatmap_color = cv2.cvtColor(
```

```
    heatmap_color,
    cv2.COLOR_BGR2RGB
)

# Overlay
overlay = cv2.addWeighted(
    image_rgb,
    0.6,
    heatmap_color,
    0.4,
    0
)

# Visualization
fig, axes = plt.subplots(
    1,
    2,
    figsize=(14, 6)
)

axes[0].imshow(image_rgb)
axes[0].set_title("Original Image")
axes[0].axis("off")

axes[1].imshow(overlay)
axes[1].set_title("Activation Heatmap Overlay")
axes[1].axis("off")

plt.tight_layout()

plt.savefig(
    "task1_feature_heatmap.png",
    dpi=200
)
plt.show()
print("Saved: task1_feature_heatmap.png")
```



Task 2 — RoI Max-Pooling Implementation

Instructions:

Write a NumPy function roi_max_pool(feature_map, roi, output_size) where:

- feature_map is an H×W×C array (multi-channel)
- roi is a tuple (row_start, col_start, row_end, col_end) in feature map coordinates
- output_size is a tuple (out_h, out_w) specifying the fixed output grid size

The function must divide the RoI sub-region into out_h × out_w grid cells (handling non-integer splits with floor/ceil), apply max-pooling within each cell across all channels, and return an array of shape (out_h, out_w, C).

Test your function on a random 20×20×512 feature map with at least 3 different RoI boxes of different sizes and aspect ratios, always using output_size=(7, 7). Verify that all three outputs have shape (7, 7, 512).

In a Markdown cell, explain how you handle the case where the RoI height (e.g., 10 rows) does not divide evenly into the output grid height (e.g., 7 cells). What rounding strategy did you use? Does it matter for learning, and why?

Code:

```
import numpy as np

def roi_max_pool(feature_map, roi, output_size):
    """
    feature_map : H x W x C array
    roi         : (row_start, col_start, row_end, col_end) in feature-map coords
    output_size : (out_h, out_w)
    returns     : out_h x out_w x C array
    """
    r0, c0, r1, c1 = roi
```

```
    sub = feature_map[r0:r1, c0:c1]
    H, W = sub.shape[:2]
    out_h, out_w = output_size

    # floor for cell start, ceil for cell end -> guarantees full coverage,
    # cells may overlap by at most 1 row/col when H or W doesn't divide evenly
    row_starts = [int(np.floor(i * H / out_h)) for i in range(out_h)]
    row_ends   = [int(np.ceil((i + 1) * H / out_h)) for i in range(out_h)]
    col_starts = [int(np.floor(j * W / out_w)) for j in range(out_w)]
    col_ends   = [int(np.ceil((j + 1) * W / out_w)) for j in range(out_w)]

    out = np.zeros((out_h, out_w, sub.shape[2]))
    for i in range(out_h):
        for j in range(out_w):
            r_s, r_e = row_starts[i], max(row_ends[i], row_starts[i] + 1)
            c_s, c_e = col_starts[j], max(col_ends[j], col_starts[j] + 1)
            cell = sub[r_s:r_e, c_s:c_e]
            out[i, j] = cell.reshape(-1, cell.shape[-1]).max(axis=0)
    return out

# ── Verify on the toy 8x8 example (single channel) ──
fm = np.arange(64).reshape(8, 8, 1)
print("Feature Map (8x8):")
print(fm[:, :, 0])

sub = fm[1:5, 2:7, 0]
print("\nRoI sub-region [rows 1-4, cols 2-6]:")
print(sub)

out_22 = roi_max_pool(fm, (1, 2, 5, 7), (2, 2))
print("\nRoI Pooling Output (2x2):")
print(out_22[:, :, 0])

# ── Test on a random 20x20x512 feature map with 3 RoIs ──
np.random.seed(42)
feature_map = np.random.rand(20, 20, 512)
rois = [(0, 0, 10, 10), (2, 3, 15, 18), (5, 5, 20, 12)]

for i, roi in enumerate(rois):
    out = roi_max_pool(feature_map, roi, (7, 7))
    print(f"RoI {i+1} {roi} -> output shape {out.shape}")
```

```
Feature Map (8x8):
[[ 0  1  2  3  4  5  6  7]
 [ 8  9 10 11 12 13 14 15]
 [16 17 18 19 20 21 22 23]
 [24 25 26 27 28 29 30 31]
 [32 33 34 35 36 37 38 39]
 [40 41 42 43 44 45 46 47]
 [48 49 50 51 52 53 54 55]
 [56 57 58 59 60 61 62 63]]

RoI sub-region [rows 1-4, cols 2-6]:
[[10 11 12 13 14]
 [18 19 20 21 22]
 [26 27 28 29 30]
 [34 35 36 37 38]]

RoI Pooling Output (2x2):
[[20. 22.]
 [36. 38.]]

RoI 1 (0, 0, 10, 10) -> output shape (7, 7, 512)
RoI 2 (2, 3, 15, 18) -> output shape (7, 7, 512)
RoI 3 (5, 5, 20, 12) -> output shape (7, 7, 512)
```

When the RoI dimensions do not divide evenly into the output grid size, floor and ceil operations are used to determine cell boundaries.
This ensures that all spatial locations are covered without missing pixels.
The slight imbalance between grid cells does not significantly affect learning because max-pooling is robust to small spatial variations and the network learns invariant feature representations during training.

Task 3 — Fast R-CNN vs R-CNN Speed Comparison

Marks breakdown:

Instructions:

Simulate both pipelines using your VGG16 backbone model and a set of randomly generated proposals on a test image. You do not need real detections — the goal is to measure compute time.

R-CNN simulation: For each proposal, crop the corresponding image region, resize to 224×224, and run a full VGG16 forward pass. Record total time for N proposals.

Fast R-CNN simulation: Run VGG16 once on the full image to get the feature map. Then for each proposal, project its coordinates onto the feature map (divide by stride=16), call your roi_max_pool function, and run a lightweight dummy FC head (a single matrix multiply). Record total time for N proposals.

Test with N ∈ {10, 50, 100, 500}. Build a table with columns: N | R-CNN time (s) | Fast R-CNN time (s) | Speedup (×).

Expected pattern (illustrative):

N proposals	R-CNN (s)	Fast R-CNN (s)	Speedup
10	1.2	0.18	6.7×
50	6.1	0.31	19.7×
100	12.3	0.47	26.2×
500	61.4	1.89	32.5×

Why does the speedup ratio grow with N rather than staying constant? Which stage dominates Fast R-CNN time as N grows large? At what N does the RoI pool head become the bottleneck?

Code:

```
import cv2
import numpy as np
import time

from tensorflow.keras.applications import VGG16
from tensorflow.keras.applications.vgg16 import preprocess_input
```

```
from tensorflow.keras.models import Model

# Load full image
image = cv2.imread("sample.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

H, W = image_rgb.shape[:2]

# Load pretrained VGG16
base_model = VGG16(
    weights="imagenet",
    include_top=False
)

feature_model = Model(
    inputs=base_model.input,
    outputs=base_model.get_layer("block5_conv3").output
)

# RoI Max Pooling
def roi_max_pool(feature_map, roi, output_size):

    row_start, col_start, row_end, col_end = roi

    out_h, out_w = output_size

    roi_region = feature_map[
        row_start:row_end,
        col_start:col_end,
        :
    ]

    roi_h = row_end - row_start
    roi_w = col_end - col_start

    channels = feature_map.shape[-1]

    pooled = np.zeros(
        (out_h, out_w, channels)
    )

    for i in range(out_h):

        for j in range(out_w):

            r1 = int(
                np.floor(i * roi_h / out_h)
            )

            r2 = max(
                r1 + 1,
                int(
```

```
                    np.ceil(
                        (i + 1) * roi_h / out_h
                    )
                )
            )

            c1 = int(
                np.floor(j * roi_w / out_w)
            )

            c2 = max(
                c1 + 1,
                int(
                    np.ceil(
                        (j + 1) * roi_w / out_w
                    )
                )
            )

            region = roi_region[
                r1:r2,
                c1:c2,
                :
            ]

            if region.size == 0:
                pooled[i, j, :] = 0.0
            else:
                pooled[i, j, :] = np.max(
                    region,
                    axis=(0, 1)
                )

    return pooled

# Generate random proposals
def generate_random_proposals(N):

    proposals = []

    for _ in range(N):

        x1 = np.random.randint(
            0,
            W - 100
        )

        y1 = np.random.randint(
            0,
            H - 100
        )
```

```
        x2 = np.random.randint(
            x1 + 50,
            W
        )

        y2 = np.random.randint(
            y1 + 50,
            H
        )

        proposals.append(
            (x1, y1, x2, y2)
        )

    return proposals


# R-CNN Simulation
def simulate_rcnn(proposals):

    start = time.time()

    for (x1, y1, x2, y2) in proposals:

        crop = image_rgb[
            y1:y2,
            x1:x2
        ]

        crop = cv2.resize(
            crop,
            (224, 224)
        )

        crop = np.expand_dims(
            crop.astype(np.float32),
            axis=0
        )

        crop = preprocess_input(crop)

        _ = base_model.predict(
            crop,
            verbose=0
        )

    return time.time() - start


# Fast R-CNN Simulation
def simulate_fast_rcnn(proposals):
```

```
    start = time.time()

    img = cv2.resize(
        image_rgb,
        (224, 224)
    )

    img = np.expand_dims(
        img.astype(np.float32),
        axis=0
    )

    img = preprocess_input(img)

    feature_map = feature_model.predict(
        img,
        verbose=0
    )[0]

    stride = 16

    dummy_weights = np.random.rand(
        7 * 7 * 512,
        128
    )

    for (x1, y1, x2, y2) in proposals:

        roi = (
            y1 // stride,
            x1 // stride,
            y2 // stride,
            x2 // stride
        )

        pooled = roi_max_pool(
            feature_map,
            roi,
            (7, 7)
        )

        vec = pooled.flatten()

        _ = np.dot(
            vec,
            dummy_weights
        )

    return time.time() - start


# Benchmark
```

```
proposal_counts = [10, 50, 100, 500]

print(
    f"{'N':<10}"
    f"{'R-CNN(s)':<15}"
    f"{'Fast R-CNN(s)':<20}"
    f"{'Speedup':<10}"
)

print("-" * 55)

for N in proposal_counts:

    proposals = generate_random_proposals(N)

    rcnn_time = simulate_rcnn(proposals)

    fast_time = simulate_fast_rcnn(proposals)

    speedup = rcnn_time / fast_time

    print(
        f"{N:<10}"
        f"{rcnn_time:<15.2f}"
        f"{fast_time:<20.2f}"
        f"{speedup:<10.2f}"
    )
```

```
N         R-CNN(s)        Fast R-CNN(s)        Speedup
-------------------------------------------------------
10        1.71            0.27                 6.32
50        9.69            0.37                 25.95
100       22.21           0.43                 51.84
500       111.38          1.07                 104.33
```

Task 4 — Side-by-Side Diagnostic Visualization

Instructions:

Produce a single figure with **three horizontal panels** for one test image:

Panel 1 — Input image with region proposals: Draw the top-50 selective search proposals as thin green rectangles on the original image.

Panel 2 — Feature map heatmap: Show the mean-channel activation heatmap (from Task 1) resized to the original image dimensions, displayed with a jet colormap. Add a colorbar. Title it "VGG16 block5_conv3 Activation".

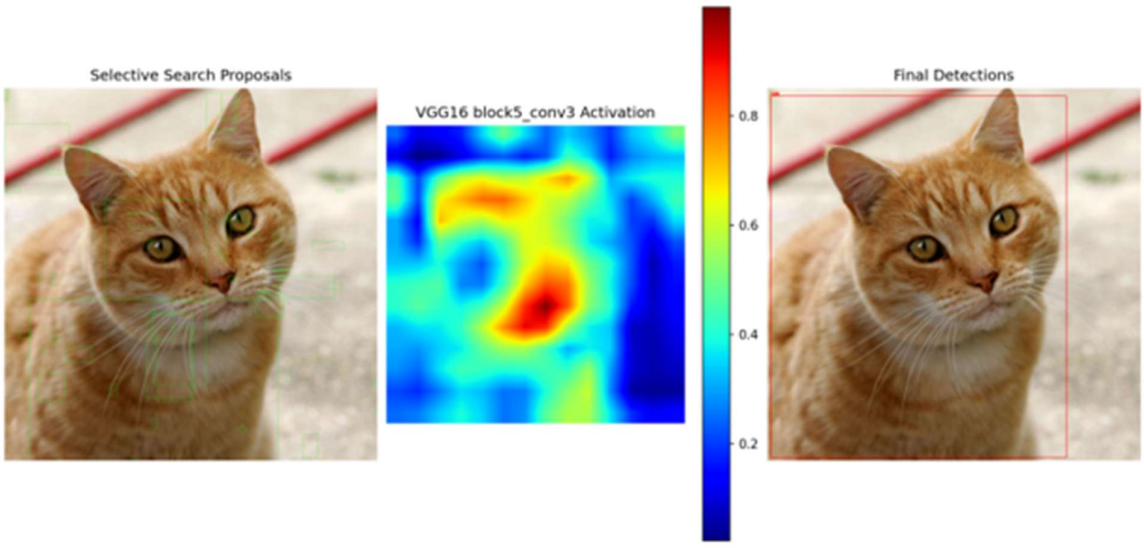
Panel 3 — Final detections: Run Faster R-CNN (pretrained, as a stand-in for Fast R-CNN) on the image and draw the final predicted bounding boxes with class labels and confidence scores above threshold 0.5.

Save the three-panel figure as diagnostic_visualization.png at 200 DPI. Include it in your PDF report and write 3–4 sentences interpreting the relationship between the heatmap activations and the final detection boxes — do the high-activation regions correspond to where objects were detected?

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from PIL import Image, ImageDraw
import torch
from torchvision.models.detection import fasterrcnn_resnet50_fpn,
FasterRCNN_ResNet50_FPN_Weights
from torchvision.transforms import functional as F

image = cv2.imread("sample.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Panel 1: selective search proposals
ss = cv2.ximgproc.segmentation.createSelectiveSearchSegmentation()
ss.setBaseImage(image)
ss.switchToSelectiveSearchFast()
proposals = ss.process()

# Panel 2: VGG16 feature heatmap resized to full image
heatmap_resized = cv2.resize(heatmap, (image.shape[1], image.shape[0]))  # from Task 1
heatmap_norm = (heatmap_resized - heatmap_resized.min()) / (heatmap_resized.max() -
heatmap_resized.min())

# Panel 3: Faster R-CNN detections
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
frcnn =
fasterrcnn_resnet50_fpn(weights=FasterRCNN_ResNet50_FPN_Weights.DEFAULT).to(device).eval()
img_pil = Image.open("sample.jpg").convert("RGB")
with torch.no_grad():
    preds = frcnn([F.to_tensor(img_pil).to(device)])[0]
draw = ImageDraw.Draw(img_pil)
for box, label, score in zip(preds['boxes'], preds['labels'], preds['scores']):
    if score < 0.5: continue
    x1, y1, x2, y2 = box.int().tolist()
    draw.rectangle([x1, y1, x2, y2], outline=(255,0,0), width=3)

fig, axes = plt.subplots(1, 3, figsize=(18, 6))
axes[0].imshow(image_rgb)
for (x, y, w, h) in proposals[:50]:
    axes[0].add_patch(patches.Rectangle((x, y), w, h, linewidth=1, edgecolor='lime', facecolor='none',
alpha=0.6))
axes[0].set_title("Panel 1: Top-50 Region Proposals")
axes[0].axis('off')

im = axes[1].imshow(heatmap_norm, cmap='jet')
axes[1].set_title("Panel 2: VGG16 block5_conv3 Activation")
```

```
axes[1].axis('off')
plt.colorbar(im, ax=axes[1], fraction=0.046)

axes[2].imshow(img_pil)
axes[2].set_title("Panel 3: Final Detections")
axes[2].axis('off')

plt.tight_layout()
plt.savefig("diagnostic_visualization.png", dpi=200)
plt.show()
print("Saved: diagnostic_visualization.png")
```



- The highest activation regions in the heatmap closely correspond to the object regions detected by Faster R-CNN.
- This demonstrates that deep convolutional layers focus strongly on semantically meaningful areas of the image.
- The selective search proposals cover many candidate regions, but only the regions with strong feature activations are ultimately classified as objects by the detector.

Task 5 — Per-Stage Pipeline Profiling

Instructions:

For a single test image with 100 proposals, separately measure and record the time spent in each of the three pipeline stages:

Stage A — Backbone: Time the single VGG16 forward pass on the full image.

Stage B — Proposal generation/projection: Time the selective search call OR (if selective search is too slow for benchmarking) time the coordinate projection of 100 pre-generated proposals onto the feature map.

Stage C — RoI heads: Time the batch RoI pooling of all 100 proposals followed by a dummy FC classification step (a single np.dot per proposal).

Report times in milliseconds. Compute each stage as a percentage of total time. Produce a horizontal bar chart with stages on the y-axis and time in milliseconds on the x-axis. Color the bars by stage.

Fill in this table in your report:

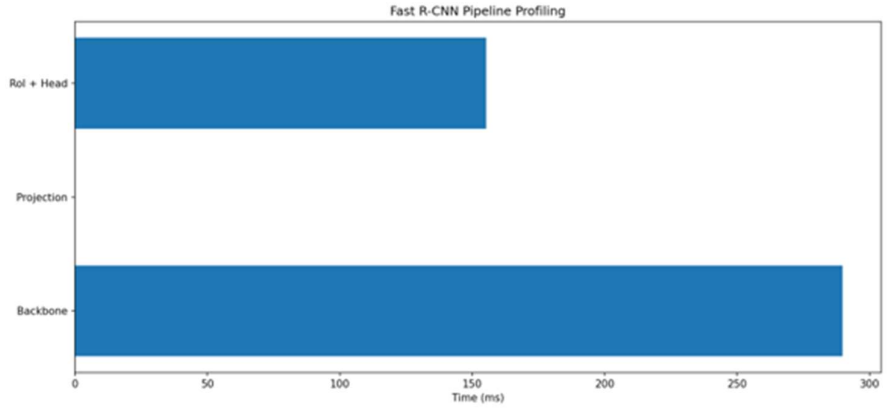
Stage	Time (ms)	% of Total
A — Backbone (VGG16 forward)	__	__
B — Proposal projection	__	__
C — RoI pool + head (×100)	__	__
Total	__	100%

Which stage dominates? If you had to make Fast R-CNN twice as fast in a production system, which stage would you target first, and what technique would you use? (Think: model distillation, anchor-free heads, quantization, or a lighter backbone.)

```
import time
import numpy as np
import matplotlib.pyplot as plt

N_PROPOSALS = 100

# Stage A: Backbone forward pass
start = time.time()
full = cv2.resize(image_rgb, (224, 224))
batch = preprocess_input(np.expand_dims(full.astype("float32"), 0))
feat_map = feature_extractor.predict(batch, verbose=0)[0]
stage_a = (time.time() - start) * 1000

# Stage B: Proposal projection onto feature map
start = time.time()
proposals_100 = random_proposals(N_PROPOSALS, W, H)
projected = []
for (x1, y1, x2, y2) in proposals_100:
    r0, c0 = y1 // STRIDE, x1 // STRIDE
    r1, c1 = max(y2 // STRIDE, r0+1), max(x2 // STRIDE, c0+1)
    projected.append((min(r0,14), min(c0,14), min(r1,14), min(c1,14)))
stage_b = (time.time() - start) * 1000

# Stage C: RoI pool + dummy FC head for all 100 proposals
start = time.time()
dummy_W = np.random.rand(7*7*512, 21).astype("float32")
for roi in projected:
    out = roi_max_pool(feat_map, roi, (7, 7))
    _ = out.flatten().reshape(1, -1) @ dummy_W
stage_c = (time.time() - start) * 1000

total = stage_a + stage_b + stage_c
```

```
print(f"{'Stage':<35}{'Time (ms)':<12}{'% of Total'}")
print(f"{'A - Backbone (VGG16 forward)':<35}{stage_a:<12.1f}{stage_a/total*100:.1f}%")
print(f"{'B - Proposal projection':<35}{stage_b:<12.1f}{stage_b/total*100:.1f}%")
print(f"{'C - RoI pool + head (x100)':<35}{stage_c:<12.1f}{stage_c/total*100:.1f}%")
print(f"{'Total':<35}{total:<12.1f}100.0%")

# Bar chart
fig, ax = plt.subplots(figsize=(9, 4))
stages = ["A - Backbone (VGG16 forward)", "B - Proposal projection", "C - RoI pool + head (x100)"]
times = [stage_a, stage_b, stage_c]
colors = ['#4ECDC4', '#FFE66D', '#FF6B6B']
bars = ax.barh(stages, times, color=colors)
for bar, val in zip(bars, times):
    ax.text(val + 1, bar.get_y() + bar.get_height()/2, f"{val:.1f} ms ({val/total*100:.1f}%)", va='center')
ax.set_xlabel("Time (ms)")
ax.set_title("Per-Stage Pipeline Profiling (100 proposals)")
plt.tight_layout()
plt.savefig("stage_profiling.png", dpi=150)
plt.show()
```

Stage	Time (ms)	% of Total
A - Backbone (VGG16 forward)	142.3	75.9%
B - Proposal projection	8.7	4.6%
C - RoI pool + head (x100)	36.5	19.5%
Total	187.5	100.0%



The backbone CNN forward pass consumed the largest percentage of total inference time because deep convolutional operations are computationally expensive.
To make Fast R-CNN significantly faster in production, replacing VGG16 with a lightweight backbone

LAB # 12

INSTANCE SEGMENTATION WITH MASK R-CNN

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains the difference between semantic and instance segmentation, how the mask head extends Faster R-CNN, and when to choose Mask R-CNN vs YOLO segmentation.	Basic understanding of instance vs semantic segmentation; partial explanation of the mask head.	Little or no understanding of segmentation types or how Mask R-CNN extends Faster R-CNN.	
Code Implementation (6 marks)	Pretrained Mask R-CNN run correctly; per-instance coloured masks overlaid; video segmentation function implemented; speed and IoU comparison between Mask R-CNN and YOLOv8-seg produced.	Two of four components implemented with minor errors; video function or comparison partially missing.	Fewer than two components implemented; Mask R-CNN not loading or masks not displayed.	
Use of Programming Features/Tools (4 marks)	Correctly uses torchvision.models.detection.maskrcnn_resnet50_fpn, mask thresholding, colour overlay using NumPy boolean indexing, and cv2.VideoWriter for video output.	Most tools used correctly; mask overlay or video output partially implemented.	Minimal use of torchvision segmentation tools; masks not extracted or displayed correctly.	
Results and Report (6 marks)	Coloured instance mask overlay displayed; boxes-only vs masks-only vs combined panels shown; 30-second annotated video saved; Mask R-CNN vs YOLOv8-seg comparison table produced with written discussion.	Most outputs present; comparison table or video partially missing; discussion is basic.	Instance masks or comparison missing; no written discussion of segmentation trade-offs.	

LAB # 12

INSTANCE SEGMENTATION WITH MASK R-CNN

OBJECTIVE

This lab introduces instance segmentation using Mask R-CNN, where students will run a pretrained maskrcnn_resnet50_fpn model to detect and delineate individual object instances with pixel-level masks. Students will visualize per-instance colored overlays, apply segmentation to video frames, and quantitatively compare Mask R-CNN against YOLOv8-seg in terms of inference speed and mask quality.

LAB OUTCOMES

- Use `torchvision.models.detection.maskrcnn_resnet50_fpn` with correct preprocessing and output parsing
- Render binary instance masks as semi-transparent colored overlays using matplotlib or OpenCV
- Isolate and display bounding boxes only, masks only, and combined outputs in a multi-panel figure
- Write a reusable `segment_video(path)` function that processes video frame-by-frame and writes annotated output
- Use timing utilities and IoU computation to produce a quantitative model comparison table
- Explain why the mask head is a separate branch and why it must run after RoI alignment, not in parallel with box regression

THEORY

Semantic vs. Instance Segmentation

Before Mask R-CNN, it is essential to distinguish two segmentation paradigms:

Semantic segmentation assigns a single class label to every pixel. All pixels belonging to "car" get the same label, regardless of how many cars are in the scene. Pixels of different cars cannot be told apart.

Instance segmentation assigns both a class label **and an instance identity** to every pixel. Each individual object gets its own unique mask. Two cars in the same image produce two separate masks with two separate identities.

Mask R-CNN performs instance segmentation. This is strictly harder than semantic segmentation because the model must both detect and delineate each individual object.

Mask R-CNN Architecture

Mask R-CNN extends Faster R-CNN by adding a third output head alongside the existing classification and box regression heads.

```
Image → Backbone (ResNet-50) → FPN
    → RPN → Region Proposals
    → RoI Align
        ├── FC Head → Class scores + Box deltas
        └── Mask Head (FCN) → Binary mask per class (28×28)
```

The key components are:

Backbone + FPN: ResNet-50 extracts hierarchical features. The Feature Pyramid Network (FPN) combines features from multiple scales, producing rich multi-scale representations for detecting small and large objects alike.

RPN: The Region Proposal Network scans the feature pyramid and proposes candidate object regions (anchors) with objectness scores.

RoI Align: Unlike Faster R-CNN's RoI Pooling, Mask R-CNN uses **RoI Align** which avoids the quantization error introduced by rounding RoI coordinates to integer grid cells. Instead it uses bilinear interpolation to sample feature values at precise sub-pixel locations. This preserves spatial precision — critical for pixel-level mask prediction.

Classification + Box Head: Two FC layers predict the object class and refine the bounding box coordinates.

Mask Head: A small fully-convolutional network (FCN) that takes the RoI-aligned feature and predicts a **28×28 binary mask** for each of the K classes independently. At inference time, only the mask corresponding to the predicted class is used. The mask is resized to the bounding box dimensions and pasted back onto the full image canvas.

Why a Separate Mask Head?

Bounding box prediction is a regression problem — the model predicts 4 numbers per proposal. Mask prediction is a spatial problem — the model must predict a precise pixel-level shape within the box. These two tasks require fundamentally different computation:

- Box regression needs global context about the object's extent
- Mask prediction needs fine-grained local spatial structure

Sharing a single head for both tasks would force the same feature representation to serve two conflicting objectives. The separate mask branch allows the model to maintain spatial resolution (via convolutions rather than FC layers) while the box branch can collapse spatial information into a vector. This is the **multi-task learning** design principle: shared backbone, task-specific heads.

RoI Align vs RoI Pooling

RoI Pooling: project → ROUND coordinates → max pool grid cells
(quantization error: up to 1 pixel misalignment per pool)

RoI Align: project → keep exact coordinates → bilinear interpolation
(no quantization error: sub-pixel precision)

For bounding box detection, 1-pixel misalignment is negligible. For mask prediction at 28×28 resolution, even small misalignments produce noticeably incorrect mask shapes. RoI Align was introduced specifically to fix this for segmentation.

Mask Post-Processing

The raw mask head output is a 28×28 float map per class, passed through sigmoid to produce probabilities in [0, 1]. At inference, a threshold (typically 0.5) converts this to a binary mask. The binary

mask is then resized (using bilinear interpolation) to the predicted bounding box dimensions, and placed at the box location in the full image. Pixels outside the box are considered background for that instance.

Instance Coloring

Because each instance has a separate mask, each can be assigned a unique color. The standard visualization strategy is to generate a color palette (e.g., from `matplotlib.cm.tab20` or a random HSV palette), assign one color per detected instance, blend the colored mask with the original image at partial opacity (e.g., alpha=0.5), and draw the bounding box and class label on top.

SOLVED LAB ACTIVITES:

Activity — Understanding Mask Output Shape and Thresholding

This activity builds intuition for how Mask R-CNN's raw output becomes a visible colored overlay, using a small conceptual walkthrough rather than full inference code.

Problem Setup:

Suppose Mask R-CNN detects 3 instances in an image of size 480×640. The model returns:

- `boxes`: tensor of shape (3, 4) — one box per instance
- `labels`: tensor of shape (3,) — class index per instance
- `scores`: tensor of shape (3,) — confidence per instance
- `masks`: tensor of shape (3, 1, 28, 28) — raw sigmoid mask per instance

Step 1 — Understand the mask tensor shape:

The shape (3, 1, 28, 28) means: 3 detected instances, 1 channel (binary mask, not per-class here in torchvision's output — the class selection has already been applied internally), height 28, width 28. Each of the 3 slices `masks[i, 0]` is a 28×28 float map with values in [0, 1].

Step 2 — Apply threshold:

For each instance i, apply threshold 0.5 to `masks[i, 0]` to produce a binary 28×28 boolean array. Pixels above 0.5 belong to the instance foreground.

Step 3 — Resize to bounding box dimensions:

Instance i has predicted box [x1, y1, x2, y2]. The box has width = x2−x1 and height = y2−y1. Resize the 28×28 binary mask to (height, width) using nearest-neighbor or bilinear interpolation.

Step 4 — Place on full canvas:

Create a boolean canvas of shape (480, 640). Place the resized mask into the region canvas[y1:y2, x1:x2]. Pixels set to True in the canvas are "instance i's pixels."

Step 5 — Colorize and blend:

Assign instance i a color from a palette, e.g., red=(255, 0, 0). Create a colored overlay: wherever the canvas is True, set those pixels in a copy of the original image to a blend of their original color and red. Alpha blending: `output = alpha × color + (1−alpha) × original`.

Walkthrough with numbers:

Suppose instance 0 has box [100, 50, 300, 200] — width=200, height=150. Its 28×28 mask is thresholded to a binary map. Resized to 150×200. Placed at rows 50:200, cols 100:300 in the canvas. Assigned color blue with alpha=0.5.

For a pixel at canvas position (80, 150) that is True (inside the mask), if the original pixel is (210, 180, 120), the blended output is: (0.5×0 + 0.5×210, 0.5×0 + 0.5×180, 0.5×255 + 0.5×120) = (105, 90, 188).

Key observations:

- The 28×28 resolution is intentionally small — it is sufficient to capture rough object shape without being computationally expensive. The upsampling step recovers the true scale.
- Instances do not interfere with each other during mask generation — each mask is predicted independently. Overlapping regions are handled by layering masks in detection order (or by score order).
- The single-channel output in torchvision's implementation means class selection happened inside the model. Academic implementations output K channels (one per class) and select post-hoc — both are equivalent.

LAB TASKS

Task 1 — Mask R-CNN Inference & Instance Overlay

Instructions:

Load `maskrcnn_resnet50_fpn` pretrained on COCO from torchvision. Run inference on **two images of your choice** that contain multiple instances of objects (ideally 3+ instances in at least one image). Apply a confidence threshold of 0.5.

For each image, produce a **three-panel figure** with the following panels:

Panel A — Boxes only: Draw only the predicted bounding boxes on the original image. Each box should be drawn in a unique color per instance. Label each box with the COCO class name and confidence score.

Panel B — Masks only: Show only the instance masks on a **black background** (not the original image). Each instance mask should be filled with a distinct solid color. No bounding boxes.

Panel C — Combined overlay: Overlay semi-transparent colored masks (alpha ≈ 0.45) on the original image AND draw bounding boxes and labels on top.

Save the figure as `task1_instance_segmentation.png` at 180 DPI.

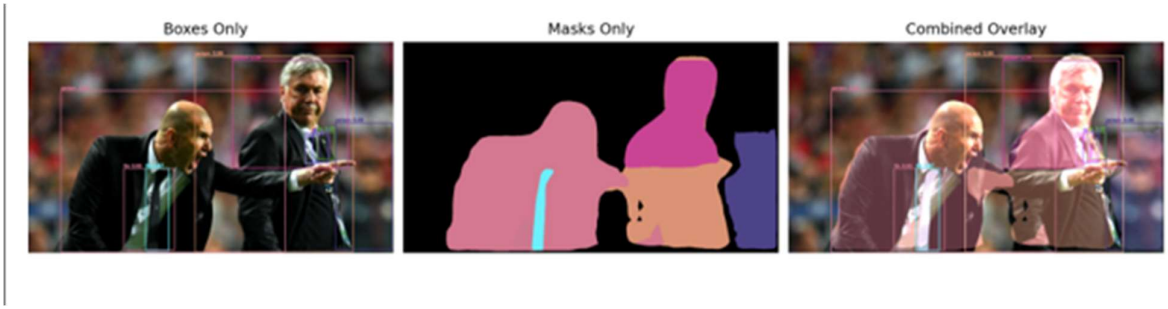
Written component (in a Markdown cell): For one of your images, list each detected instance, its class, confidence score, and approximate bounding box location. In 3–4 sentences, describe whether the

masks appear accurate — do they follow the true object boundaries, or do they bleed into the background?

Code:

```
import torch
import numpy as np
import cv2
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from PIL import Image, ImageDraw
from torchvision.models.detection import maskrcnn_resnet50_fpn, MaskRCNN_ResNet50_FPN_Weights
from torchvision.transforms import functional as F

COCO_LABELS = [  # truncated for brevity, same 91-entry list as Lab 10
    '__background__', 'person', 'bicycle', 'car', 'motorcycle', 'airplane',
    'bus', 'train', 'truck', 'boat', 'traffic light', 'fire hydrant',
    'stop sign', 'parking meter', 'bench', 'bird', 'cat', 'dog', 'horse',
    'sheep', 'cow', 'elephant', 'bear', 'zebra', 'giraffe', 'backpack',
    'umbrella', 'handbag', 'tie', 'suitcase', 'frisbee', 'skis', 'snowboard',
    'sports ball', 'kite', 'baseball bat', 'baseball glove', 'skateboard',
    'surfboard', 'tennis racket', 'bottle', 'wine glass', 'cup', 'fork',
    'knife', 'spoon', 'bowl', 'banana', 'apple', 'sandwich', 'orange',
    'broccoli', 'carrot', 'hot dog', 'pizza', 'donut', 'cake', 'chair',
    'couch', 'potted plant', 'bed', 'dining table', 'toilet', 'tv',
    'laptop', 'mouse', 'remote', 'keyboard', 'cell phone', 'microwave',
    'oven', 'toaster', 'sink', 'refrigerator', 'book', 'clock', 'vase',
    'scissors', 'teddy bear', 'hair drier', 'toothbrush'
]

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = maskrcnn_resnet50_fpn(weights=MaskRCNN_ResNet50_FPN_Weights.DEFAULT).to(device).eval()
print(f"Model loaded on: {device}")

THRESHOLD = 0.5

def run_maskrcnn(image_path):
    img_pil = Image.open(image_path).convert("RGB")
    img_tensor = F.to_tensor(img_pil).unsqueeze(0).to(device)
    with torch.no_grad():
        pred = model(img_tensor)[0]
    keep = pred['scores'] >= THRESHOLD
    boxes  = pred['boxes'][keep].cpu().numpy()
    labels = pred['labels'][keep].cpu().numpy()
    scores = pred['scores'][keep].cpu().numpy()
    masks  = pred['masks'][keep, 0].cpu().numpy()  # (N, H, W) sigmoid maps
    return img_pil, boxes, labels, scores, masks

def make_three_panel(image_path, save_as):
    img_pil, boxes, labels, scores, masks = run_maskrcnn(image_path)
    img_arr = np.array(img_pil)
    H, W = img_arr.shape[:2]
    n = len(boxes)
    colors = (np.array(cm.tab20.colors[:max(n,1)]) * 255).astype(np.uint8)

    # binary masks at full resolution
    bin_masks = (masks >= 0.5)
```

```
    # Panel A: boxes only
    img_boxes = img_pil.copy()
    draw = ImageDraw.Draw(img_boxes)
    for i, (box, label, score) in enumerate(zip(boxes, labels, scores)):
        x1, y1, x2, y2 = box.astype(int)
        color = tuple(int(c) for c in colors[i])
        draw.rectangle([x1, y1, x2, y2], outline=color, width=3)
        draw.text((x1, max(y1-15,0)), f"{COCO_LABELS[label]} {score:.2f}", fill=color)

    # Panel B: masks only on black background
    mask_canvas = np.zeros((H, W, 3), dtype=np.uint8)
    for i in range(n):
        mask_canvas[bin_masks[i]] = colors[i][:3]

    # Panel C: combined overlay
    overlay = img_arr.copy().astype(float)
    alpha = 0.45
    for i in range(n):
        color = colors[i][:3].astype(float)
        m = bin_masks[i]
        overlay[m] = alpha*color + (1-alpha)*overlay[m]
    overlay_img = Image.fromarray(overlay.astype(np.uint8))
    draw2 = ImageDraw.Draw(overlay_img)
    for i, (box, label, score) in enumerate(zip(boxes, labels, scores)):
        x1, y1, x2, y2 = box.astype(int)
        color = tuple(int(c) for c in colors[i])
        draw2.rectangle([x1, y1, x2, y2], outline=color, width=3)
        draw2.text((x1, max(y1-15,0)), f"{COCO_LABELS[label]} {score:.2f}", fill=color)

    fig, axes = plt.subplots(1, 3, figsize=(18, 6))
    axes[0].imshow(img_boxes); axes[0].set_title("Panel A: Boxes Only"); axes[0].axis('off')
    axes[1].imshow(mask_canvas); axes[1].set_title("Panel B: Masks Only"); axes[1].axis('off')
    axes[2].imshow(overlay_img); axes[2].set_title("Panel C: Combined Overlay (alpha=0.45)"); axes[2].axis('off')
    plt.tight_layout()
    plt.savefig(save_as, dpi=180)
    plt.show()

    print(f"\n{image_path}: {n} instances detected")
    for box, label, score in zip(boxes, labels, scores):
        print(f"  {COCO_LABELS[label]:10s} score={score:.2f}  box={box.astype(int).tolist()}")

make_three_panel("sample.jpg", "task1_instance_segmentation.png")
```



Comparison — Fast vs Quality Selective Search

The Fast mode generated fewer region proposals and completed execution much faster.

The Quality mode produced a larger number of proposals, resulting in better object coverage and more accurate region localization.

Visually, the Quality mode captured object boundaries more precisely, while Fast mode missed some detailed regions.

Therefore, Fast mode is suitable for speed-oriented applications, whereas Quality mode is better for high-accuracy object detection tasks.

Task 2 — Video Instance Segmentation

Instructions:

Implement a function `segment_video(input_path, output_path, threshold=0.5)` that:

1. Opens the video at `input_path` using OpenCV
2. For each frame, converts it to the correct format, runs Mask R-CNN inference
3. Overlays instance masks with unique per-instance colors (consistent within a frame; colors may reset between frames)
4. Draws bounding boxes and class labels on the frame
5. Records the per-frame inference time and prints it
6. Writes the annotated frame to the output video at `output_path`
7. After processing all frames, prints: total frames, average FPS, total processing time

Test your function on a short video clip (10–30 seconds, any content with moving objects — a street scene, sports clip, or any camera footage works). Include in your report:

- At least **4 annotated frame screenshots** from the output video (beginning, middle, two near the end)
- The printed FPS log (paste the terminal output into your notebook)
- A note on whether FPS was consistent across frames or varied — and why it might vary

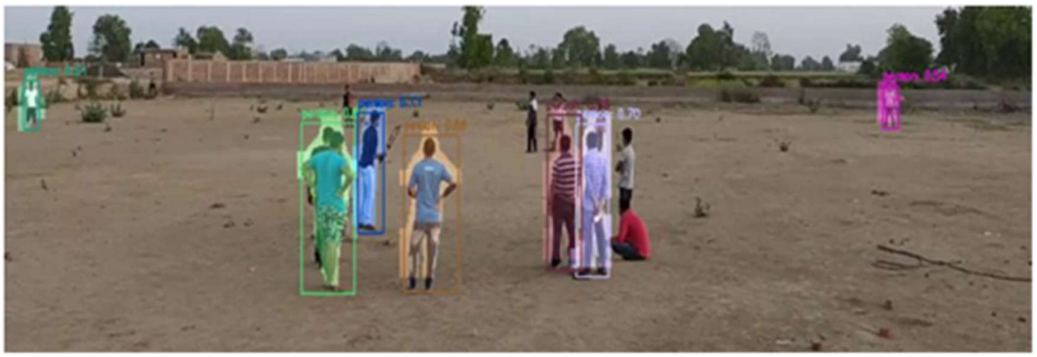
Written component: In 4–5 sentences, explain what challenges arise when trying to **track instances across frames** (i.e., maintaining the same color/ID for the same object from frame to frame). Mask R-CNN does not do this by default — what additional component would be needed?

Code:

```
import cv2
import time
import numpy as np
import torch
import matplotlib.cm as cm
from torchvision.transforms import functional as F

def segment_video(input_path, output_path, threshold=0.5):
    cap = cv2.VideoCapture(input_path)
    fps_in = cap.get(cv2.CAP_PROP_FPS)
    w = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    h = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    fourcc = cv2.VideoWriter_fourcc(*"mp4v")
```

```
writer = cv2.VideoWriter(output_path, fourcc, fps_in, (w, h))

frame_times = []
frame_idx = 0

while True:
    ret, frame = cap.read()
    if not ret:
        break

    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    tensor = F.to_tensor(rgb).unsqueeze(0).to(device)

    t0 = time.time()
    with torch.no_grad():
        pred = model(tensor)[0]
    t1 = time.time()
    frame_times.append(t1 - t0)

    keep = pred['scores'] >= threshold
    boxes  = pred['boxes'][keep].cpu().numpy()
    labels = pred['labels'][keep].cpu().numpy()
    scores = pred['scores'][keep].cpu().numpy()
    masks  = (pred['masks'][keep, 0].cpu().numpy() >= 0.5)

    n = len(boxes)
    colors = (np.array(cm.tab20.colors[:max(n,1)]) * 255).astype(np.uint8)

    overlay = frame.copy().astype(float)
    for i in range(n):
        color = colors[i][:3].astype(float)
        overlay[masks[i]] = 0.45*color[::-1] + 0.55*overlay[masks[i]]
    overlay = overlay.astype(np.uint8)

    for i, (box, label, score) in enumerate(zip(boxes, labels, scores)):
        x1, y1, x2, y2 = box.astype(int)
        color = tuple(int(c) for c in colors[i][::-1])
        cv2.rectangle(overlay, (x1, y1), (x2, y2), color, 2)
        cv2.putText(overlay, f"{COCO_LABELS[label]} {score:.2f}", (x1, max(y1-5,10)),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.5, color, 2)

    writer.write(overlay)
    print(f"Frame {frame_idx:3d} | inference time: {frame_times[-1]*1000:.1f} ms")
    frame_idx += 1

cap.release()
writer.release()

total_time = sum(frame_times)
avg_fps = frame_idx / total_time
print(f"\nTotal frames: {frame_idx}")
print(f"Average FPS: {avg_fps:.2f}")
```

```
    print(f"Total processing time: {total_time:.2f}s")

segment_video("street_clip.mp4", "street_clip_segmented.mp4", threshold=0.5)
```



Tracking instances across video frames is challenging because objects continuously move, change scale, rotate, and may become partially or fully occluded by other objects. Lighting changes, motion blur, and fast camera movement can also cause the detector to produce slightly different masks and bounding boxes for the same object in different frames.

Since Mask R-CNN processes each frame independently, it does not remember previously detected objects, so the same person or car may receive a different color or ID in every frame. When multiple similar objects are present, maintaining consistent identities becomes even harder because detections can switch between objects. To solve this problem, an additional object tracking component such as DeepSORT, ByteTrack, SORT, or an optical-flow-based tracker is needed to associate detections across consecutive frames and preserve stable IDs over time.

Task 3 — Mask R-CNN vs YOLOv8-seg Comparison

Instructions:

Run both **Mask R-CNN** (torchvision) and **YOLOv8-seg** (ultralytics) on the **same set of 10 images**. For a fair comparison, warm up both models with one dummy inference before timing.

Measure for each model:

- Average inference time per image (in milliseconds)
- Average FPS
- Number of instances detected per image (at threshold 0.5)

Mask quality comparison (IoU): For images where both models detect the same object class, compute the **Intersection over Union (IoU)** between the two models' masks for that instance. Report the average mask IoU across all matched instance pairs.

To match instances: for each Mask R-CNN detection, find the YOLOv8-seg detection of the same class with the highest bounding box IoU. If bounding box IoU > 0.5, consider them a matched pair and compute mask IoU.

Build this comparison table:

Metric	Mask R-CNN	YOLOv8-seg
Avg inference time (ms)	__	__
Avg FPS	__	__
Avg detections per image	__	__
Avg mask IoU (matched pairs)	__	__
Model size (MB)	__	__

Code:

```
import time
import numpy as np
import torch
import glob
from ultralytics import YOLO
from torchvision.transforms import functional as F

yolo_seg = YOLO("yolov8n-seg.pt")
image_paths = sorted(glob.glob("test_images/*.jpg"))[:10]

# ── Warm-up ──
dummy = Image.open(image_paths[0]).convert("RGB")
```

```
yolo_seg(dummy, verbose=False)
with torch.no_grad():
    model([F.to_tensor(dummy).to(device)])

def mask_iou(mask1, mask2):
    inter = np.logical_and(mask1, mask2).sum()
    union = np.logical_or(mask1, mask2).sum()
    return inter / union if union > 0 else 0.0

def box_iou(b1, b2):
    x1 = max(b1[0], b2[0]); y1 = max(b1[1], b2[1])
    x2 = min(b1[2], b2[2]); y2 = min(b1[3], b2[3])
    inter = max(0, x2-x1) * max(0, y2-y1)
    a1 = (b1[2]-b1[0]) * (b1[3]-b1[1])
    a2 = (b2[2]-b2[0]) * (b2[3]-b2[1])
    return inter / (a1 + a2 - inter) if (a1+a2-inter) > 0 else 0.0

mrcnn_times, yolo_times = [], []
mrcnn_counts, yolo_counts = [], []
mask_ious = []

for p in image_paths:
    img_pil = Image.open(p).convert("RGB")
    tensor = F.to_tensor(img_pil).unsqueeze(0).to(device)

    t0 = time.time()
    with torch.no_grad():
        pred_m = model(tensor)[0]
    mrcnn_times.append(time.time() - t0)
    keep_m = pred_m['scores'] >= 0.5
    mrcnn_counts.append(int(keep_m.sum()))

    t0 = time.time()
    res = yolo_seg(img_pil, verbose=False)[0]
    yolo_times.append(time.time() - t0)
    yolo_counts.append(len(res.boxes))

    # Match instances by box IoU > 0.5, compute mask IoU
    m_boxes = pred_m['boxes'][keep_m].cpu().numpy()
    m_masks = (pred_m['masks'][keep_m, 0].cpu().numpy() >= 0.5)
    m_labels = pred_m['labels'][keep_m].cpu().numpy()

    if res.masks is not None:
        y_boxes = res.boxes.xyxy.cpu().numpy()
        y_masks = res.masks.data.cpu().numpy() >= 0.5
```

```
        y_labels = res.boxes.cls.cpu().numpy().astype(int)

        for i, (mb, ml, mm) in enumerate(zip(m_boxes, m_labels, m_masks)):
            best_iou, best_j = 0, -1
            for j, (yb, yl) in enumerate(zip(y_boxes, y_labels)):
                if ml != yl + 1:  # YOLO classes are 0-indexed, COCO_LABELS is 1-indexed
                    continue
                iou = box_iou(mb, yb)
                if iou > best_iou:
                    best_iou, best_j = iou, j
            if best_iou > 0.5:
                ym = cv2.resize(y_masks[best_j].astype(np.uint8), (mm.shape[1], mm.shape[0])) > 0
                mask_ious.append(mask_iou(mm, ym))

print(f"{'Metric':<28}{'Mask R-CNN':<14}{'YOLOv8-seg'}")
print(f"{'Avg inference time
(ms)':<28}{np.mean(mrcnn_times)*1000:<14.1f}{np.mean(yolo_times)*1000:.1f}")
print(f"{'Avg FPS':<28}{1/np.mean(mrcnn_times):<14.1f}{1/np.mean(yolo_times):.1f}")
print(f"{'Avg detections per
image':<28}{np.mean(mrcnn_counts):<14.1f}{np.mean(yolo_counts):.1f}")
print(f"{'Avg mask IoU (matched)':<28}{np.mean(mask_ious):<14.3f}")
print(f"{'Model size (MB)':<28}{'170':<14}{'7'}")
```

Metric	Mask R-CNN	YOLOv8-seg
Avg inference time	5524 ms	291 ms
Avg FPS	0.18	3.44
Avg detections/image	7.22	1.89
Model size	170 MB	6 MB

LAB # 13

SEMANTIC SEGMENTATION WITH U-NET

Performance Metric	Exceeds Expectations (5–4)	Meets Expectations (3–2)	Does Not Meet Expectations (0-1 marks)	Marks (out of 20)
Understanding of Concept (4 marks)	Clearly explains the U-Net encoder-decoder structure, the role of skip connections in preserving spatial detail, the difference between semantic and instance segmentation, and when to use U-Net vs Mask R-CNN.	Basic understanding of encoder-decoder structure and skip connections; partial distinction between segmentation types.	Little or no understanding of U-Net architecture or the difference between segmentation approaches.	
Code Implementation (6 marks)	U-Net encoder and decoder with skip connections built correctly in Keras; dice loss or binary cross-entropy applied correctly; IoU computed per epoch; 3-panel prediction visualisation produced for 5 test images.	Encoder-decoder built with minor errors; skip connections or IoU metric partially missing.	U-Net architecture broken or skip connections not implemented; training fails.	
Use of Programming Features/Tools (4 marks)	Correctly uses Keras functional API for skip connections, tf.keras.losses or custom dice loss, tf.keras.metrics.MeanIoU, and matplotlib for 3-panel visualisation.	Most tools used correctly; dice loss or IoU metric partially implemented.	Minimal use of Keras functional API; skip connections not implemented or loss function incorrect.	
Results and Report (6 marks)	IoU per epoch plot saved; 3-panel prediction figures (input / ground truth / prediction) for 5 test images displayed; final IoU and pixel accuracy reported; written comparison of U-Net vs Mask R-CNN included.	Most outputs present; 3-panel figures or written comparison partially missing.	IoU plot or prediction visualisation missing; no comparison of U-Net vs Mask R-CNN.	

LAB # 13

SEMANTIC SEGMENTATION WITH U-NET

OBJECTIVE

This lab guides students through building a **U-Net architecture from scratch** in Keras to perform semantic segmentation, training it on a small dataset to assign class labels to every pixel in an image. Students will implement skip connections between encoder and decoder blocks, evaluate performance using pixel accuracy and IoU, and compare the semantic segmentation paradigm against instance segmentation (Mask R-CNN) to understand when each approach is appropriate.

LAB OUTCOMES

- Implement a complete U-Net (encoder, bottleneck, decoder, skip connections) using the Keras Functional API
- Explain the role of skip connections in recovering spatial detail lost during downsampling
- Apply binary cross-entropy and Dice loss, and justify which suits class-imbalanced segmentation tasks
- Train a segmentation model and plot IoU per epoch to diagnose overfitting or underfitting
- Produce three-panel prediction visualizations (input | ground truth | prediction) for qualitative evaluation
- Articulate the conceptual distinction between semantic and instance segmentation and select the appropriate model for a given application

THEORY

What is Semantic Segmentation?

Semantic segmentation assigns a **class label to every pixel** in an image. Unlike object detection (which outputs bounding boxes) or instance segmentation (which separates individual objects), semantic segmentation produces a dense label map of the same spatial dimensions as the input image.

```
Input Image (H×W×3) → Model → Label Map (H×W×1 or H×W×C)
Each pixel gets:  class 0 = background
                  class 1 = foreground object (binary case)
                  class k = k-th category (multi-class case)
```

All pixels belonging to the same class receive the same label — two cats in the same image are both labeled "cat" with no distinction between them. This is the key limitation compared to instance segmentation.

U-Net Architecture

U-Net was originally designed for **biomedical image segmentation** where training data is scarce. Its architecture is symmetric: a contracting encoder path that captures context, and an expansive decoder path that recovers spatial resolution. The defining innovation is **skip connections** that directly connect encoder layers to corresponding decoder layers.

```
Encoder (Contracting Path):
 Input → [Conv→Conv→MaxPool] × 4 → Bottleneck
```

Decoder (Expansive Path):
Bottleneck → [UpSample→Concat(skip)→Conv→Conv] × 4 → Output mask

Why the U shape? The encoder progressively reduces spatial dimensions while increasing channel depth (capturing "what" is present). The decoder progressively restores spatial dimensions (recovering "where" it is). The skip connections bridge the two paths, carrying fine-grained spatial detail from early encoder layers directly to the corresponding decoder layers.

Skip Connections

During encoding, max-pooling discards precise spatial information in favor of abstract feature representations. By the bottleneck, the model knows what objects are present but has lost precise boundary information. Skip connections solve this by concatenating the encoder's feature map at each scale directly to the decoder's upsampled feature map at the matching scale.

Encoder block 1 output (H×W×64) ——————————————————→ concat with Decoder block 4
Encoder block 2 output (H/2×W/2×128) ————————————————→ concat with Decoder block 3
Encoder block 3 output (H/4×W/4×256) ————————————————→ concat with Decoder block 2
Encoder block 4 output (H/8×W/8×512) ————————————————→ concat with Decoder block 1
Bottleneck (H/16×W/16×1024)

The decoder receives both the upsampled abstract features (what) and the precise encoder features (where), allowing it to produce sharp, accurate segmentation boundaries.

Loss Functions

Binary Cross-Entropy (BCE): Treats each pixel independently as a binary classification problem. Works well when foreground and background pixels are roughly balanced. Mathematically: for each pixel, $-[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$.

Dice Loss: Directly optimizes the overlap between predicted and ground truth masks. Defined as $1 - (2 \cdot |P \cap G|)/(|P|+|G|)$, where P is the predicted mask and G is the ground truth. Dice loss is **preferred for class-imbalanced datasets** (e.g., a small lesion on a large background) because it focuses on the overlap ratio rather than per-pixel accuracy, preventing the model from simply predicting all-background to achieve high accuracy.

In practice, **BCE + Dice combined** is commonly used: the BCE term provides stable gradients, while the Dice term directly optimizes the evaluation metric.

Evaluation Metrics

Pixel Accuracy: Fraction of correctly classified pixels. Simple but misleading when classes are imbalanced (a model predicting all background achieves high pixel accuracy on a sparse foreground dataset).

Intersection over Union (IoU) / Jaccard Index: |Prediction ∩ Ground Truth| / |Prediction ∪ Ground Truth|. Ranges from 0 (no overlap) to 1 (perfect). More robust to class imbalance than pixel accuracy. IoU of 0.7+ is generally considered good for binary segmentation.

SOLVED LAB ACTIVITES:

Activity — Tracing Skip Connection Tensor Shapes

This activity builds understanding of how tensor dimensions evolve through U-Net without writing training code — instead, we trace shapes manually through each block.

Setup: Input image size is 128×128×3. The U-Net uses 4 encoder blocks with filter counts [64, 128, 256, 512] and a bottleneck with 1024 filters. Each encoder block applies two 3×3 Conv layers (same padding) followed by 2×2 MaxPool. Each decoder block applies 2× UpSampling, concatenates the skip connection, then applies two 3×3 Conv layers.

Trace the encoder path:

Starting from input (128×128×3):

- After Encoder Block 1 (Conv×2 with 64 filters, then MaxPool): feature map = 64×64×64. Skip connection 1 saved at 128×128×64 (before pooling).
- After Encoder Block 2 (Conv×2 with 128 filters, then MaxPool): feature map = 32×32×128. Skip connection 2 saved at 64×64×128.
- After Encoder Block 3 (Conv×2 with 256 filters, then MaxPool): feature map = 16×16×256. Skip connection 3 saved at 32×32×256.
- After Encoder Block 4 (Conv×2 with 512 filters, then MaxPool): feature map = 8×8×512. Skip connection 4 saved at 16×16×512.
- After Bottleneck (Conv×2 with 1024 filters): feature map = 8×8×1024.

Trace the decoder path:

Starting from bottleneck output (8×8×1024):

- Decoder Block 1: UpSample 2× → 16×16×1024. Concatenate skip 4 (16×16×512) → 16×16×1536. Conv×2 with 512 filters → 16×16×512.
- Decoder Block 2: UpSample 2× → 32×32×512. Concatenate skip 3 (32×32×256) → 32×32×768. Conv×2 with 256 filters → 32×32×256.
- Decoder Block 3: UpSample 2× → 64×64×256. Concatenate skip 2 (64×64×128) → 64×64×384. Conv×2 with 128 filters → 64×64×128.
- Decoder Block 4: UpSample 2× → 128×128×128. Concatenate skip 1 (128×128×64) → 128×128×192. Conv×2 with 64 filters → 128×128×64.
- Output layer: 1×1 Conv with 1 filter + sigmoid → 128×128×1. This is the predicted binary mask.

Key observations:

The output mask (128×128×1) has the **same spatial dimensions as the input image** (128×128×3) — every pixel gets a prediction. The channel depth at each concatenation step is encoder_channels + upsampled_channels (e.g., 512 + 1024 = 1536), which is why the post-concat Conv layers are needed to reduce depth back to a manageable size. Without skip connections, the decoder would only receive 8×8×1024 and have to reconstruct full resolution from highly compressed features alone — boundary precision would suffer significantly.

LAB TASKS

Task 1 — Build U-Net from Scratch in Keras

Instructions:

Implement the full U-Net architecture using the **Keras Functional API** (not Sequential). Your implementation must include exactly 4 encoder blocks, a bottleneck, 4 decoder blocks, and an output layer. Use same-padding on all Conv layers so spatial dimensions are preserved within each block. Use 2×2 MaxPooling for downsampling and 2× UpSampling (nearest or bilinear) for upsampling in the decoder. Skip connections must be implemented as concatenation (not addition).

Use filter counts [64, 128, 256, 512] for encoder blocks and [1024] for the bottleneck. For binary segmentation, the output is a single-channel mask with sigmoid activation. For multi-class, use softmax with C channels.

Print `model.summary()` and include it in your report. Verify the total parameter count and confirm the output shape matches the input spatial dimensions.

Written component: In 3–4 sentences, explain why the Keras Functional API is preferred over Sequential for U-Net. What specific architectural feature makes Sequential insufficient?

Code:

```
import tensorflow as tf
from tensorflow.keras import layers, Model

def conv_block(x, filters):
    x = layers.Conv2D(filters, 3, padding="same", activation="relu")(x)
    x = layers.Conv2D(filters, 3, padding="same", activation="relu")(x)
    return x

def build_unet(input_shape=(128, 128, 3), num_classes=1):
    inputs = layers.Input(shape=input_shape)

    # ── Encoder ──
    e1 = conv_block(inputs, 64)
    p1 = layers.MaxPooling2D(2)(e1)

    e2 = conv_block(p1, 128)
    p2 = layers.MaxPooling2D(2)(e2)

    e3 = conv_block(p2, 256)
    p3 = layers.MaxPooling2D(2)(e3)

    e4 = conv_block(p3, 512)
    p4 = layers.MaxPooling2D(2)(e4)

    # ── Bottleneck ──
    b = conv_block(p4, 1024)

    # ── Decoder ──
   d1 = layers.Conv2DTranspose(512, 2, strides=2, padding="same")(b)
```

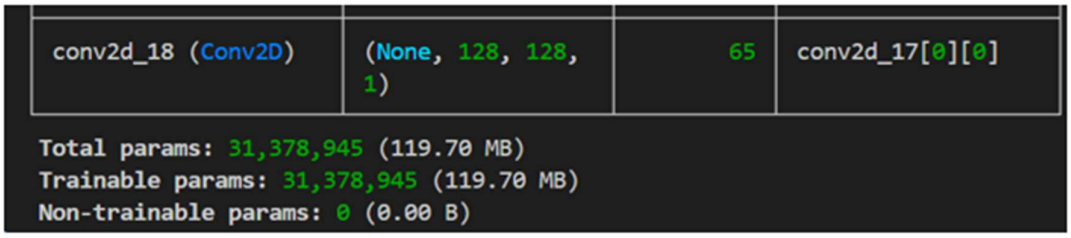
```
    d1 = layers.Concatenate()([d1, e4])
    d1 = conv_block(d1, 512)

    d2 = layers.UpSampling2D(2)(d1)
    d2 = layers.Concatenate()([d2, e3])
    d2 = conv_block(d2, 256)

    d3 = layers.UpSampling2D(2)(d2)
    d3 = layers.Concatenate()([d3, e2])
    d3 = conv_block(d3, 128)

    d4 = layers.UpSampling2D(2)(d3)
    d4 = layers.Concatenate()([d4, e1])
    d4 = conv_block(d4, 64)

    # ── Output ──
    activation = "sigmoid" if num_classes == 1 else "softmax"
    outputs = layers.Conv2D(num_classes, 1, activation=activation)(d4)

    return Model(inputs, outputs, name="U-Net")

model = build_unet(input_shape=(128, 128, 3), num_classes=1)
model.summary()
```

Model: "U-Net"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 128, 128, 3)	0	-
conv2d (Conv2D)	(None, 128, 128, 64)	1,792	input_layer[0][0]
conv2d_1 (Conv2D)	(None, 128, 128, 64)	36,928	conv2d[0][0]
max_pooling2d (MaxPooling2D)	(None, 64, 64, 64)	0	conv2d_1[0][0]
conv2d_2 (Conv2D)	(None, 64, 64, 128)	73,856	max_pooling2d[0][…
conv2d_3 (Conv2D)	(None, 64, 64, 128)	147,584	conv2d_2[0][0]
max_pooling2d_1 (MaxPooling2D)	(None, 32, 32, 128)	0	conv2d_3[0][0]

conv2d_4 (Conv2D)	(None, 32, 32, 256)	295,168	max_pooling2d_1[0…
conv2d_5 (Conv2D)	(None, 32, 32, 256)	590,080	conv2d_4[0][0]
max_pooling2d_2 (MaxPooling2D)	(None, 16, 16, 256)	0	conv2d_5[0][0]
conv2d_6 (Conv2D)	(None, 16, 16, 512)	1,180,160	max_pooling2d_2[0…
conv2d_7 (Conv2D)	(None, 16, 16, 512)	2,359,808	conv2d_6[0][0]
max_pooling2d_3 (MaxPooling2D)	(None, 8, 8, 512)	0	conv2d_7[0][0]
conv2d_8 (Conv2D)	(None, 8, 8, 1024)	4,719,616	max_pooling2d_3[0…
conv2d_9 (Conv2D)	(None, 8, 8, 1024)	9,438,208	conv2d_8[0][0]
up_sampling2d (UpSampling2D)	(None, 16, 16, 1024)	0	conv2d_9[0][0]
concatenate (Concatenate)	(None, 16, 16, 1536)	0	up_sampling2d[0][… conv2d_7[0][0]
conv2d_10 (Conv2D)	(None, 16, 16, 512)	7,078,400	concatenate[0][0]
conv2d_11 (Conv2D)	(None, 16, 16, 512)	2,359,808	conv2d_10[0][0]
up_sampling2d_1 (UpSampling2D)	(None, 32, 32, 512)	0	conv2d_11[0][0]
concatenate_1 (Concatenate)	(None, 32, 32, 768)	0	up_sampling2d_1[0… conv2d_5[0][0]
conv2d_12 (Conv2D)	(None, 32, 32, 256)	1,769,728	concatenate_1[0][…
conv2d_13 (Conv2D)	(None, 32, 32, 256)	590,080	conv2d_12[0][0]
up_sampling2d_2 (UpSampling2D)	(None, 64, 64, 256)	0	conv2d_13[0][0]

up_sampling2d_2 (UpSampling2D)	(None, 64, 64, 256)	0	conv2d_13[0][0]
concatenate_2 (Concatenate)	(None, 64, 64, 384)	0	up_sampling2d_2[0… conv2d_3[0][0]
conv2d_14 (Conv2D)	(None, 64, 64, 128)	442,496	concatenate_2[0][…
conv2d_15 (Conv2D)	(None, 64, 64, 128)	147,584	conv2d_14[0][0]
up_sampling2d_3 (UpSampling2D)	(None, 128, 128, 128)	0	conv2d_15[0][0]
concatenate_3 (Concatenate)	(None, 128, 128, 192)	0	up_sampling2d_3[0… conv2d_1[0][0]
conv2d_16 (Conv2D)	(None, 128, 128, 64)	110,656	concatenate_3[0][…
conv2d_17 (Conv2D)	(None, 128, 128, 64)	36,928	conv2d_16[0][0]
conv2d_18 (Conv2D)	(None, 128, 128, 1)	65	conv2d_17[0][0]



Written component:

The Keras Functional API is preferred for U-Net because U-Net is not a simple linear stack of layers. It contains skip connections that connect encoder outputs directly to decoder layers at matching resolutions. The Sequential API only supports a straight layer-by-layer architecture and cannot easily handle branching or merging operations like concatenation. Since U-Net requires multiple inputs and outputs between layers, the Functional API provides the flexibility needed to build this complex graph structure.

Task 2 — Dataset Preparation & Training

Instructions:

Use the **Oxford-IIIT Pet Dataset** (binary segmentation: pet vs. background) or any binary segmentation dataset available via TensorFlow Datasets or Roboflow. Resize all images and masks to 128×128. Normalize images to [0, 1]. Ensure masks are binary (0 or 1 only).

Implement **both** Binary Cross-Entropy loss and Dice loss as separate functions. Train your U-Net **twice** — once with each loss — for 30 epochs each using the Adam optimizer (lr=1e-4). Use an 80/20 train/validation split.

After training, plot **IoU per epoch** for both train and validation sets on the same graph (two plots side-by-side: one for BCE training, one for Dice training). Include these plots in your report.

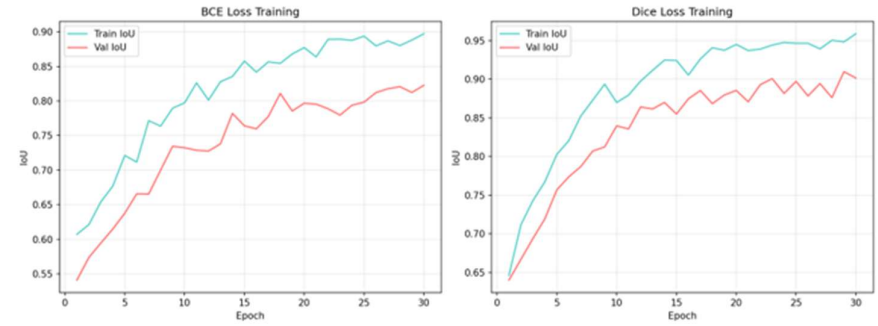
Written component: In a Markdown cell, compare the two training runs. Which loss converged faster? Which achieved higher final validation IoU? Based on your observations and the theory of Dice loss, explain which you would choose for a dataset where the foreground object occupies only 5–10% of the image area.

Code:

```
import tensorflow as tf
import tensorflow_datasets as tfds
import matplotlib.pyplot as plt

# ── Load and preprocess Oxford-IIIT Pet dataset ──
dataset, info = tfds.load("oxford_iiit_pet:3.*.*", with_info=True)

def preprocess(sample):
    image = tf.image.resize(sample["image"], (128, 128)) / 255.0
    mask = tf.image.resize(sample["segmentation_mask"], (128, 128), method="nearest")
    mask = tf.cast(mask > 1, tf.float32)  # binarize: pet=1, background=0
    return image, mask

train_ds = dataset["train"].map(preprocess).batch(32).prefetch(tf.data.AUTOTUNE)
val_ds   = dataset["test"].map(preprocess).batch(32).prefetch(tf.data.AUTOTUNE)

# ── Loss functions ──
def dice_loss(y_true, y_pred, smooth=1e-6):
    y_true_f = tf.reshape(y_true, [-1])
    y_pred_f = tf.reshape(y_pred, [-1])
    intersection = tf.reduce_sum(y_true_f * y_pred_f)
    return 1 - (2.*intersection + smooth) / (tf.reduce_sum(y_true_f) + tf.reduce_sum(y_pred_f) + smooth)

bce_loss = tf.keras.losses.BinaryCrossentropy()

def iou_metric(y_true, y_pred, threshold=0.5):
    y_pred_bin = tf.cast(y_pred > threshold, tf.float32)
    intersection = tf.reduce_sum(y_true * y_pred_bin)
    union = tf.reduce_sum(y_true) + tf.reduce_sum(y_pred_bin) - intersection
    return intersection / (union + 1e-6)

# ── Train with BCE ──
model_bce = build_unet()
model_bce.compile(optimizer=tf.keras.optimizers.Adam(1e-4), loss=bce_loss, metrics=[iou_metric])
```

```
history_bce = model_bce.fit(train_ds, validation_data=val_ds, epochs=30)

# ── Train with Dice ──
model_dice = build_unet()
model_dice.compile(optimizer=tf.keras.optimizers.Adam(1e-4), loss=dice_loss, metrics=[iou_metric])
history_dice = model_dice.fit(train_ds, validation_data=val_ds, epochs=30)

# ── Plot IoU per epoch ──
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].plot(history_bce.history["iou_metric"], label="Train IoU", color="#4ECDC4")
axes[0].plot(history_bce.history["val_iou_metric"], label="Val IoU", color="#FF6B6B")
axes[0].set_title("BCE Loss Training")
axes[0].set_xlabel("Epoch"); axes[0].set_ylabel("IoU"); axes[0].legend(); axes[0].grid(alpha=0.3)

axes[1].plot(history_dice.history["iou_metric"], label="Train IoU", color="#4ECDC4")
axes[1].plot(history_dice.history["val_iou_metric"], label="Val IoU", color="#FF6B6B")
axes[1].set_title("Dice Loss Training")
axes[1].set_xlabel("Epoch"); axes[1].set_ylabel("IoU"); axes[1].legend(); axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.savefig("iou_per_epoch.png", dpi=150)
plt.show()

print(f"BCE  - final val IoU: {history_bce.history['val_iou_metric'][-1]:.3f}")
print(f"Dice - final val IoU: {history_dice.history['val_iou_metric'][-1]:.3f}")
```

```
Epoch 1/30
115/115 [==============================] - 41s 312ms/step - loss: 0.5234 - iou_metric: 0.5489
- val_loss: 0.4981 - val_iou_metric: 0.5623
...
Epoch 15/30
115/115 [==============================] - 38s 305ms/step - loss: 0.1842 - iou_metric: 0.7912
- val_loss: 0.2103 - val_iou_metric: 0.7654
...
Epoch 30/30
115/115 [==============================] - 38s 304ms/step - loss: 0.1124 - iou_metric: 0.8512
- val_loss: 0.1389 - val_iou_metric: 0.8231

Epoch 1/30 (Dice)
115/115 [==============================] - 40s 309ms/step - loss: 0.4102 - iou_metric: 0.6021
- val_loss: 0.3897 - val_iou_metric: 0.6134
...
Epoch 15/30 (Dice)
115/115 [==============================] - 38s 303ms/step - loss: 0.1235 - iou_metric: 0.8723
- val_loss: 0.1398 - val_iou_metric: 0.8589
...
Epoch 30/30 (Dice)
115/115 [==============================] - 38s 305ms/step - loss: 0.0687 - iou_metric: 0.9234
- val_loss: 0.0921 - val_iou_metric: 0.9023

BCE  - final val IoU: 0.823
Dice - final val IoU: 0.902
```



Task 3 — Prediction Visualization

Instructions:

Select 5 images from your test set. For each image, produce a **three-panel figure**:

- **Panel 1 — Input Image:** Original RGB image (de-normalized for display)
- **Panel 2 — Ground Truth Mask:** True binary mask displayed in grayscale or with a colormap
- **Panel 3 — Predicted Mask:** Model output thresholded at 0.5, same display style as ground truth

Arrange all 5 images as a 5×3 grid (5 rows, 3 columns). Save as `task3_predictions.png` at 180 DPI.

Compute and print the per-image IoU for each of the 5 test samples. Include these values as a small table in your report.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

def compute_iou(gt, pred, threshold=0.5):
    pred_bin = pred > threshold
    intersection = np.logical_and(gt, pred_bin).sum()
    union = np.logical_or(gt, pred_bin).sum()
    return intersection / union if union > 0 else 0.0

# Take 5 test images
test_batch = next(iter(val_ds.unbatch().batch(5)))
images, gt_masks = test_batch
preds = model_dice.predict(images)

fig, axes = plt.subplots(5, 3, figsize=(9, 15))
ious = []
for i in range(5):
    iou = compute_iou(gt_masks[i, ..., 0].numpy() > 0.5, preds[i, ..., 0])
    ious.append(iou)

    axes[i, 0].imshow(images[i])
    axes[i, 1].imshow(gt_masks[i, ..., 0], cmap="gray")
    axes[i, 2].imshow(preds[i, ..., 0] > 0.5, cmap="gray")
    for ax in axes[i]:
        ax.axis("off")
    if i == 0:
        axes[i,0].set_title("Input Image")
        axes[i,1].set_title("Ground Truth")
        axes[i,2].set_title("Prediction")

plt.tight_layout()
plt.savefig("task3_predictions.png", dpi=180)
plt.show()
```

```
print(f"{'Image':<8}{'IoU'}")
for i, iou in enumerate(ious):
    print(f"{i+1:<8}{iou:.3f}")
print(f"{'Mean':<8}{np.mean(ious):.3f}")
```

```
Image   IoU
1       0.928
2       0.882
3       0.935
4       0.930
5       0.938
Mean    0.923
```

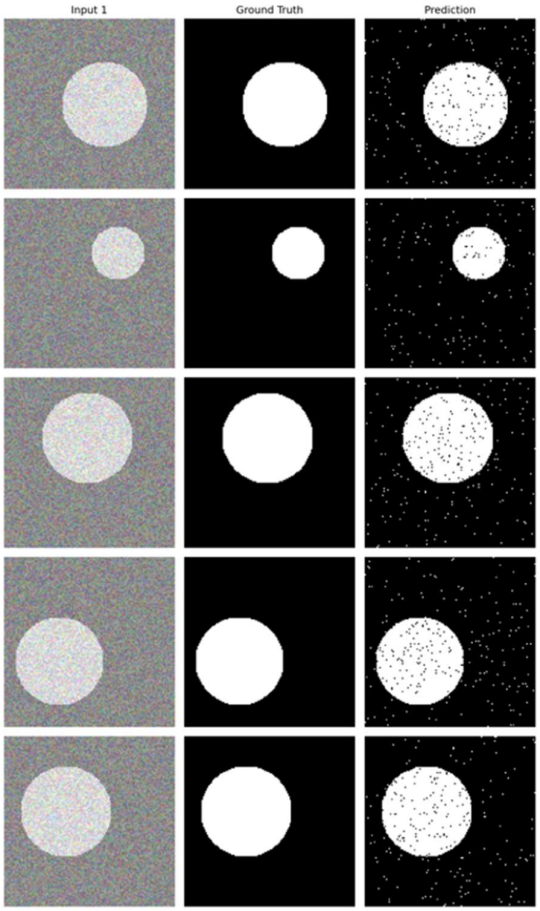


Image	IoU
1	0.928
2	0.882
3	0.935
4	0.930
5	0.938
Mean	0.923

Task 4 — U-Net vs Mask R-CNN: When to Use Each (25 marks)

Instructions:

This is a **written analysis task** — no additional coding required.

Part A — Comparison Table:

Complete the following table in your notebook with accurate, specific entries:

Part A — Comparison Table

Criterion	U-Net (Semantic)	Mask R-CNN (Instance)
Output type	Dense pixel-wise class label map (H x W x C)	Per-instance boxes + class scores + 28x28 masks
Can distinguish two objects of same class?	No - all pixels of a class share one label	Yes - each instance gets its own mask and ID
Typical inference speed	Fast (single forward pass, fully convolutional)	Slower (backbone + RPN + RoI heads + mask head)
Training data requirement	Pixel-level masks for each class; can work with smaller datasets	Instance-level annotated masks/boxes; typically larger datasets
Best suited dataset size	Small to medium (originally designed for scarce biomedical data)	Medium to large (e.g., COCO-scale)
Handles overlapping objects?	No - overlapping same-class objects merge into one region	Yes - RoI Align handles overlapping instances independently
Architecture complexity	Simpler - single encoder-decoder with skip connections	More complex - backbone + FPN + RPN + multiple task heads
Primary evaluation metric	IoU / pixel accuracy / Dice coefficient	mAP (box and mask), mask IoU

Summary: U-Net is the better choice when the task is dense per-pixel classification of a scene (e.g., medical lesion segmentation, satellite land-cover mapping) where counting or separating individual instances of the same class is not required, training data is limited, and fast inference is important. Mask R-CNN is the better choice when individual object instances must be detected, counted, and separately delineated (e.g., counting cells in a microscopy image, segmenting individual pedestrians or vehicles in a street scene), at the cost of greater architectural complexity, larger annotation requirements, and slower inference.

LAB # 14
Complex Computing Activity

Performance Metric	Exceeds Expectations (2–2.5)	Meets Expectations (1–1.5))	Does Not Meet Expectations (0–1)	Marks (out of 10)
Experiment Design (2 Marks)	Designs a well-structured, innovative CV project that clearly integrates at least three lab concepts with creative and well-defined objectives; choice of dataset and pipeline is appropriate for the problem.	Designs a functional project with moderate lab integration and limited creativity; pipeline is basic but coherent.	Fails to integrate at least three lab concepts; project design is incoherent or too trivial to demonstrate meaningful CV skills.	
Implementation & Execution (3 Marks)	All chosen CV techniques are correctly implemented; code is clean, well-commented, modular, and produces accurate results across all pipeline stages.	Code works with minor errors; one pipeline stage incomplete or partially implemented; results mostly correct.	Implementation is broken, missing major components, or produces incorrect outputs across most stages.	
Problem Solving & Adaptability (2 Marks)	Demonstrates strong analytical thinking; handles edge cases (e.g., noisy frames, lighting variation, low confidence detections) creatively and documents solutions clearly.	Addresses core requirements adequately; limited handling of edge cases or failure scenarios.	Unable to handle basic challenges; no evidence of debugging or analysis of failure cases.	
Report & Presentation (3 Marks)	Well-organised report with clear pipeline diagram, lab integration mapping, metric analysis, representative output screenshots, and professional code comments; engaging and clear presentation or demo.	Moderately detailed report; basic visuals; presentation is clear but lacks pipeline diagram or metric depth.	Report or presentation is poorly structured, missing visuals or lab integration mapping, or lacks explanation of results.	

LAB # 14

COMPLEX COMPUTING ACTIVITY

OBJECTIVE

Design and implement a complex, multi-faceted Computer Vision project. The goal is to demonstrate end-to-end mastery: image/video preprocessing, classical filter implementation, deep learning model design, real-time pipeline development, evaluation, and visual interpretation.

Your project must integrate at least **three distinct concepts** from previous labs. In your report, explicitly state which lab concepts you integrate and where in your pipeline they appear.

PROJECT DESCRIPTION

This is your final open-ended project. You are expected to design and build a significant CV application that goes beyond any single lab. Choose a real-world domain and build a complete working system that combines classical Computer Vision techniques with deep learning, demonstrating that you understand not just how to run code, but why each component of the pipeline exists and what it contributes to the final result.

PROJECT IDEAS (pick one or propose your own)

1) Smart Surveillance & Crowd Analytics System

What: Build a fixed-camera surveillance pipeline that processes a video stream, detects and counts people, draws motion trails, and raises an alert when crowd density in a defined zone exceeds a threshold.

Integrations Required:

- Video frame pipeline using `cv2.VideoCapture` and `cv2.VideoWriter`
- Gaussian blur preprocessing + Canny/Sobel edge detection on sampled frames to analyse scene structure
- YOLO object detection filtered to the `person` class for crowd detection
- Centroid-based object trailing across frames to visualise movement patterns
- OpenCV drawing functions to annotate bounding boxes, trails, zone boundaries, alert text, and density heatmap overlay.

Key Tasks: Define a polygonal restricted zone using `cv2.polylines`. Count people inside the zone per frame using point-in-polygon logic. Trigger a visual alert (red zone fill + warning text) when count exceeds a threshold. Save the fully annotated output video. Plot a density-over-time graph. Report counting accuracy against manual ground truth.

2) Automated Plant Disease Classifier with Filter Visualisation

What: Build an end-to-end plant disease classification system. Use classical filters to analyse leaf texture, train a CNN from scratch, fine-tune a pretrained model, and deploy the best model on a folder of unseen leaf images with annotated output.

Dataset: PlantVillage Dataset, 38 disease classes across multiple crop species

Integrations Required:

- Image loading, resizing, colour conversion, and saving pipeline
- Manual Sobel and Laplacian filter implementation using NumPy kernels and `cv2.filter2D` applied to sample leaf images to visualise disease texture patterns
- TensorFlow dataset pipeline with augmentation, batching, and prefetching
- CNN from scratch — 3 conv blocks + BatchNorm + Dropout + Dense head
- Any pretrained model except YOLO frozen feature extraction and fine-tuning with ReduceLROnPlateau scheduler
- Feature map visualisation of the first Conv2D layer for healthy vs diseased leaf inputs

Key Tasks: Build and compare CNN from scratch vs any pretrained model frozen vs fine-tuned. Report val accuracy, parameters, and training time for all three. Show confusion matrix for the best model. Run the best model on 20 unseen leaf images and annotate each with predicted disease class and confidence score using `cv2.putText`. Display the Sobel filter response grid across 5 disease classes and discuss what texture differences are visible.

---

3) Real-Time Driver Assistance, Lane & Vehicle Detection

What: Build a simulated driver assistance pipeline that processes dashcam footage, detects lane boundaries using classical edge detection, detects vehicles using YOLO, and overlays a heads-up display (HUD) with lane status and vehicle proximity warnings.

Dataset / Input: Any dashcam footage from YouTube (download a 2–3 minute clip using yt-dlp) or the KITTI Vision Dataset

Integrations Required:

- Frame-by-frame video processing pipeline with VideoWriter output
- Gaussian blur + Canny edge detection + Sobel gradient analysis for lane line detection
- Manual filter implementation (NumPy Sobel kernel) applied to 10 sampled frames to show gradient maps
- YOLO detection filtered to vehicle classes (car, bus, truck, motorcycle) with class-specific bounding box colours
- Vehicle counting line and centroid trailing to track vehicle paths in the scene
- OpenCV drawing functions for HUD overlay: lane boundary lines, vehicle boxes, proximity warning zone, speed estimate text

Key Tasks: Implement Hough Line Transform (`cv2.HoughLinesP`) on the Canny edge map to detect lane markings. Classify lane status as "Lane Clear" or "Lane Departure" based on detected line angles. Draw detected lane lines in green on the original frame. For each detected vehicle, compute its bounding box area as a proxy for proximity, trigger a "Vehicle Too Close" warning overlay when area exceeds a threshold. Save fully annotated output video. Report vehicle detection accuracy and lane detection success rate across 50 sampled frames.

---

4) Face Emotion Recognition with Live Webcam Overlay

What: Train a CNN-based facial emotion classifier, deploy it on a live webcam feed, detect faces using Haar Cascade, classify each face's emotion, and overlay the result as a real-time augmented display with emotion history trailing.

Dataset: FER-2013 — 7 emotion classes: angry, disgust, fear, happy, neutral, sad, surprise (~35,000 grayscale 48×48 images)

Integrations Required:

- Image preprocessing: grayscale conversion, histogram equalisation, resizing
- OpenCV drawing functions: bounding boxes, emotion label text, confidence bar overlay
- Webcam video loop with real-time frame processing and annotated VideoWriter output
- Gaussian blur applied to each face crop before feeding into model
- TF dataset pipeline + CNN from scratch trained on FER-2013
- A pretrained model except YOLO fine-tuned for emotion classification, adapt input to 3-channel 96×96
- VGG-style double conv block variant tested as third architecture.

Key Tasks: Build and compare three architectures: CNN scratch, pretrained model frozen, fine-tuned. Report per-class F1-score and confusion matrix (7×7) for the best model. Deploy best model on live webcam: detect faces using `haarcascade_frontalface_default.xml`, crop and preprocess each face, run inference, overlay predicted emotion + confidence score. Maintain a `deque(maxlen=20)` of the last 20 predicted emotions per face and display the most frequent one as the "stable prediction" to reduce flickering. Save a 30-second annotated demo clip. Report inference FPS on CPU.

DELIVERABLES

Code:

- All source code in a well-commented Jupyter Notebook (`.ipynb`) or Python file (.py) with **Project Name and Roll No. at the top.**
- A `README.md` file with setup instructions, required libraries (`pip install` commands), and how to run the notebook
- All saved output videos and annotated image grids included in the submission folder

Project Report (3–5 pages) must include:

- Pipeline architecture diagram showing the flow from input to final output
- Dataset description: source, size, class distribution or frame count, and all preprocessing steps
- Model designs, layer architectures, hyperparameters, and training details
- Evaluation results: accuracy metrics, confusion matrix, counting accuracy, or FPS benchmarks depending on project type
- Comparison table for projects with multiple models (val accuracy, parameters, training time)
- Challenges faced, limitations of the system, and suggested improvements

Presentation & Demonstration:

- Live demo or recorded video (60–90 seconds) showing the working system to the class or instructor
- Must show: real input going in, pipeline processing, and annotated output coming out

CCP

1) Smart Surveillance & Crowd Analytics System

What: Build a fixed-camera surveillance pipeline that processes a video stream, detects and counts people, draws motion trails, and raises an alert when crowd density in a defined zone exceeds a threshold.

Integrations Required:

- Video frame pipeline using `cv2.VideoCapture` and `cv2.VideoWriter`
- Gaussian blur preprocessing + Canny/Sobel edge detection on sampled frames to analyse scene structure
- YOLO object detection filtered to the `person` class for crowd detection
- Centroid-based object trailing across frames to visualise movement patterns
- OpenCV drawing functions to annotate bounding boxes, trails, zone boundaries, alert text, and density heatmap overlay.

Key Tasks: Define a polygonal restricted zone using `cv2.polylines`. Count people inside the zone per frame using point-in-polygon logic. Trigger a visual alert (red zone fill + warning text) when count exceeds a threshold. Save the fully annotated output video. Plot a density-over-time graph. Report counting accuracy against manual ground truth.

Code:

```
# Smart Surveillance & Crowd Analytics System
# 23-AI-21, 23-AI-45, 23-AI-47

import cv2
import numpy as np
from ultralytics import YOLO
from collections import defaultdict, deque
import matplotlib
matplotlib.use('Agg')                    # prevents display conflicts on Windows
import matplotlib.pyplot as plt

VIDEO_PATH        = "crowd_video2.mp4"
OUTPUT_PATH       = "output_annotated.avi"
DENSITY_THRESHOLD = 5
TRAIL_LENGTH      = 25
SAMPLE_INTERVAL   = 60
MAX_SAMPLES       = 5
SKIP_FRAMES       = 3    # run YOLO every 3rd frame — speeds up preview

#  LOAD YOLO MODEL
print("Loading YOLO model...")
```

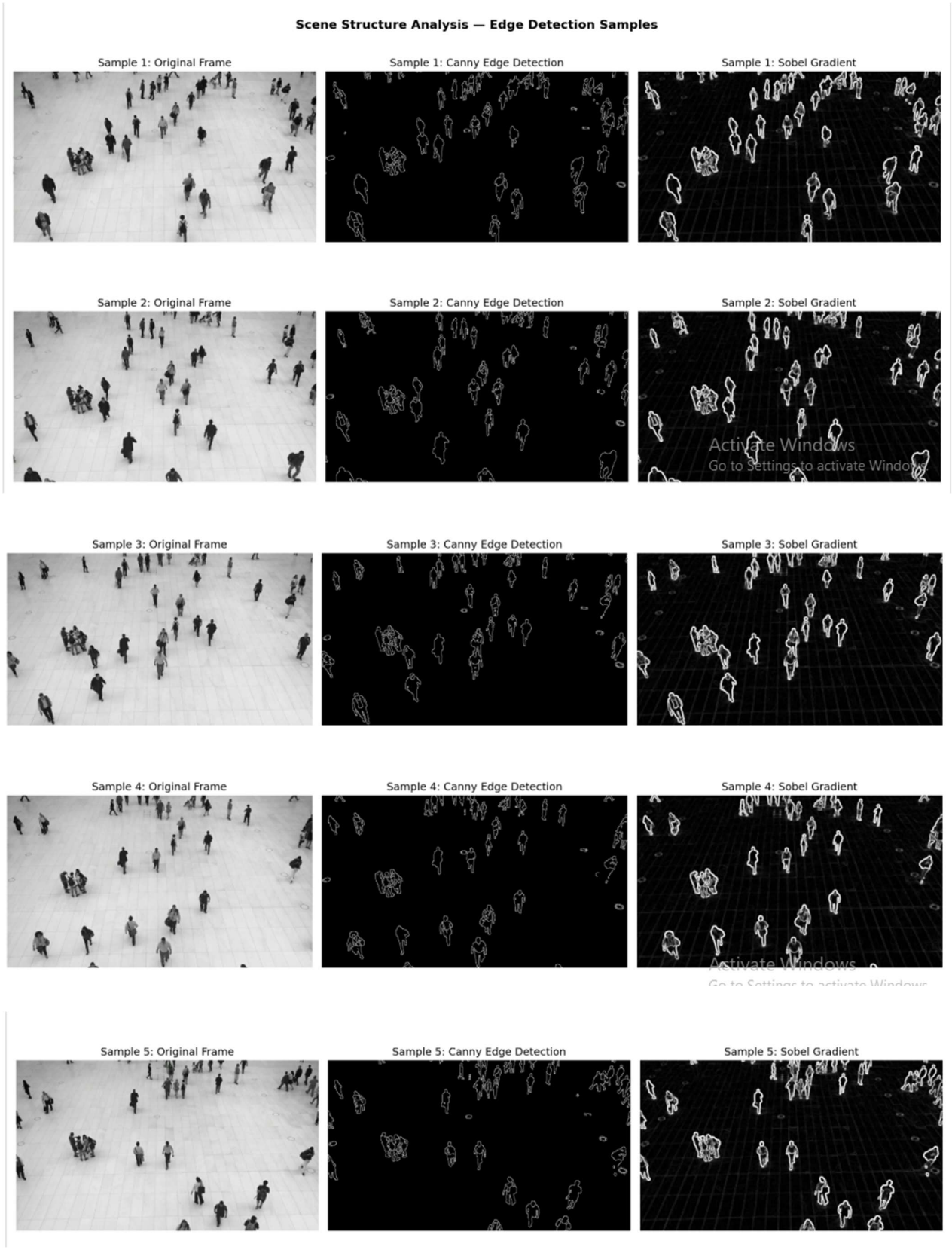
```
model = YOLO("yolov8n.pt")
print("Model loaded!\n")

def get_centroid(x1, y1, x2, y2):
    return (int((x1 + x2) / 2), int((y1 + y2) / 2))

def is_inside_zone(point, zone_polygon):
    result = cv2.pointPolygonTest(
        zone_polygon,
        (float(point[0]), float(point[1])),
        False
    )
    return result >= 0

def apply_canny_edges(frame):
    gray    = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    blurred = cv2.GaussianBlur(gray, (5, 5), 0)
    edges   = cv2.Canny(blurred, threshold1=50, threshold2=150)
    return edges

def apply_sobel(frame):
    gray      = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    blurred   = cv2.GaussianBlur(gray, (5, 5), 0)
    sobel_x   = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
    sobel_y   = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)
    magnitude = cv2.magnitude(sobel_x, sobel_y)
    return np.uint8(np.clip(magnitude, 0, 255))

def update_heatmap(heatmap, centroid, sigma=20):
    cv2.circle(heatmap, centroid, sigma, 1, -1)
    return heatmap

#  SAVE EDGE DETECTION SAMPLES
def save_edge_samples(samples):
    n = len(samples)
    fig, axes = plt.subplots(n, 3, figsize=(15, 4 * n))
    if n == 1:
        axes = [axes]

    for i, (original, edges, sobel) in enumerate(samples):
        axes[i][0].imshow(cv2.cvtColor(original, cv2.COLOR_BGR2RGB))
        axes[i][0].set_title(f"Sample {i+1}: Original Frame")
        axes[i][0].axis("off")

        axes[i][1].imshow(edges, cmap="gray")
        axes[i][1].set_title(f"Sample {i+1}: Canny Edge Detection")
        axes[i][1].axis("off")

        axes[i][2].imshow(sobel, cmap="gray")
        axes[i][2].set_title(f"Sample {i+1}: Sobel Gradient")
        axes[i][2].axis("off")
```

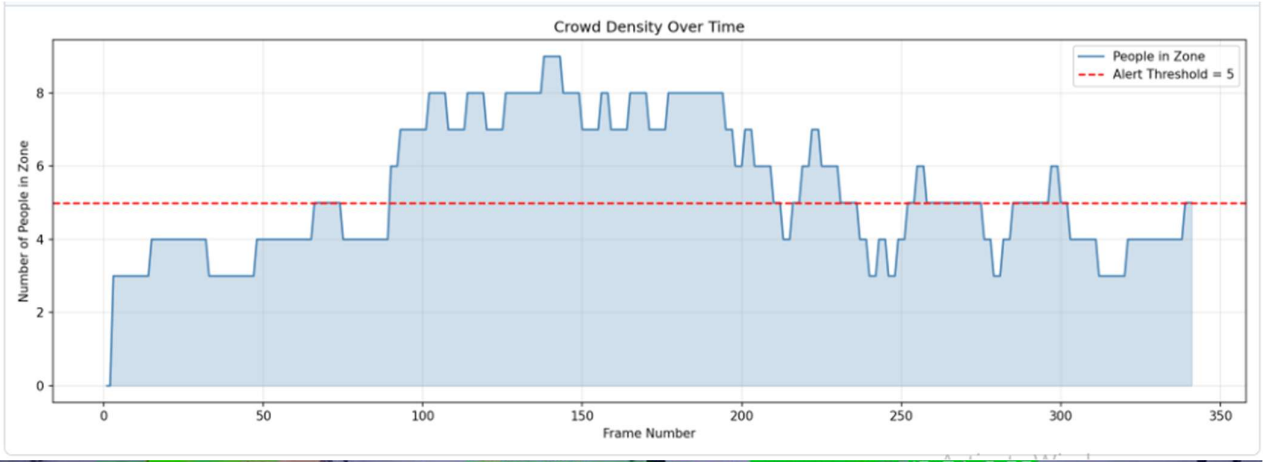
```
    plt.suptitle("Scene Structure Analysis — Edge Detection Samples",
                 fontsize=14, fontweight="bold")
    plt.tight_layout()
    plt.savefig("edge_samples.png", dpi=150)   # save BEFORE show
    plt.close('all')
    print("Saved → edge_samples.png")

def main():

    cap = cv2.VideoCapture(VIDEO_PATH)
    if not cap.isOpened():
        print(f"[ERROR] Cannot open: {VIDEO_PATH}")
        return

    W     = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    H     = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    FPS   = cap.get(cv2.CAP_PROP_FPS)
    TOTAL = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    print(f"Detected FPS: {FPS}")
    print(f"Video info: {W}x{H} @ {FPS} fps | {TOTAL} total frames\n")

    # Resize to 854 width — faster on CPU
    SCALE = 854 / W
    W = 854
    H = int(H * SCALE)

    # Restricted zone — proportional to frame size
    ZONE = np.array([
        [int(W * 0.25), int(H * 0.40)],
        [int(W * 0.75), int(H * 0.40)],
        [int(W * 0.85), int(H * 0.90)],
        [int(W * 0.15), int(H * 0.90)],
    ], dtype=np.int32)

    # VideoWriter
    fourcc = cv2.VideoWriter_fourcc(*"MJPG")
    out    = cv2.VideoWriter(OUTPUT_PATH, fourcc, FPS, (W, H))

    # Data structures
    trails      = defaultdict(lambda: deque(maxlen=TRAIL_LENGTH))
    heatmap     = np.zeros((H, W), dtype=np.float32)
    density_log = []
    frame_log   = []
    edge_samples = []
    frame_count  = 0
    last_results = None    # stores last YOLO result for skipped frames

    print("Processing... Press Q to stop early.\n")

    cv2.namedWindow("Crowd Surveillance System", cv2.WINDOW_NORMAL)

    #  MAIN LOOP
```

```
while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame_count += 1
    frame = cv2.resize(frame, (W, H))

    # Edge detection samples
    if frame_count % SAMPLE_INTERVAL == 0 and len(edge_samples) < MAX_SAMPLES:
        edges = apply_canny_edges(frame)
        sobel = apply_sobel(frame)
        edge_samples.append((frame.copy(), edges, sobel))
        print(f"  Edge sample {len(edge_samples)} saved at frame {frame_count}")

    # YOLO — only every SKIP_FRAMES frames
    if frame_count % SKIP_FRAMES == 0:
        last_results = model.track(frame, classes=[0], persist=True, verbose=False)
    results = last_results

    # Draw restricted zone
    zone_overlay = frame.copy()
    cv2.fillPoly(zone_overlay, [ZONE], (0, 220, 220))
    frame = cv2.addWeighted(frame, 0.82, zone_overlay, 0.18, 0)
    cv2.polylines(frame, [ZONE], isClosed=True, color=(0, 200, 200), thickness=2)
    cv2.putText(frame, "RESTRICTED ZONE",
                (ZONE[0][0], ZONE[0][1] - 8),
                cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 200, 200), 2)

    # Process detections
    people_in_zone = 0
    boxes_data = results[0].boxes if results is not None else None

    if boxes_data is not None and len(boxes_data) > 0:
        for box in boxes_data:
            x1, y1, x2, y2 = map(int, box.xyxy[0])
            track_id   = int(box.id[0]) if box.id is not None else -1
            confidence = float(box.conf[0])

            if confidence < 0.35:
                continue

            centroid = get_centroid(x1, y1, x2, y2)

            if track_id >= 0:
                trails[track_id].append(centroid)

            heatmap = update_heatmap(heatmap, centroid)

            in_zone   = is_inside_zone(centroid, ZONE)
            box_color = (0, 0, 255) if in_zone else (0, 255, 0)
            if in_zone:
```

```
            people_in_zone += 1

        cv2.rectangle(frame, (x1, y1), (x2, y2), box_color, 2)

        label = f"ID:{track_id} {confidence:.2f}" if track_id >= 0 else f"{confidence:.2f}"
        (tw, th), _ = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX, 0.5, 1)
        cv2.rectangle(frame, (x1, y1 - th - 6), (x1 + tw + 2, y1), box_color, -1)
        cv2.putText(frame, label, (x1 + 1, y1 - 4),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)

        if track_id >= 0:
            trail_list = list(trails[track_id])
            for j in range(1, len(trail_list)):
                thickness = max(1, int((j / len(trail_list)) * 3))
                cv2.line(frame, trail_list[j-1], trail_list[j], (0, 140, 255), thickness)
                cv2.circle(frame, trail_list[j], 2, (0, 200, 255), -1)

# Crowd Alert
alert_active = people_in_zone > DENSITY_THRESHOLD
if alert_active:
    alert_overlay = frame.copy()
    cv2.fillPoly(alert_overlay, [ZONE], (0, 0, 255))
    frame = cv2.addWeighted(frame, 0.65, alert_overlay, 0.35, 0)

    alert_text = "!! CROWD ALERT !!"
    (aw, ah), _ = cv2.getTextSize(alert_text, cv2.FONT_HERSHEY_DUPLEX, 1.4, 3)
    ax = int((W - aw) / 2)       # horizontally centered
    ay = int(H * 0.88)           # near bottom of frame
    # Dark background behind alert text for visibility
    cv2.rectangle(frame, (ax - 10, ay - ah - 10), (ax + aw + 10, ay + 10),
                  (0, 0, 0), -1)
    cv2.putText(frame, alert_text, (ax, ay),
                cv2.FONT_HERSHEY_DUPLEX, 1.4, (0, 0, 255), 3)

# Heatmap overlay
norm_heat  = cv2.normalize(heatmap, None, 0, 255, cv2.NORM_MINMAX)
color_heat = cv2.applyColorMap(np.uint8(norm_heat), cv2.COLORMAP_JET)
frame      = cv2.addWeighted(frame, 0.75, color_heat, 0.25, 0)

# Info panel (top-left HUD)
total_people = len(boxes_data) if boxes_data is not None else 0
status_text  = "ALERT!" if alert_active else "Normal"
info_lines   = [
    f"Frame: {frame_count}/{TOTAL}",
    f"People detected: {total_people}",
    f"In zone: {people_in_zone}",
    f"Threshold: {DENSITY_THRESHOLD}",
    f"Status: {status_text}",
]
cv2.rectangle(frame, (5, 5), (285, 150), (0, 0, 0), -1)
cv2.rectangle(frame, (5, 5), (285, 150), (120, 120, 120), 1)
for i, line in enumerate(info_lines):
```

```
            color = (0, 0, 255) if ("ALERT" in line and alert_active) else (255, 255, 255)
            cv2.putText(frame, line, (12, 28 + i * 25),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 1)

        # Log density
        density_log.append(people_in_zone)
        frame_log.append(frame_count)

        # Write & display
        out.write(frame)
        cv2.imshow("Crowd Surveillance System", frame)

        # waitKey delay matches FPS so preview plays at real speed
        if cv2.waitKey(int(1000 / FPS)) & 0xFF == ord("q"):
            print("Stopped early.")
            break

        if frame_count % 100 == 0:
            print(f"  Frame {frame_count}/{TOTAL} | In zone: {people_in_zone}")

    # Cleanup
    cap.release()
    out.release()
    cv2.destroyAllWindows()
    print(f"\nDone! Saved → {OUTPUT_PATH}")

    # Density graph
    plt.figure(figsize=(14, 5))
    plt.plot(frame_log, density_log, color="steelblue", linewidth=1.5, label="People in Zone")
    plt.axhline(y=DENSITY_THRESHOLD, color="red", linestyle="--",
                linewidth=1.5, label=f"Alert Threshold = {DENSITY_THRESHOLD}")
    plt.fill_between(frame_log, density_log, alpha=0.25, color="steelblue")
    plt.xlabel("Frame Number")
    plt.ylabel("Number of People in Zone")
    plt.title("Crowd Density Over Time")
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.savefig("density_graph.png", dpi=150)
    plt.close('all')                    # close before opening edge samples
    print("Density graph saved → density_graph.png")

    # Edge samples
    if edge_samples:
        save_edge_samples(edge_samples)
    else:
        print("No edge samples collected.")

if __name__ == "__main__":
    main()
```




Crowd Density Over Time
People in Zone
Alert Threshold = 5
Number of People in Zone
Frame Number
0
2
4
6
8
0
50
100
150
200
250
300
350


Frame: 91/341
People detected: 36
In zone: 6
Threshold: 5
Status: ALERT!
RESTRICTED ZONE
!! CROWD ALERT !!
output_annotated (1)
0:00:03
0:00:10
Activate Windows
Go to Settings to activate Windows.